

The Price of Delay: infrastructure lock-in and the carbon tax lock-in premium in the oil sector

Léo Jean*

September 2026

[Click here for the latest version](#)

Abstract

A carbon tax high enough to prevent the development of an oil field is not necessarily high enough to stop its production once the initial fixed cost is sunk. Delaying climate policy therefore commits emissions that the delayed policy can no longer remove. I measure this lock-in on 25,336 oil fields, giving each field separate breakevens for development and production instead of the single cost per barrel of existing supply models. A 1.5°C-aligned carbon tax delayed from 2016 to 2030 overshoots the oil carbon budget by 19%, and nearly a quarter of this overshoot is committed beyond 2030 by fields that a timely tax would have deterred. A planner setting the 2030 tax with a single cost per barrel treats these fields as if they could still be deterred, leaving a residual overshoot of 17 GtCO_{2eq}, 12% of the remaining budget. Closing this gap requires a lock-in premium of \$3.6/tCO_{2eq}. Locked-in fields are resilient to the premium, which strands production in the Middle East and Russia instead, while Brazil and Africa end up producing more than under a timely policy. Because lock-in is highly concentrated, early targeted restrictions on a few hundred projects remove a large share of its cost.

KEYWORDS: Climate change, fossil fuels, oil, carbon tax, climate policy delay, carbon lock-in, stranded assets, infrastructure lock-in, supply-side climate policy.

JEL CODES: Q35, Q52, Q54, Q58, L71.

*Paris School of Economics, ENS - PSL. Email: leo.jean@psemail.eu.

1 Introduction

In 2015, the Paris Agreement set the objective of holding global warming well below 2°C and pursuing efforts to limit it to 1.5°C . Meeting this objective requires leaving a large share of known oil reserves in the ground (McGlade and Ekins, 2015; Welsby et al., 2021), and carbon pricing is widely regarded as the most effective way to get there. Yet in the decade since, carbon prices have remained limited in coverage and far below the levels these objectives require (Metcalf, 2019). Oil companies do anticipate some form of policy, but they screen new projects with internal carbon prices well below what the objectives imply, often only to stress-test existing assets (Ahluwalia, 2017). Recent evidence suggests why. Firms invest according to the carbon price they find credible, and under policy uncertainty this price sits far below the announced one (Dolls et al., 2026). The period since Paris is therefore best described as a delay in carbon pricing, and one that investors largely did not expect to end.

A delay of this kind has two consequences for the oil sector. The first is overproduction: fields already in operation keep producing barrels that a timely tax would have retired. The second is investment: firms develop new fields that a timely tax would have deterred. The difference between the two is the timing of excess emissions. Costly fields are immediately phased out once the tax is implemented, and the associated overproduction stops. By contrast, investments cannot be recovered, and newly built fields can continue producing for decades if their operating costs are low enough. In other words, a tax sufficient to deter an oil project is not necessarily sufficient to stop its production once upfront capital expenses are sunk. A delay therefore leaves behind a stock of committed emissions that the delayed tax cannot remove, and the tax must rise beyond what overproduction alone would justify in order to meet the same carbon budget.

Most models of oil supply do not capture this second consequence. Any model that represents supply by a single curve assumes by construction that a field occupies the same position on that curve whether or not its development capital has already been spent. So do the supply blocks of the integrated assessment models used to price the delay of climate policy (Grubb et al., 2021). More recently, the emerging literature using asset-level models of oil supply makes the same assumption by using a single production cost per barrel for each asset, encompassing both fixed development costs and operating costs (Heal and Schlenker, 2019; Asker et al., 2019; Coulomb et al., 2026). Under this convention, any tax high enough to have deterred a project is also high enough to shut it down once it is built, making lock-in impossible. This means that any model using one of these methods when assessing the cost of inaction will only account for the overproduction happening during the delay, and overlook entirely the new emissions the delay locks in.

This paper measures what the delay since Paris costs in the oil sector once both overproduction and lock-in effects are accounted for, and investigates the role played by the latter and its redistributive consequences.

I develop a simple model of oil extraction, in which a field sells its oil at a constant price, and a constant carbon tax is deferred. Before development, a field must cover both its fixed development cost and its operating costs, rising with age. Once the capital is sunk, only operating costs remain, and the tax level it can survive jumps upward. This jump is the *lock-in zone*: the range of carbon taxes high enough to have deterred the development of the field, but too low to stop its production once the field exists. The size of this lock-in zone is dictated by capital intensity, and its position by operating costs and carbon intensity. If the tax falls in this zone, and the field is young enough to cover its operating costs at implementation date, then the field commits emissions post-delay that would not have existed in a timely policy world. Summing these emissions gives the cost of the delay attributed to lock-in. I then define the tax needed after the delay to make up for both overproduction and these overcommitted emissions, and compare it to the tax a planner using one single production cost per field (hereafter a naive planner) would set. The difference between the two taxes is the *carbon tax lock-in premium*, and the emissions left unabated at the naive tax constitute the *residual overshoot*. I extend the model to a tax increasing over time and show that the same conditions hold, with the development threshold now depending on the year at which a field was developed.

I then take the model to asset-level data from Rystad Energy’s UCube database, covering 25,336 oil fields with positive oil reserves in 2016. For each field, I retrieve its development cost, its operating costs per barrel rising over time, and estimate its life-cycle carbon intensity covering extraction, refining and final use. Investments observed up to 2025 and Rystad’s baseline forecast thereafter provide the counterfactual without a tax. The reference oil price is constant at \$80/bbl and the discount rate is 10%.

In a first exercise, I compare two scenarios: one in which a \$57.2/tCO₂eq carbon tax, aligned with the oil carbon budget set by [Welsby et al. \(2021\)](#), is implemented in 2016, and then rises at a 2% rate per year; a second in which the same tax is delayed to 2030, starting at the level the timely tax would have reached in that year. The delay overshoots the oil carbon budget by 66.7 GtCO₂eq, 19% of the budget and about four years of global oil emissions at their current rate.¹ Intensive and extensive margins provide 62% and 38% of this overshoot respectively. More interestingly, almost two thirds of the extensive margin originates from emissions released after the tax is implemented. Shale and oil sands account

¹Total lifecycle emissions of oil amount to 16.5 GtCO₂eq in 2025, using estimated carbon intensities and Rystad data on production.

for most of the overproduction before 2030, while offshore fields concentrate three quarters of post-2030 emissions, and deepwater alone 39%. The model explains this: shale projects are cheap to develop and expensive to operate, so their lock-in zones are narrow and low, while deepwater projects are capital-intensive and cheap to operate once built.

In a second exercise, I focus on the tax needed in 2030 to make up for the delay, and how far from it the naive planner ends up by overlooking locked-in emissions. Meeting the 144.9 GtCO₂eq left in the budget in 2030 requires a tax of \$70.5, while the naive planner sets \$66.8. The lock-in premium is therefore \$3.6/tCO₂eq, 5% of the tax². At the naive tax, fields locked-in because of the delay entail a residual overshoot of 17.1 GtCO₂eq, 12% of the remaining budget. The premium and the residual overshoot resist to realistic expectations of future climate policy by investors: four fifths remain when investors screen their projects with half the required tax, the level closest to the internal carbon prices firms report ([Ahluwalia, 2017](#)).

The first comes from the fact that lock-in is concentrated and resilient. The 17.1 GtCO₂eq committed beyond 2030 come from 694 fields, 41 of which carry half of the total. Raising the tax to the level the budget requires leaves 92% of the stock in place. The premium therefore strands production in other fields, which means that the redistribution of stranded assets entailed by the delay is twofold. First, quantities overproduced during the delay needs to be compensated by the stranding of other assets after it (through the increase in tax from \$57.2 to \$66.8). Then, locked-in emissions induced by the delay, because they are resilient to higher taxes, require stranding additional assets to meet the budget (through the increase in tax from \$66.8 to \$70.5). North America gains through both channels, since its high-cost shale and oil sands fields overproduce during the delay and its deepwater projects lock in production afterwards. The Middle East and Russia lose through both, because a higher tax retires their low-cost fields earlier without affecting them during the delay. More interestingly, lock-in can reverse the incidence of the delay: Brazil and Africa lose on overproduction, gain on lock-in, and the second effect dominates, leaving them with more cumulative production than under a timely tax.

This concentration makes targeted policy effective at alleviating the cost from lock-in. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 40% and brings the premium down to \$2.5, while a ban on new shale approvals leaves both close to their initial level. Banning the fifty projects that carry the most overcommitted emissions cuts the residual overshoot by more than half. Acting later costs most of the gain. Applied

²All tax levels in the paper are expressed in 2016-anchored level. The tax increasing by 2% annually, these tax levels correspond respectively to \$88.1 and \$93 in 2030. Additionally, since there is no demand side, the tax is not passed through to consumers, and should thus be interpreted as the level of tax borne by producers.

from 2025, the deepwater ban cuts the residual overshoot by 11%, since most of the projects concerned have been approved by then.

This paper relates to several strands of literature.

A first strand uses asset-level data to study the oil market. [Asker et al. \(2019\)](#) quantify the misallocation of extraction across fields and show that OPEC's market power accounts for a large share of it, and [De Canniere \(2024\)](#) estimates a dynamic cartel-fringe model to study how oil supply responds to a carbon tax. [Asker et al. \(2024\)](#) and [Coulomb et al. \(2026\)](#) turn to the emissions side and measure the excess emissions caused by extracting carbon-intensive barrels before cleaner ones. These papers work with a single production cost per barrel for each field, development costs being spread over the field's lifetime. This is a reasonable simplification when the question is how a given quantity of oil is allocated across fields. It is not when the question is the timing of policy, because for a field developed before the policy, part of that cost is already sunk when the policy arrives. I build on the same data and the same field-level approach, and separate development costs from operating costs, which is enough for lock-in to appear.

Treating investment as a commitment of emissions toward the future is not new in the theoretical literature. In [Venables \(2014\)](#), oil supply results from a sequence of field openings, each requiring a fixed investment, after which the field delivers a quantity that cannot be freely adjusted, so that prices affect the timing of openings rather than the flow from existing fields. [Okullo and Reynès \(2016\)](#) use a similar structure, with production, capacity investment and reserve development as distinct decisions, to study OPEC behaviour, and [Kellogg \(2024\)](#) uses a calibrated model with costly drilling and declining wells production to show that an anticipated fall in demand reduces investment more than it accelerates extraction. The empirical support for this view originates from [Anderson et al. \(2018\)](#), who document that producers respond to prices by drilling more or fewer wells rather than by adjusting the flow from wells already drilled. [Wachtmeister and Höök \(2020\)](#) take this logic to data by comparing typical investment and production profiles of giant conventional fields and tight oil wells, and find that high upfront capital and low operating costs keep production going once the capital is spent, so that conventional fields lock in far more future production than tight oil. Their exercise uses representative profiles with a common capital cost per barrel and no discounting, and remains illustrative. I apply the same logic to actual fields, with their own development costs, operating costs, carbon intensities and production paths, and ask what follows when climate policy arrives late.

The third strand studies the consequences of delaying climate policy, mainly through the use of integrated assessment models (IAMs). Most of these studies conclude that a delay until 2030 raises mitigation costs substantially while leaving the objectives attainable ([Riahi](#)

et al., 2015; Bertram et al., 2015; Luderer et al., 2013; Kotlikoff et al., 2021), although the decarbonisation rates required after 2030 in these pathways are themselves uncertain (Van Vuuren et al., 2016). Reviewing twenty-eight such models, Grubb et al. (2021) find that inertia, induced innovation and path dependence are at best partially represented, and that models of the DICE family treat abatement in each period as independent of past abatement. IAMs with the richest representation of inertia and learning (IMACLIM-R, REMIND and WITCH), compared in the RECIPE project (?), are also those finding that a delay until 2030 puts the 450 ppm objective out of reach (Jakob et al., 2012; Luderer et al., 2013; Bertram et al., 2015). However, in these models, inertia relates to the fixed investment costs on the demand side (power generation mainly), while fossil fuels supply enters as a simple cost curve increasing in cumulative extraction. This paper addresses two shortcomings of this literature: it introduces inertia on the supply side, and identifies the mechanisms underlying lock-in effects and the specific investments leading to it, mechanisms that the complexity of IAMs does not allow to understand (Grubb et al., 2021).

The fourth literature focuses on carbon lock-in. Unruh (2000) introduced the concept and Seto et al. (2016) distinguish its infrastructural, institutional and behavioural forms, the first being the subject of this paper. Following the commitment accounting of Davis and Socolow (2014), Tong et al. (2019) show that existing energy infrastructure already commits most of the 1.5°C budget, and Erickson et al. (2015) and Trout et al. (2022) apply the same accounting to fossil fuel supply, the latter finding that fields already producing or under construction commit more emissions than a 1.5°C budget allows. Draeger et al. (2026) go one step further and impose the forecast production of committed oil projects as a floor in an integrated assessment model, which shows how much room they leave to other producers. These studies measure a stock, not a cost, and without a counterfactual they cannot say which part of it exists because policy was delayed. Comparing a timely tax with a delayed one separates the committed emissions the delay created from those any policy would have faced, and puts a price on the first. It also turns the committed stock into something policy can act on, since I identify the fields that carry it, the year each commitment became irreversible, the tax necessary to retire each one, and the targeted policies able to reach them.

The fifth literature studies stranded assets, which is the other side of the same accounting, since committed emissions somewhere implies stranded assets somewhere else. This literature has established the existence of unburnable reserves (Meinshausen et al., 2009; McCollum et al., 2014) and their uneven distribution across countries (McGlade and Ekins, 2015; Welsby et al., 2021). It has shown that the distribution depends on extraction costs and carbon intensities (Coulomb et al., 2026), traced the corresponding losses to their ultimate owners (Mercure et al., 2018; Semieniuk et al., 2022), and examined what producers do when they

expect the policy (Coulomb and Henriët, 2018; van der Ploeg and Rezai, 2020) and what compensation for halted projects would involve (Paletta et al., 2025; Draeger et al., 2026). In all of these studies, policy arrives at a given date and stranding is computed on a supply curve that carries no memory of investment. Delaying the policy changes the distribution through two channels that this paper separates: overproduction and locked-in emissions, both leading to a redistribution of stranded assets. The timing of climate policy, and not only its level, thus decides who pays for the transition.

Finally, the paper contributes to the case made for supply-side climate policy, on grounds of abatement certainty, administrative cost, public support and commitment value (Erickson et al., 2015; Green and Denniss, 2018; Green et al., 2024; Harstad and Holtsmark, 2024). Measuring lock-in field by field makes it possible to evaluate specific restrictions rather than the overall principle of it. I report what each one removes, at which tax, on whom the cost falls, how much is lost by acting in 2025 rather than in 2016, and why a badly targeted ban aimed at capital-light projects, or a tax on development capital, leaves the problem untouched or makes it worse.

The rest of the paper is organised as follows. Section 2 presents the simplified model of field development and production under a delayed carbon tax. It defines the lock-in zone of a field, shows its determining factors, decomposes the overshoot caused by the delay into emissions released before and after implementation, and defines the residual overshoot and the lock-in premium as two measures of the same error. Section 3 describes the data and methods used in the empirical exercise, as well as the assumptions made on costs, prices, carbon intensities and fiscal regimes. Section 4 reports the overshoot and its decomposition by supply segment and region, the emissions committed by new projects, the lock-in premium and the residual overshoot, and their sensitivity to anticipations, assumptions and the design of the tax. Section 5 locates the locked-in fields, shows how lock-in changes the incidence of the delay across countries and firms, and evaluates targeted policies. Section 6 concludes.

2 A simplified model of investment and lock-in under delayed carbon taxation

This section presents a simple model of investment in oil fields when a carbon tax is implemented late. It serves two purposes: defining the objects measured in the empirical part, and showing which field characteristics determine their size. The *lock-in zone* of a field is the range of carbon taxes that would have prevented its development but cannot stop its production once the development cost is sunk. *Committed emissions* at a given tax are the

future emissions of existing fields that survive this tax, and *overcommitted emissions* are the committed emissions of fields that a timely tax would have deterred. The *lock-in premium* is the additional tax required after the delay to offset overcommitted emissions, on top of the increase needed to compensate for overproduction during the delay. I use a constant tax to keep the model analytically tractable, and show in Section 2.6 that the main results carry over to an increasing tax. Closed-form expressions, which require no discounting and a parametric cohort of fields, are in Appendix A.1.

Production profiles are fixed by geology and by the initial design of the field, so that the tax affects when a field starts and stops producing, but not how much it produces in a given year. Okullo and Reynès (2016) model supply in the same way: production is bounded by installed capacity and by a geological limit on the share of reserves extracted each year, and capacity only expands through investment. Anderson et al. (2018) show that production from existing wells hardly responds to prices, while drilling does. With fixed profiles and development dates taken from the data, producers who expect the tax cannot accelerate extraction before it is implemented, which rules out the green paradox and makes the total cost of the delay a lower bound. Expecting the tax can also deter investment during the delay, which reduces lock-in, as shown in Section 2.5.

2.1 Setup

A carbon tax τ is levied on producers. Under the timely policy it is in place from 2016; under the delayed policy it is implemented at date Δ . From Δ onwards, both policies apply the same tax.

A field is characterised by a development cost C , paid when the field is developed; a production flow $\bar{q}$; a carbon intensity θ , so that it emits $\theta\bar{q}$ per period; and a selling price p , constant over time.³ Its operating cost per barrel increases with its age a , counted from the start of production,

$$c(a) = c_0 + \gamma a, \quad \gamma > 0. \quad (1)$$

Operating costs rise as a field gets depleted because reservoir pressure declines and more water is produced with the oil, which requires more energy to lift each barrel (Chermak and Patrick, 1995; Leighty and Lin, 2012), and the production breakevens computed from the data increase accordingly across production stages (Section 3). Capital intensity $\kappa \equiv C/\bar{q}$ is the development cost per barrel of annual capacity. Firms discount profits at rate δ , while emissions are not discounted.

³The selling price is the common reference price minus a discount that depends on the characteristics of the oil (Section 3.3).

2.2 Two breakevens and the lock-in zone

Once developed, a field of age a earns $p - c_0 - \gamma a - \theta\tau$ per barrel. It keeps producing as long as this margin is positive, that is as long as the tax does not exceed its *production breakeven* $\bar{\tau}_P(a)$, and it is phased out at age $a^*(\tau)$:

$$\bar{\tau}_P(a) \equiv \frac{p - c_0 - \gamma a}{\theta}, \quad a^*(\tau) = \frac{p - c_0 - \theta\tau}{\gamma}. \quad (2)$$

The production breakeven only depends on operating costs, since the development cost is sunk once the field produces. It is highest at start-up, $\bar{\tau}_P(0) = (p - c_0)/\theta$, and falls by γ/θ each year.

Before development, investors decide to invest in a field if its net present value is positive:

$$-C + \int_0^{a^*(\tau)} e^{-\delta a} \bar{q} (p - c_0 - \theta\tau - \gamma a) da \geq 0. \quad (3)$$

Equivalently, the discounted profit per barrel of capacity must cover the capital intensity of the field,

$$\bar{v}_\delta(\tau) \equiv \int_0^{a^*(\tau)} e^{-\delta a} (p - c_0 - \theta\tau - \gamma a) da \geq \kappa. \quad (4)$$

The discounted profit $\bar{v}_\delta(\tau)$ decreases with the tax, so the condition holds if and only if the tax does not exceed the *development breakeven* $\bar{\tau}_D$, defined by

$$\bar{v}_\delta(\bar{\tau}_D) = \kappa. \quad (5)$$

$\bar{\tau}_D$ is the model counterpart of the maximum viable tax computed field by field in Section 3. Since $\kappa > 0$, $\bar{\tau}_D < \bar{\tau}_P(0)$: the taxes between the two breakevens would have prevented the development of the field but do not stop its production once it exists. This interval is the field's lock-in zone. I describe it by its depth, its width and its height, which Figure 1 illustrates. Panel (a) plots the two breakevens against capital intensity for a field of a given age at implementation, and panel (b) plots them against the age at implementation for a field of a given capital intensity. In both panels, region II is the set of taxes at which the field is locked in.

Depth. The depth of the zone is the distance between the two breakevens at start-up, $D \equiv \bar{\tau}_P(0) - \bar{\tau}_D$. Expressed per barrel, $\theta D = p - c_0 - \theta\bar{\tau}_D$ is the margin at start-up at which the field just recovers its development cost, and from (4) it solves

$$\int_0^{\theta D/\gamma} e^{-\delta a} (\theta D - \gamma a) da = \kappa. \quad (6)$$

The depth reflects how a field's costs split between fixed and variable costs. The production breakeven only has to cover variable costs, while the development breakeven also has to cover the fixed development cost, so the larger the fixed cost per barrel of capacity, the further apart the two breakevens. In panel (a), the development breakeven falls as capital intensity rises and region II opens up; in panel (b), a more capital-intensive field has a lower horizontal line $\bar{\tau}_D$ and a taller region II. Expressed per barrel, the depth depends on capital intensity, on the slope of the operating cost path and on the discount rate, and neither on carbon intensity nor on the level of operating costs. Without discounting, $\theta D = \sqrt{2\gamma\kappa}$. Discounting lowers the value of developing a field without changing the decision to keep it producing, so it lowers $\bar{\tau}_D$ and deepens the zone: the undiscounted depth is a lower bound (Appendix A.1).

Width. A field only locks in emissions if it still produces when the tax arrives. In panel (b), its production breakeven falls with age with slope γ/θ , and the zone closes where it crosses the development breakeven, at age $a^*(\bar{\tau}_D) = \theta D/\gamma$. A field developed during the delay can therefore carry overcommitted emissions only if it is young enough at implementation, $a_\Delta < a^*(\bar{\tau}_D)$. The zone is wider for fields whose operating costs rise slowly, since a flatter production breakeven crosses the development breakeven later, and for capital-intensive fields, since their development breakeven is lower.

Height. The tax that meets the carbon budget must also fall inside the zone. The height of the zone is set by its top, $\bar{\tau}_P(0) = (p - c_0)/\theta$. A lower operating cost at start-up raises both breakevens by the same amount and moves the whole zone up without changing its depth; a lower carbon intensity raises both breakevens and also deepens the zone in tax units. In panel (b), this shifts both lines upwards. Oil sands illustrate why height matters: their capital intensity is close to that of deepwater fields, but their high operating costs and carbon intensity place a deep zone below the taxes the budget requires. Deepwater fields have a deep zone that sits high, while shale wells have a shallow zone that sits low.

2.3 Which fields lock in, and the overshoot they cause

Fields developed during the delay differ in their costs and in their age at implementation: a field developed at date $s \in [0, \Delta)$ is $a_\Delta = \Delta - s$ years old when the tax arrives. Its contribution to the overshoot, compared with the timely policy, depends on two questions. Would a timely tax have allowed its development? From (4), it would not if $\kappa > \bar{v}_\delta(\tau)$, that is if its discounted profit per barrel of capacity under the tax does not cover its capital intensity, or equivalently if $\tau > \bar{\tau}_D$. Fields developed under both policies are called *intensive*, and fields developed only because of the delay are called *delay-induced*. Is the field still

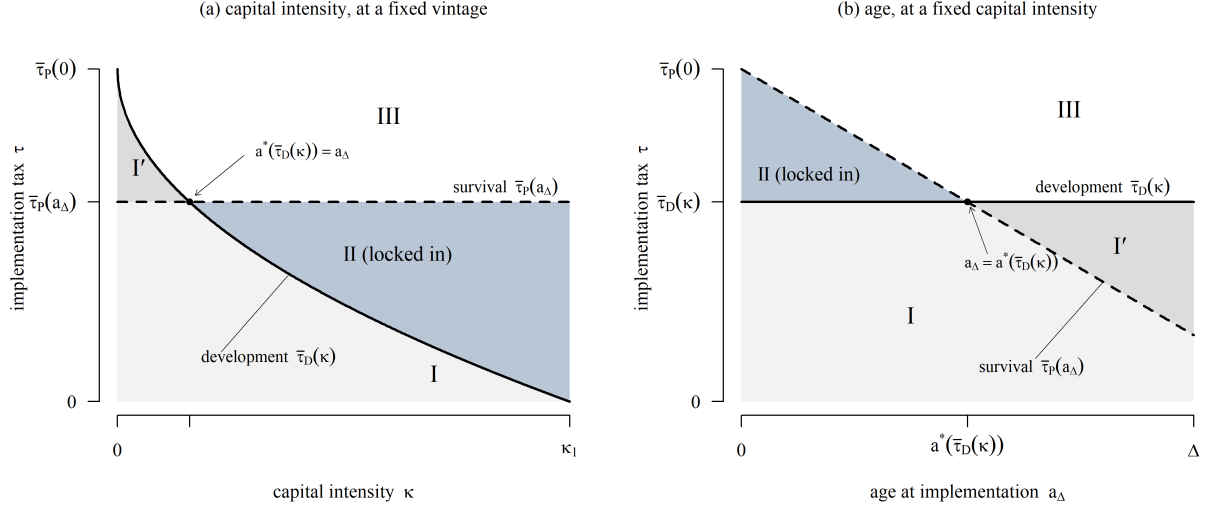


Figure 1: Which fields lock in.

Notes: The implementation tax is on the vertical axis. The development boundary is the development breakeven $\bar{\tau}_D$, the survival boundary the production breakeven at implementation $\bar{\tau}_P(a_\Delta)$, and only region II carries emissions after implementation. (a) At a given age at implementation, the lock-in zone deepens with capital intensity, and the two boundaries cross where $a_\Delta = a^*(\bar{\tau}_D)$. (b) At a given capital intensity, the survival boundary falls with age and the zone closes at $a_\Delta = a^*(\bar{\tau}_D)$. Drawn without discounting, with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $a_\Delta = 4$ in (a) and $\kappa = 8$ in (b).

producing at implementation? It is if $a_\Delta < a^*(\tau)$. The two questions give four types of fields, which correspond to the four regions of Figure 1. The table reports their excess emissions compared with the timely policy, per field and in units of $\theta\bar{q}$:

Type	Condition	released during the delay	overcommitted after Δ
I	$\kappa \leq \bar{v}_\delta(\tau)$, $a_\Delta < a^*(\tau)$	0	0
I'	$\kappa \leq \bar{v}_\delta(\tau)$, $a_\Delta \geq a^*(\tau)$	$a_\Delta - a^*$	0
II	$\kappa > \bar{v}_\delta(\tau)$, $a_\Delta < a^*(\tau)$	a_Δ	$a^* - a_\Delta$
III	$\kappa > \bar{v}_\delta(\tau)$, $a_\Delta \geq a^*(\tau)$	a_Δ	0

Only Type II fields emit after implementation what a timely policy would have avoided. They are capital-intensive and cheap to operate, like deepwater projects. Type III fields are also delay-induced, but their operating costs are too high for them to survive the tax at implementation. Some are cheap to develop, like shale wells, and others are not, like oil sands. Type I' fields would have been developed under both policies, but the timely tax would have stopped them before Δ , because they are cheap to develop but costly to operate. The table also applies to fields already producing in 2016, whose development cost is sunk and which can be treated as having $\kappa = 0$. They are of Type I' if the timely tax would have stopped them before Δ , and of Type I otherwise. Type I fields, typically the onshore conventional fields of the Middle East, are cheap to operate and cheap to develop or already

developed, and they produce under both policies.

Summing over the fields developed during the delay splits the overshoot into three parts: the overshoot at the intensive margin, released before Δ by Type I' fields, the overshoot at the extensive margin released before Δ by Type II and III fields, and the *lock-in wedge*

$$W(\tau) \equiv \sum_{\text{Type II at } \tau} \theta \bar{q} (a^*(\tau) - a_\Delta), \quad (7)$$

the overcommitted emissions of the cohort at tax τ . The wedge is defined for any tax, independently of the budget. It is zero when there is no tax, since no field is then delay-induced, and it is zero again at taxes high enough to stop every field at implementation. In between, a higher tax brings more fields into their lock-in zone, but shortens the remaining life of those already in it, so the wedge first increases and then decreases with the tax. Appendix A.1 gives its closed form for a parametric cohort.

2.4 The residual overshoot and the lock-in premium

Let B_Δ denote the carbon budget left at implementation, that is the budget minus the emissions released during the delay. The tax that meets this budget depends on the supply curve used to compute it.

The naive planner. The naive planner described in the introduction assigns each field a single cost per barrel, which includes its development cost. For this planner, a field developed during the delay stops producing as soon as the tax exceeds its development breakeven. Its lock-in zone disappears: in the table above, Type II fields become Type III fields, and in Figure 1, region II merges with region III. The naive planner therefore expects no overcommitted emissions after Δ .⁴ Let $E(\tau)$ be the emissions of the cohort after implementation, that is the sum of $\theta \bar{q}(a^*(\tau) - a_\Delta)$ over all fields still producing at Δ , and $E^N(\tau)$ the same emissions as computed by the naive planner, which leaves out Type II fields. Then

$$E(\tau) - E^N(\tau) = W(\tau). \quad (8)$$

E does not depend on the discount rate nor on the distribution of capital intensity, since emissions after implementation only depend on the fields that exist. The tax τ^* that meets the budget, defined by $E(\tau^*) = B_\Delta$, is therefore independent of both.

⁴I only apply the single cost per barrel to the decision of delay-induced fields to produce after Δ . Applying it to the producing life of every field would shorten these lives in the planner's calculation, lower its estimate of emissions further and widen the gap with actual emissions, so this choice is conservative.

The residual overshoot and the premium. Let τ^N be the tax that meets the budget according to the naive planner, $E^N(\tau^N) = B_\Delta$, which I call the naive tax. Implementing τ^N leaves a *residual overshoot* of the budget

$$\Omega^N \equiv E(\tau^N) - B_\Delta = W(\tau^N). \quad (9)$$

Meeting the budget requires increasing the tax by the *lock-in premium*

$$\pi \equiv \tau^* - \tau^N. \quad (10)$$

The wedge is a function of the tax, and the residual overshoot is its value at τ^N . The difference matters in the data, because the wedge takes different values at the timely tax, at τ^N and at τ^* , and only its value at τ^N is what the naive planner leaves unabated. The residual overshoot and the premium measure the same error, in tonnes and in dollars. To a first approximation,

$$\pi \approx \frac{\Omega^N}{|E'(\tau^N)|}, \quad (11)$$

where $|E'|$ is the reduction in emissions after implementation obtained with one more dollar of tax. For a given residual overshoot, the premium is small when emissions respond strongly to the tax and large when they respond weakly. Section 4.3 reports both numbers and the slope that relates them.

Two properties of the premium hold in general. Appendix A.1 gives a closed form, without discounting and with capital intensity uniformly distributed, in which they are explicit. First, $\pi > 0$ if the delay led to the development of any field the tax τ^* would have deterred. Second, the premium increases with the discount rate. A higher δ lowers $\bar{v}_\delta(\tau)$, so that more fields are delay-induced and E^N falls while E is unchanged: τ^N decreases, τ^* does not change, and both W and π increase. The undiscounted case is therefore a lower bound, and the sensitivity analysis of Section 4.4 shows the same ordering with discount rates of 5% and 15%.

Who pays for the delay. Locked-in fields hardly respond to the premium. When their zones are deep, their production breakevens lie well above τ^N , so an additional dollar of tax does not stop them, and the budget is met by stranding production on fields that a timely policy would have left in place. The same holds for overproduction. Emissions released during the delay reduce the budget left at Δ , so that even with the right supply curve, the tax after the delay must be higher than the timely one, and this increase strands production that a timely policy would not have stranded. The delay thus shifts stranding through two channels: overproduction, which raises the tax to τ^N , and lock-in, which raises it further to

τ^* . Section 5 measures both.

2.5 Anticipation

So far, investors do not anticipate the tax during the delay. Suppose instead that they screen projects with an anticipated tax $\tau^A \in [0, \tau]$, which can combine a probability of implementation, an expected implementation date and an expected level. The baseline corresponds to $\tau^A = 0$ and full anticipation to $\tau^A = \tau$. The only change concerns which fields are developed during the delay: a project is developed if $\kappa \leq \bar{v}_\delta(\tau^A)$, that is if $\tau^A \leq \bar{\tau}_D$. The delay-induced fields are then those whose development breakeven lies between the anticipated and the actual tax, $\tau^A \leq \bar{\tau}_D < \tau$. The four types, the decomposition of the overshoot, the wedge and the premium are defined as before.

Anticipation removes from the cohort the fields whose lock-in zone reaches lowest, that is whose development breakeven $\bar{\tau}_D = \bar{\tau}_P(0) - D$ is below τ^A . Both the depth and the height of the zone matter here. A field with a deep zone can still be developed under anticipation if its zone sits high enough, as for a deepwater project that is cheap to operate and relatively clean, while a field with a shallower zone can be deterred if its zone sits low. The premium remains positive as long as some fields developed during the delay have a development breakeven between τ^A and τ^* , and disappears when investors screen with the tax the budget turns out to require. The anticipation scenarios of Section 4.4 follow this pattern: the premium falls as anticipation increases and only disappears at full anticipation.

2.6 Increasing tax and field heterogeneity

The empirical analysis departs from the simple model in four ways. Each of them affects the results in a direction the model predicts.

An increasing tax. In the empirical analysis, the tax increases at rate r . Under a constant tax, the tax only enters the development decision through the margin at start-up, which is why the depth of the zone does not depend on the tax. With an increasing tax, it also enters through the path of the tax after start-up, and the development breakeven depends on the date of development, $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{ra_\Delta}$, where $\bar{\tau}_D^0$ is the development breakeven of a field developed at Δ . Fields developed early in the delay would have faced a lower tax under the timely policy, and are therefore more likely to have been developed anyway. Figure 2 shows the resulting partition. The production breakeven, the four types and the identity $W = E - E^N$ are unchanged, but the closed forms are lost. Appendix A.2 gives the details.

An increasing tax affects lock-in in two opposite ways. It raises the development breakeven

of young fields, so more of them become delay-induced, and it shortens the life of every locked-in field after implementation. After Δ , the zone now closes from both sides: the production breakeven falls by γ/θ each year while the tax rises by about $r\tau e^{r\Delta}$. The two effects offset each other, so that lock-in and the premium are of similar size under a constant and an increasing tax. The total cost of the delay differs much more. A tax that meets the same budget while rising over time starts at a lower level, so a timely policy retires fewer fields before Δ , and the overproduction added by the delay before implementation is smaller. Section 4.4 quantifies both, and Appendix A.2 illustrates the mechanism on a synthetic cohort.

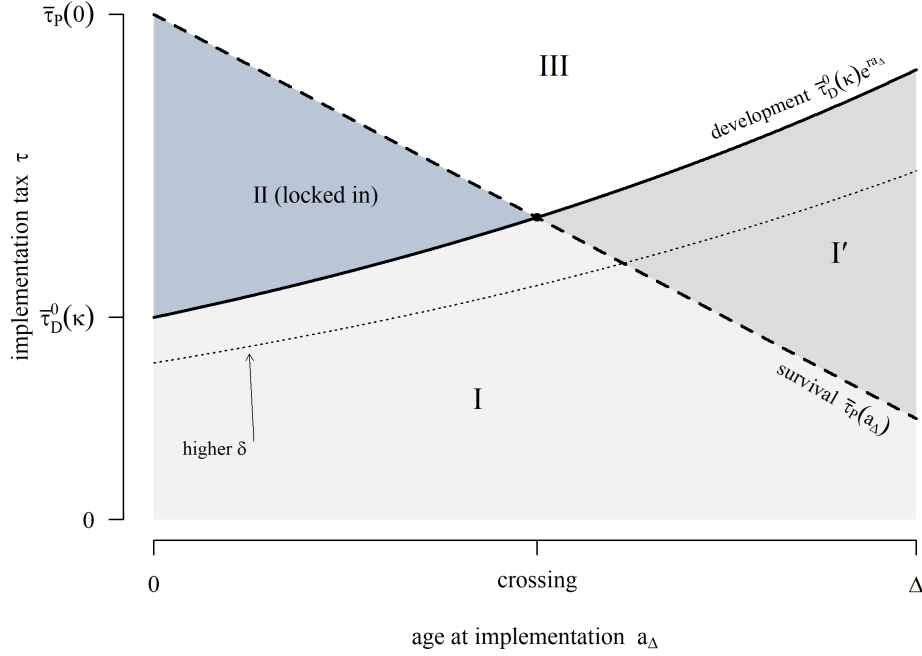


Figure 2: The taxonomy under an increasing tax, at a given capital intensity.

Notes: The survival boundary $\bar{\tau}_P(a_\Delta) = (p - c_0 - \gamma a_\Delta)/\theta$ is the same as in Figure 1. The development boundary is $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{ra_\Delta}$, increasing with age because fields developed early in the delay would have faced a lower tax under a timely policy. Region II, the set of locked-in fields, lies between the two boundaries and closes where they cross. The dotted line shows the development boundary at a higher discount rate: δ shifts it down and enlarges region II, r tilts it, and neither moves the survival boundary. Drawn with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $r = 0.10$, $\bar{\tau}_D^0 = 4$.

Heterogeneous production, cost paths and development periods. Three other features of the data affect the results in one direction each. First, the overcommitted emissions of a locked-in field, $\theta\bar{q}(a^* - a_\Delta)$, increase with $\bar{q}$, whereas whether a field is locked in does not depend on $\bar{q}$. The size of fields therefore changes the amount of lock-in but not which fields carry it, which is why the concentration documented in Section 5.1 reflects the size of a few projects. Second, capital-intensive offshore projects also have lower operating costs

and flatter cost paths than shale. A lower c_0 raises the zone and a lower γ widens it, so heterogeneity in operating costs reinforces the concentration of lock-in in deepwater fields. Third, developing a field takes several years, and some fields are only partly built when the tax arrives. Appendix A.3 adds a development period: a partly built field is completed if its production value covers the development cost it still has to pay, and abandoned otherwise. Since a development period can only add fields to the locked-in set, the model without it gives a lower bound.

The empirical analysis includes the increasing tax and these three features. The next section describes the data and how the model is implemented at the field level.

3 Data and methods

This section describes the data, the construction of the breakevens for each field, and the assumptions the empirical model rests on.

3.1 Oil field data and sample

The analysis relies on proprietary asset-level data from Rystad Energy. Rystad’s UCube database covers the vast majority of oil and gas production in existing fields worldwide since 1900, together with fields expected to be discovered and developed up to 2100, over 100,000 assets in total across oil, gas, LNG and biofuels. The analysis is conducted at the asset level, the smallest geographic scale available in the database, which typically corresponds to a field or to a portion of a field. For each asset, the data contain annual production by oil type and costs by category (operating expenses, split into production, transportation and SG&A opex; capital expenses for facilities and wells; exploration expenses; and government take), together with administrative, geographic and geological characteristics. Production and costs are observed up to 2025 and forecast by Rystad thereafter. Appendix B reproduces Rystad’s definition of each cost item.

The estimation sample keeps oil-producing assets discovered by 2025 with positive remaining oil reserves in 2016.⁵ This leaves 25,336 assets, of which 16,746 were already producing in 2016. Reserves in this paper are oil resources as defined by Rystad: all economically recoverable oil over the lifetime of the asset, which is typically larger than 2P reserves because it includes reserve growth. Oil is any type of crude (bitumen, extra heavy, heavy, regular, light or sour), together with condensate and natural gas liquids, which are counted

⁵Gas assets are excluded. Assets that Rystad models as not yet discovered are excluded from the main sample because their production, costs and reserves are entirely modelled; the sensitivity analysis of Section 4.4 adds them back, with their exploration expenses treated as development costs.

as oil because (i) they are liquids and sometimes mixed with crude oil and (ii) their price is set mainly by the oil price.

Table 1 describes the sample. Panel A splits it by vintage: fields already producing in 2016, on which the delay acts only through overproduction; the delay cohort, the 3,136 fields on which some development takes place between 2016 and 2029, and which are the only fields a delay can lock in; fields approved in 2030 or later; and fields never developed on the baseline path (without carbon tax). Panel B splits the delay cohort by supply segment, and Panel C splits the full sample by region. For each group, the table reports the medians of capital intensity κ (development capex per barrel of peak annual capacity), of the two breakevens $\bar{\tau}_D$ and $\bar{\tau}_P(0)$, and of the depth of the lock-in zone $\bar{\tau}_P(0) - \bar{\tau}_D$. Medians are weighted by 2016 reserves, so that they describe the typical barrel rather than the typical field.

	Fields	Reserves 2016 (Gbbl)	Emissions post-2016 (GtCO ₂ eq)	Emissions post-2030 (GtCO ₂ eq)	κ (\$/bbl)	$\bar{\tau}_D$	$\bar{\tau}_P(0)$	$\bar{\tau}_P(0) - \bar{\tau}_D$
<i>Panel A. All fields, by vintage</i>								
Producing by 2016	16,746	1,072.9	479.4	286.6	40.8	—	215.1	—
Delay cohort	3,136	225.4	103.9	77.2	89.0	68.6	117.3	32.8
Approved 2030 or later	2,259	208.3	89.0	89.0	71.8	54.5	79.2	23.7
Never developed	3,195	48.8	23.3	15.5	10.5	−23.5	−3.7	88.0
Total	25,336	1,555.4	695.6	468.2	49.6	57.0	146.9	25.6
<i>Panel B. Delay cohort, by supply segment</i>								
Onshore conventional	1,427	77.4	35.9	26.7	73.4	80.4	121.3	28.9
Offshore shelf	544	56.2	24.2	18.5	92.8	58.6	124.2	46.8
Offshore midwater	206	12.8	6.0	4.3	128.7	79.9	116.2	28.8
Offshore deepwater	203	44.1	20.5	13.9	120.6	63.1	110.4	29.9
Shale and tight oil	708	27.7	13.1	10.6	2.7	34.0	45.7	7.4
Oil sands	48	7.2	4.1	3.2	102.1	34.5	72.4	23.6
<i>Panel C. All fields, by region</i>								
Middle East	1,540	711.7	295.9	225.0	35.9	59.4	230.3	19.9
North America	9,670	366.4	175.3	113.7	5.8	17.5	49.9	21.0
Russia	2,967	147.2	66.8	41.1	89.0	68.7	157.7	30.0
Asia	3,730	115.7	53.8	30.4	101.5	52.0	167.3	45.5
South America	2,393	102.5	49.1	30.9	107.6	49.0	106.0	36.5
Africa	2,116	79.2	39.5	20.8	107.5	51.3	163.2	35.2
Europe	2,242	27.9	13.0	5.3	108.7	39.1	160.1	51.7
Oceania	678	4.9	2.1	1.2	76.4	79.9	152.7	26.5

Table 1: Summary statistics.

Notes: Medians are weighted by 2016 reserves and are not additive: the depth column is computed field by field. Taxes are 2016-anchored levels of the increasing path, in \$/tCO₂eq. The delay cohort is the set of fields on which some development takes place during the delay (start-up after 2016 and approval by 2029).

Three features of the table matter for what follows. First, the delay cohort is more than twice as capital-intensive as the fields already producing in 2016, with a median κ of \$89 per

barrel of capacity against \$41, and its lock-in zone is correspondingly deeper, \$33 per tonne against a full-sample median of \$26. Second, within the cohort, the depth of the lock-in zone follows capital intensity across segments, from \$7 for shale, at a capital intensity of \$3 per barrel, to \$29–47 for the offshore segments, at \$93–129. The two conditions of Section 2.2 can be read side by side. Shale has a narrow zone that sits low, deepwater a wide zone that sits high, and oil sands a zone as wide as the offshore ones (with a capital intensity of \$102) that sits low ($\bar{\tau}_P(0) = 72$), because of the segment’s high operating costs and carbon intensity. Third, the regional panel reads the same way. North America, where shale and oil sands are concentrated, has a median production breakeven of \$50, so most of its output is retired by any tax in the range studied here, while South America and the Middle East have medians of \$106 and \$230. The regional medians describe where a region’s production sits relative to the tax, not where its lock-in sits, which Section 4.1 discusses.

3.2 Field-level carbon intensities

Rystad’s UCube data contain no carbon intensity. I estimate a lifecycle carbon intensity for each asset by combining upstream, midstream and downstream emissions. Appendix C gives the full procedure. For upstream emissions, I use the input data of Masnadi et al. (2018) for 737 oil fields, matched to 1,136 Rystad assets, and run the Oil Production Greenhouse Gas Emissions Estimator (OPGEE). The main source of upstream heterogeneity is flaring, which is not reported at the asset level. I use flared volumes derived from the satellite observations of NOAA’s VIIRS instruments over 2012–2024 (Elvidge et al., 2015), matched to Rystad assets following Ghosh et al. (2025), and predict flaring-to-oil ratios for assets which production has not started by 2025. A regression on flaring, steam-to-oil and gas-to-oil ratios, offshore status and oil type then assigns an upstream intensity to every remaining asset, including those not yet developed. For midstream and downstream emissions, I use the Petroleum Refinery Life-Cycle Inventory Model (PRELIM) and the Oil Products Emissions Module (OPEM), matching 493 publicly available crude assays to 392 Rystad crude streams, which cover 64.4% of 2024 production, and predicting intensities for unmatched assets from API gravity, sulphur content and predicted refinery configuration. Summing the three stages gives one lifecycle intensity per asset. In 2024, the production-weighted average is 466 kgCO₂eq/bbl, with a weighted standard deviation of 112. Carbon intensity enters the analysis in one place, the height of the lock-in zone $\bar{\tau}_P(0) = (p - c_0)/\theta$, and the sensitivity analysis replaces the asset-level estimates with a common value to show how much of the result depends on them.

3.3 Costs, prices and discounting

Development and production costs. Following Rystad’s split of costs, only expenses attributed to oil operations are kept. The capital expenses for drilling wells and building facilities before production starts and in its first years are the development cost of the asset, the empirical counterpart of C . Following Rystad’s own definition, all capital expenses incurred up to two years after start-up are counted as development. For modelling purposes, development expenses incurred after start-up are moved to the year before start-up and discounted accordingly, which leaves the net present value of the project unchanged. Production costs include all operating expenses (production, transportation and SG&A opex) and all other capital expenses, that is facilities and wells added more than two years after start-up.

The two-year rule matters for lock-in, because it determines how much of a field’s capital counts as sunk once production starts. It is the relevant rule for this paper. On most projects, a large part of the development cost is spent in the first years of production, on the wells drilled to reach plateau, and this capital is committed by the time a tax can reach the field. Section 4.4 tests a stricter rule, which counts only capital spent before start-up as development. It moves these two-years of capital expenses into the operating cost and narrows every lock-in zone, therefore giving a lower bound on lock-in. Under this more restrictive rule, results remain remarkably close to the main specification.

Prices. The reference Brent price used in the empirical analysis is constant at \$80/bbl. What this paper measures is driven by investment decisions taken during the delay, so the relevant price is the long-run price investors use to screen those projects, not the price of any particular year. The price level matters for the results, since it sets the pool of assets that are viable without carbon tax. Changing it moves the tax that meets the budget more than it moves the overshoot, because both the timely and the delayed policy are re-calibrated on the same budget, and because few projects need more than \$80/bbl to be developed. Two arguments support this level. The first is the behaviour of the oil market. Its stability is guaranteed mainly by the three core countries of OPEC, Saudi Arabia, Kuwait and the United Arab Emirates (Kellogg, 2024), and the IMF reports fiscal breakeven prices, the price at which an exporting country balances its budget, of \$80/bbl in the recent past and \$84/bbl in projection for Saudi Arabia, and lower for the other two (International Monetary Fund, 2024). Given the influence these countries exert on the price, their fiscal breakeven is a reasonable anchor for a long-run price. The second is that \$80/bbl keeps the investment margin studied here active. At much lower prices, few new projects are viable with or without a carbon tax, and the question of what a delay lets through does not arise. One

may object that the introduction of a worldwide carbon tax would itself change long-run price expectations. The price does not respond to the tax in the model, so the tax levels reported in the paper should be interpreted as the levels borne by producers. Section 4.4 reports a variant in which part of the tax is passed on to consumers. Beyond this, the countries that anchor the price are also those a carbon tax affects least, since their fields have both low costs and low carbon intensities. Assuming producing countries capture the carbon tax on their domestic production, the revenue they draw from oil production falls only moderately under the tax.

Price volatility is not modelled either, because the investments studied here are long-lived and screened on long-run price expectations. OPEC's use of its spare capacity reduces price volatility and supports investors' confidence in a stable market (Pierru et al., 2018; Fattouh, 2007). Berntsen et al. (2018) find that volatility has no significant effect on oil and gas investment in Norway once the expected price level is accounted for, and Tang et al. (2017) argue that volatility changes the timing of investment rather than the decision to invest. The sensitivity analysis uses Brent prices of \$60 and \$100/bbl, which move the residual overshoot and the premium in opposite directions (Section 4.4).

Asset-specific price discounts. Reference prices such as Brent describe one particular crude stream, and each asset sells its oil at Rystad's long-term discount to that reference, which depends on crude quality. Every oil type sells at a discount on average, which reflects the high quality of the crudes constituting the Brent stream, and heavier oils, which are more difficult to refine into the same products, carry larger discounts; NGL is a special case and sells at roughly half the price of oil. The reserve-weighted average discount is 12.9%, so that with a reference price of \$80/bbl the average selling price in the model is closer to \$70. Appendix Table 7 reports the discounts by oil type, and the sensitivity analysis replaces them with a common selling price across assets.

Discounting. Net present values are computed at a 10% discount rate. This is the most common rate used in the oil and gas industry to screen projects, the rate at which the industry reports its breakeven prices, and the one Rystad uses (Bailey et al., 2000; Jahn et al., 2008; Wood Mackenzie, 2013). The model of Section 2.2 says in which direction this choice matters. A higher rate lowers the value of developing a field without changing the value of keeping it in production, so it widens every lock-in zone and raises both the premium and the residual overshoot. The sensitivity analysis reports results at 5% and 15%.

3.4 From the model to the data

The empirical model is a sequence of yearly decisions taken for each asset independently of the others: whether to invest in the development of an undeveloped asset, and whether to keep an active asset in production.

Investment decision. An asset is developed if the expected net present value of the project is non-negative, given what investors anticipate about the carbon tax τ :

$$\sum_{t=T_1}^T \frac{P(1-d_i)q_{it} - c_{it}}{(1+\delta)^{t-T_0}} - C_{iT_0} - \sum_{t=T_\tau}^T \frac{\tau \theta_i q_{it} (1+r)^{t-2016}}{(1+\delta)^{t-T_0}} \geq 0, \quad (12)$$

where T_1 and T are the first and last years of production, T_τ the anticipated year of implementation, $P(1-d_i)$ the reference price net of the asset's discount, C_{iT_0} the development cost expensed one year before start-up, c_{it} production costs, θ_i carbon intensity and r the increase rate of the tax. Investors choose T to maximise the value of the project. Rearranging (12) gives the maximum anticipated tax $\bar{\tau}_i$ consistent with a non-negative net present value, the empirical counterpart of the development breakeven $\bar{\tau}_D(\kappa)$ of equation (5).

Development often takes several years, and can stop and resume, as it did on many projects after the 2020 price collapse. The maximum viable tax is therefore computed at each development stage, with C_{iT_0} replaced by the development expenses that remains to be spent, and the asset is developed only if the anticipated tax is below this threshold at every stage. The threshold rises as the remaining cost falls, so an asset that clears it when development starts keeps clearing it as long as anticipations do not change. When the tax is implemented, an asset under development may no longer clear the threshold at its remaining development cost. Development then stops, the capital already spent is lost and the infrastructure is abandoned. This remaining-cost threshold is the empirical counterpart of the completion boundary of Appendix A.3. For a given tax, the year in which an asset's remaining cost falls below it is the year of investment responsible for its lock-in: from that year on, the asset would be completed even if the tax were implemented immediately.

Production decision. Once developed, an asset produces in a given year if the price net of the tax covers its production cost. Production costs are computed as breakeven prices by stage of production, and rise over the life of the asset: I compute the last year of production that minimises the breakeven price, which closes the first stage, and repeat the operation on the remaining years to identify the following stages. For assets developed after 2015, the average number of stages is 5.8 and the average stage lasts 5.2 years, the first stage usually being the longest. The breakeven of the current stage is the empirical counterpart of $\bar{\tau}_P(a)$

in equation (2): it rises with age because operating costs rise and because capital expenses per barrel rise as production declines. An asset is phased out in the first year in which the tax exceeds this breakeven, and does not come back once phased out.

Investment and production paths. The path of investment and production of each asset is the one observed up to 2025 and forecast by Rystad thereafter. Production rates depend mainly on the initial investment and cannot be changed at will afterwards, because of geological and technological constraints; changing the path would mean changing development expenses and all subsequent costs. During the delay, the tax is not, or only partially, anticipated, which makes the business-as-usual path forecast by Rystad the relevant counterfactual. A field’s production profile is therefore taken as given from its start-up year, and a later development date shifts that profile in time rather than reshaping it. The development date still matters once the tax is implemented. Under a tax rising over time, a field that starts producing later meets a higher tax at each age and is phased out younger. This is why the year in which an asset is developed matters for what it still emits after 2030. Investors who anticipate the tax could instead bring production and investment forward, the green paradox response excluded in Section 2.

3.5 Scenarios, budget and parameters

The analysis compares a scenario in which the tax is implemented in 2016, the timely policy, with one in which it is delayed to 2030 without being anticipated by investors, which is equivalent to business as usual until 2030. The tax rises from its implementation at a 2% social discount rate. Under a carbon budget, the dynamically efficient tax rises at the discount rate (Lemoine and Rudik, 2017; Dietz and Venmans, 2019). A social discount rate below market rates is justified by uncertainty about growth and interest rates in the distant future (Stern, 2007; Weitzman, 2007). The literature on declining discount rates points to social discount rates between 1% and 2% (Weitzman, 1998; Gollier, 2009), while Nordhaus (2007) argues for a higher rate, around 3%. I adopt a rate of 2%, between these approaches, and the sensitivity analysis uses rates from 1% to 6%. All tax levels in the paper are expressed at their 2016-anchored level, whatever the implementation date, so that a level of x on any axis or in any table corresponds to $1.32x$ in 2030.

Anticipation scenarios. I explore alternative anticipation scenarios in which investors screen projects either with a fraction of the eventual tax (a wrong level), or with the correct tax arriving at different dates (in 2030, or five years after the asset’s start-up). Appendix B.3 defines them, Section 2.5 summarises them with a single anticipated tax τ^A , and Section 4.4

reports the results.

The reference level of tax is the one consistent with the oil carbon budget of [Welsby et al. \(2021\)](#). Their budget of 747 Gbbl of oil over 2018–2100 is extended to 2016–2100 by adding the observed production of 2016 and 2017, and converted into emissions using the average downstream carbon intensity used in most models, including the TIAM-UCL model of [Welsby et al. \(2021\)](#), taken from [IPCC \(2006\)](#). The resulting budget is 350.3 GtCO₂eq from 2016. Under a timely policy, the tax that meets this budget starts at \$57.2/tCO₂eq in 2016. Under the delay, the budget remaining at implementation, B_{Δ} , is the initial budget net of the 205.4 GtCO₂eq released between 2016 and 2029 on the business-as-usual path, that is 144.9 GtCO₂eq⁶. Leakage towards other fossil fuels is not modelled, and the results should be read as if similar policies applied to other fuels at the same time. The sensitivity analysis varies the budget between 75% and 125% of its central value.

An asset’s excess emissions under the delay are counted on the intensive margin if the asset produces under the timely policy or was already producing in 2016, and on the extensive margin, as a delay-induced field, if it does not produce under the timely policy. Since anticipations affect only investment decisions, the anticipation scenarios differ only on the extensive margin. Table 2 summarises the parameters.

Brent price	\$80/bbl, constant; asset-specific discounts (12.9% on average)
Discount rate δ	10%
Tax increase r	2% per year; levels reported 2016-anchored
Oil carbon budget	350.3 GtCO ₂ eq from 2016 (Welsby et al., 2021)
Delay	2016–2029, tax in force from 2030
Anticipation (main case)	none
Development cost	capex up to two years after start-up
Carbon intensity	lifecycle, GWP-100, asset-level (Appendix C)
Sample	25,336 assets with oil reserves in 2016, discovered by 2025

Table 2: Parameters of the main specification.

3.6 Fiscal regimes

Government take is excluded from costs, except for taxes reported as opex by companies, which producers treat as operating costs unrelated to profitability. Profit taxes change the size of an asset’s net profitability but not its sign, and bonuses are paid before development decisions are made. The key assumption concerns royalties, which are levied on gross revenue and can therefore make a project unprofitable. I assume that, in the presence of a worldwide

⁶Note that B_{Δ} can vary when assessing anticipation scenarios and early ban policies.

carbon tax, governments renegotiate royalty rates so as to keep viable assets in production and keep collecting a share of the rent, without subsidising assets that are no longer viable. Two cases bound this assumption. On the one hand, a government at its Ramsey optimum could in principle respond to a worldwide carbon tax by adjusting fiscal regimes so as to leave the equilibrium unchanged, substituting carbon tax revenue for royalty revenue. However, fully restoring the pre-tax equilibrium would in some cases require subsidising development or production. On the other hand, contractual rigidities in existing production-sharing agreements and concession contracts may prevent renegotiation in the short term, causing the carbon tax to pile on top of the existing fiscal regime and further constrain investment and production beyond what the model predicts. The assumption made in this paper implicitly lies between these two extremes: governments are assumed to absorb the carbon tax to the extent necessary to keep viable assets in production (potentially capturing a higher rent in the process) without subsidising uneconomical ones, so that assets not resilient to the carbon tax are either phased out earlier than in a no-tax world or simply remain undeveloped. The locked-in emissions measured in the paper therefore correspond to an intermediate case, with full absorption giving an upper bound and full rigidity a lower bound.⁷ The same assumption makes Rystad’s forecast paths, which embed current fiscal terms, the relevant counterfactual without a carbon tax.

3.7 Capital intensity and the depth of lock-in in the data

The sorting result of the model, equation (6), says that the depth of a field’s lock-in zone per barrel is set by its capital intensity and the steepness of its operating cost path, and not by its carbon intensity or by the tax. Panel B of Table 1 shows the corresponding ordering in the data. Across segments of the delay cohort, the median depth of the lock-in zone rises with the median capital intensity, from shale to the offshore segments. Appendix Figure 24 shows the same ordering field by field, and two features of it matter. The ordering holds within segments as well as across them, and it survives the heterogeneity of carbon intensities and operating costs, which the model says should not enter the depth of the zone. Measured per barrel, the depth of the lock-in zone rises with capital intensity with an elasticity of 0.55, so that a field ten times more capital-intensive than another has a lock-in zone about three and a half times deeper. The fit should not be read as a test of the model, since development capital enters both the capital intensity on the horizontal axis and the full-cycle breakeven that defines depth, and part of the relation is mechanical. What the data add is the ordering, within segments as well as across them, and the size of the gaps

⁷On the interaction between extraction taxes and rent capture, see [Daubanes and Lasserre \(2023\)](#); on the response of investment to oil taxation, [Ahlvik et al. \(2022\)](#) and [Stefanski et al. \(2025\)](#); on compensation for halted upstream projects, [Paletta et al. \(2025\)](#).

between segments.

4 Results

All results in this section rely on a tax rising at 2% per year from its 2016-anchored level, under the no-anticipation scenario, with a Brent price of \$80/bbl. The tax that meets the budget under a timely policy is $\tau_{2016}^* = \$57.2/\text{tCO}_2\text{eq}$. The section first measures the overshoot caused by the delay and splits it by margin and date of emission, supply segment and region (Section 4.1). It then follows the investments of the delay cohort from the year they are made (Section 4.2), turns to the lock-in premium and the residual overshoot left by a tax set on full-cycle costs (Section 4.3). It ends with sensitivity analysis, including the constant tax (Section 4.4 and Appendix D.4).

4.1 The overshoot and where it comes from

Figure 3 reports, for each level of the tax, the overshoot of the carbon budget: the excess of cumulative emissions under the delay over cumulative emissions under the timely policy. The overshoot is split into the cells of the decomposition of Section 2.3, (i) emissions released before 2030 by fields that a timely tax would also have built but would have retired earlier (the intensive margin), (ii) emissions released before 2030 by delay-induced fields and (iii) emissions released after 2030 by delay-induced fields that the tax arriving in 2030 cannot stop. The fourth cell, intensive emissions after 2030, is zero, as the model predicts. Appendix Figure 21 draws the same bands on top of cumulative emissions under the timely policy.

The overshoot rises with the tax, slowly below about \$25 and steadily above. At low levels, a tax removes little production, during the delay or after it, because oil fields earn large rents. At the tax that meets the budget, the delay overshoots the oil carbon budget by 66.7 GtCO₂eq, 19.1% of the budget and about four years of global oil emissions at their current rate. Of this overshoot, 41.2 GtCO₂eq are released before 2030 by fields on the intensive margin, while the extensive margin represents 9.6 GtCO₂eq before 2030 and 15.9 GtCO₂eq after. Delay-induced fields, those that exist because of the delay, therefore account for 38% of the overshoot, and their emissions after 2030 alone for 24%. Three quarters of the overshoot is already in the atmosphere when the tax arrives, and no tax set in 2030 can recover it. The top band is the part a policy set in 2030 can still act on, and the rest of the paper measures what it costs.

Supply segments. The three bands come from different kinds of fields. Figure 4 repeats the decomposition by supply segment, on top of the emissions each segment would release

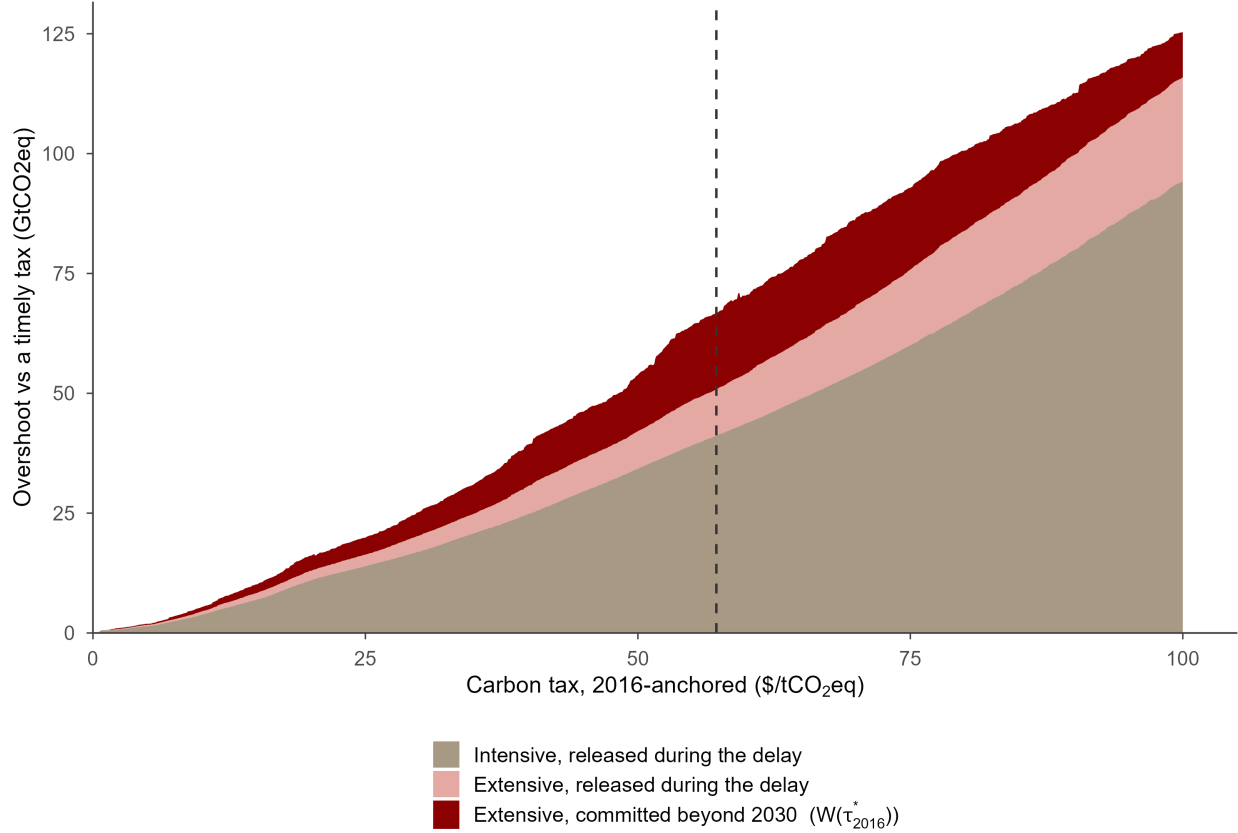


Figure 3: Carbon budget overshoot caused by the delay, by margin and date of emission.

Notes: The horizontal axis is the 2016-anchored level of the tax rising at 2% per year; the dashed line is the tax that meets the budget under a timely policy, \$57.2. Intensive emissions after 2030, the fourth cell of the decomposition, are zero and have no band. The top band read at \$57.2 is the lock-in wedge at the timely tax, $W(\tau_{2016}^*) = 15.9$ GtCO₂eq.

under the timely policy. The baseline explains a large part of the pattern. A timely tax at \$57.2 would phase out or deter most oil sands and shale projects, while about 60% of deepwater emissions and 70% of onshore conventional and shelf emissions would remain. Segments that would almost disappear under a timely tax can only overproduce during the delay; segments that would largely survive it can also lock in. Table 1 gives the median breakevens that place each segment's lock-in zone.

Shale and tight oil is the clearest case of a lock-in zone the tax passes over. The median shale barrel in the cohort costs \$3 per barrel of capacity to develop, so its zone is only \$7 wide, and it sits low, with a median production breakeven tax of \$46, well below the tax in force after 2030. Almost all of the segment's production is removed by a timely tax, and its overshoot is almost entirely on the intensive margin. Shale fields run untaxed during the delay and are shut down as soon as the tax arrives, whether or not the tax would have deterred their development. At the timely tax, shale carries 15.8 GtCO₂eq of the intensive overshoot, 38% of the total and the largest contribution of any segment, and almost none of

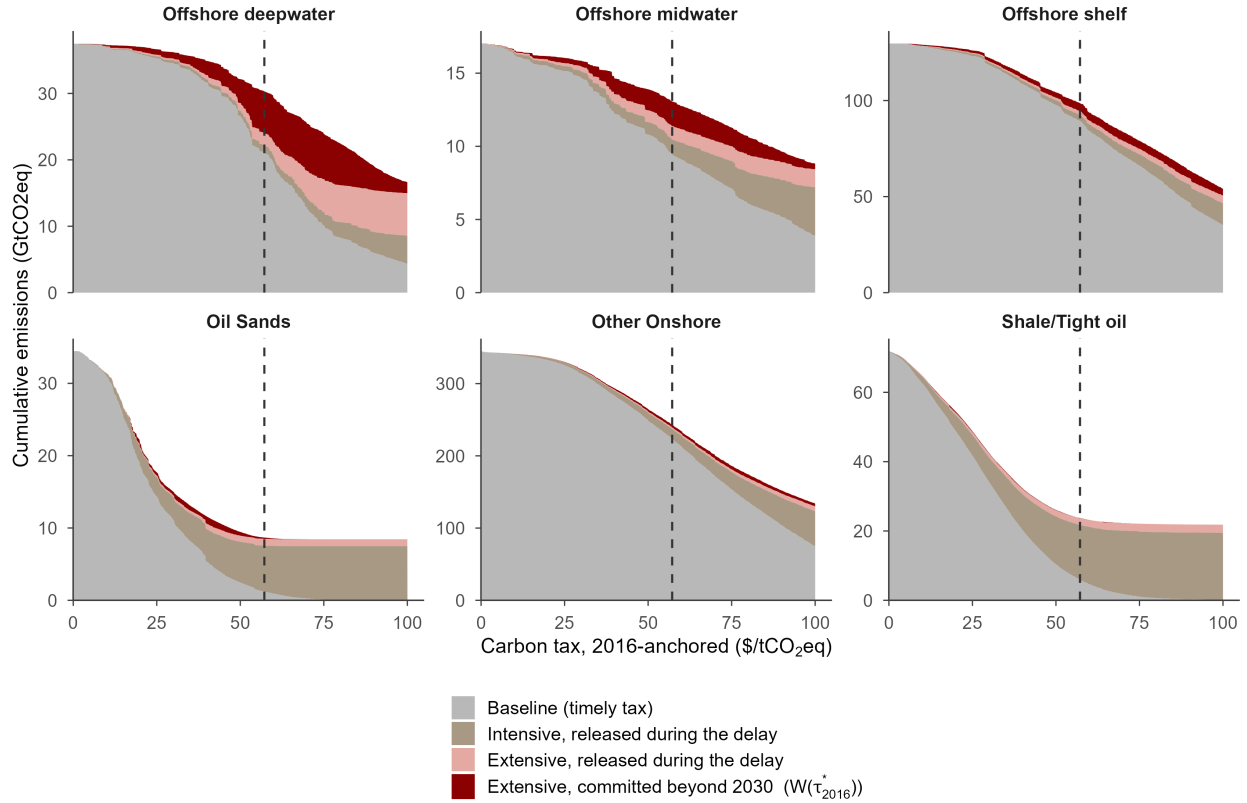


Figure 4: Overshoot caused by the delay, by supply segment.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay by margin and date of emission, tax rising at 2% per year. Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Appendix Figure 22 shows the overshoot bands alone.

the emissions released after 2030.

Oil sands reach the same outcome by a different route, and they show that the width of a lock-in zone and its position are different things. Oil sands projects are as capital-intensive as offshore ones, at \$102 per barrel of capacity, and their zone is correspondingly wide, \$24. But it sits low. The median production breakeven is \$72, and the segment's post-2030 band closes almost entirely at the timely tax level. The reason is the segment's high operating costs and carbon intensity. Oil sands barrels are the most carbon-intensive in the sample, and since the upper edge of the zone is $\bar{\tau}_P(0) = (p - c_0)/\theta$, both push the whole zone down in tax terms. Capital intensity sets how wide a lock-in zone is, operating costs and carbon intensity set where it sits, and a field needs both a wide and a high zone to be locked in.

Offshore fields meet both conditions. The median deepwater barrel has a development breakeven of \$63 and a production breakeven of \$110, and the median shelf barrel \$59 and \$124. A tax in the \$67–70 range falls inside both zones, well below their upper edge, so a delay-induced offshore field survives the tax and keeps producing for years. Offshore fields

also have low operating costs and are rarely retired early, so their intensive band stays small, while their post-2030 band keeps growing beyond the tax that meets the budget. Offshore fields carry 76% of the 15.9 GtCO₂eq released after 2030 by delay-induced fields, and deepwater fields alone 39%.

Onshore conventional is the most resilient to the tax. Because it concentrates the largest reserves, its overshoot is the largest of all segments in absolute terms, 19.3 GtCO₂eq, and comes mostly from fields already producing in 2016. The median development breakeven of the segment’s delay cohort, \$80, is above the tax, so most of these projects would also have been built under a timely policy.

Regions. The segment pattern has a geography, which drives the incidence results of Section 5. Appendix Figure 23 repeats the decomposition by region. North America, where shale and oil sands are concentrated, accounts alone for 67% of the intensive overshoot, and very little of its overshoot survives 2030. The median North American barrel has a production breakeven of \$50, so its output is retired by a tax in the \$67–70 range whatever its development history. South America is the mirror image. Its overshoot is dominated by the post-2030 band, which grows steeply between \$40 and \$60 as the deepwater developments off Brazil and Guyana enter their lock-in zones, and the region carries 5.4 GtCO₂eq of the 15.9 released after 2030, 34% of the total and the largest regional share. The Middle East is dominated by low-cost fields already in production, with a median production breakeven of \$230. Its overshoot at the timely tax is small, 4.0 GtCO₂eq, but 2.7 GtCO₂eq of it is released after 2030, from the minority of its cohort projects that a timely tax would have deterred. The other regions are affected on both margins. Asia’s overshoot at the timely tax is 9.4 GtCO₂eq, mostly on the intensive margin, Russia’s 3.4 GtCO₂eq and Europe’s 2.0 GtCO₂eq. Africa’s overshoot is 4.4 GtCO₂eq, of which 44% is released after 2030 by delay-induced fields; Section 5 shows that this is enough to reverse the sign of the delay’s effect for Africa as a whole.

4.2 Committed and overcommitted emissions

Figure 3 counts emissions in the year they are released. Lock-in is better described by the year they are committed, that is the year of the investment after which, at a given tax, the full development of the asset can no longer be deterred. Figure 5 reports, by supply segment and five-year period of the delay, the emissions committed by fields developed during the delay. It splits them between emissions that the timely tax of \$57.2 would also have committed and overcommitted emissions, which are due to the delay itself.

Two features stand out. First, overcommitted emissions are a large share of what new

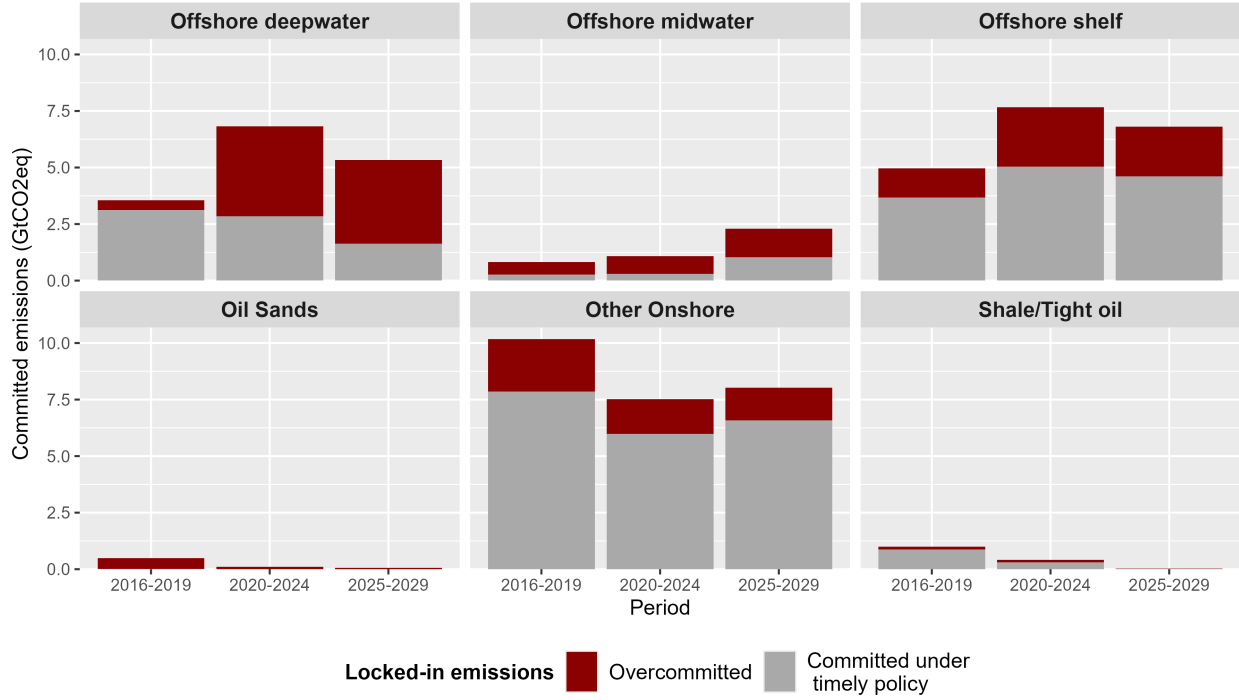


Figure 5: Committed and overcommitted emissions, by supply segment and period.

Notes: Tax rising at 2% per year from \$57.2/tCO₂eq. Each bar is the lifetime emissions committed by investments in new oil assets of the segment during the period. The grey part is committed under the timely policy as well; the red part is overcommitted, that is committed only because of the delay. Emissions are attributed to the year from which the full development of the asset can no longer be deterred at the tax shown.

projects commit: 34% of the emissions committed by new oil projects over 2016–2029 would not have been committed under a timely policy. Second, this share varies across segments in the order the model predicts (Section 3.7). Over half of the emissions committed by deepwater projects are overcommitted, and 62% of those committed by midwater projects. Onshore conventional projects commit the most in absolute terms, but only a fifth of their commitments are due to the delay. Shale commits little at the timely tax, because most of its production does not survive that tax, and oil sands commit almost nothing that a timely tax would also have committed. The time profile matters for policy. Overcommitted deepwater emissions rise from 0.4 GtCO₂eq in 2016–2019 to 4.0 GtCO₂eq in 2020–2024 and remain at 3.7 GtCO₂eq in 2025–2029, so that a large share of the lock-in measured in this paper is decided in the years just before implementation. Section 4.3 returns to this when describing how the residual overshoot builds up.

4.3 The lock-in premium and what the naive planner leaves behind

Overproduction during the delay raises the tax needed to meet the budget. The 205.4 GtCO₂eq released between 2016 and 2029, against 154.6 in the timely tax world, leave 144.9 GtCO₂eq for the rest of the century. Investment during the delay does something more, it locks in additional reserves resilient to high taxes, so that a given tax removes fewer emissions than it would have before the delay. The tax has to rise for both reasons, and a naive planner using a single full-cycle cost per asset only sees the first.

Panel (a) of Figure 6 reports post-implementation emissions under the two calibrations of Section 2.4. E^N is computed from full-cycle breakevens, as if the delay-induced fields could still be deterred, and E from their operating breakevens as of 2030. Both are compared with the remaining budget $B_\Delta = 144.9$ GtCO₂eq, drawn as a horizontal line. The line is flat because no tax is levied before 2030, so the choice of the 2030 tax does not change what has already been emitted. The tax at which E^N meets the budget, the naive tax, is $\tau_{2030}^N = \$66.8$. The tax at which E meets it is $\tau_{2030}^* = \$70.5$. The premium is the difference between the two, $\pi = \tau_{2030}^* - \tau_{2030}^N = \$3.6/\text{tCO}_2\text{eq}$, 5.4% of the naive tax.⁸ Both curves fall in small steps because emissions are counted field by field. They are close to parallel over the whole range shown, 12 to 18 GtCO₂eq apart, so the premium does not depend on reading the two curves at a particular point.

Panel (b) draws the wedge itself as a function of the tax, and it has the hump shape the model predicts. It is zero at a zero tax, where no field is delay-induced. It rises as the tax draws the capital-intensive fields of the cohort into their lock-in zone, with steps where large fields enter, then forms a plateau between about \$55 and \$80, and finally declines, as taxes that high phase out even locked-in fields after 2030. Both the naive tax and the lock-in-adjusted tax sit on this plateau, close to its highest point. This is not a choice made in the analysis, since both taxes are set by the remaining budget, and it means that a moderate error in the naive tax would change the residual overshoot very little. The plateau may look relatively narrow in tax terms, but it is wide in terms of emissions, because taxes below \$50 change cumulative production little: moving the budget by 10% in either direction leaves the residual overshoot within 1 GtCO₂eq of its central value (Section 4.4).

The residual overshoot. A tax set on full-cycle costs and implemented from 2030 would still overshoot the budget by $\Omega^N = 17.1$ GtCO₂eq, 11.8% of the budget remaining in 2030, and about one year of global oil emissions. All of it comes from delay-induced fields. Ignoring lock-in therefore leads to setting the carbon tax too low, even when the budget

⁸In 2030 dollars, the premium amounts to \$4.8, on a tax of \$93.0 against \$88.2.

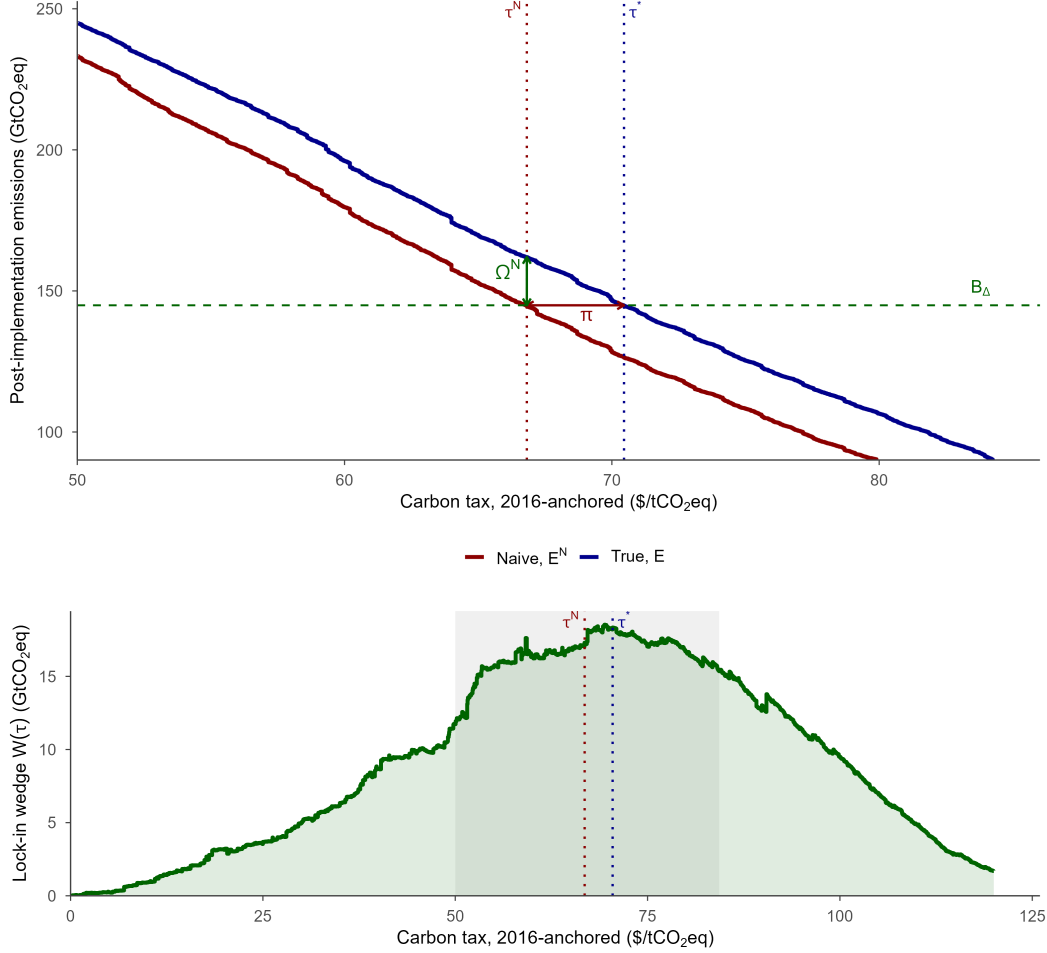


Figure 6: Full-cycle and lock-in-adjusted calibration of the 2030 tax.

Notes: (a) Post-implementation emissions computed from full-cycle (E^N) and operating (E) breakevens of the delay-induced fields, against the remaining budget $B_\Delta = 144.9$ GtCO₂eq; the dotted lines mark $\tau^N = 66.8$ and $\tau^* = 70.5$. The premium π is the horizontal distance between the two curves at the budget, the residual overshoot Ω^N the vertical distance at τ^N . (b) The lock-in wedge $W(\tau) = E(\tau) - E^N(\tau)$ as a function of the tax; the dotted lines mark the naive tax, at which the wedge gives Ω^N , and the lock-in-adjusted tax, and the shaded band is the range of taxes shown in panel (a). Taxes are 2016-anchored levels of the tax rising at 2% per year.

itself is correctly identified⁹.

The size of the premium needs to be read against the tax it adds to. Because the oil price does not respond to the tax in the model, the tax studied here is the part borne by producers, which is why the tax that meets the budget in 2016 is only \$57.2. And because it applies to the lifecycle emissions of all the oil produced worldwide, each dollar of it moves a large quantity of emissions. Near the naive tax, post-implementation emissions fall by about 4.8 GtCO₂eq per dollar. A premium of a few dollars therefore corresponds to a residual

⁹The relevant number is read at $\tau^N = 66.8$ because the full-cycle calibration solves $E^N(\tau^N) = B_\Delta$. This differs from the 15.9 GtCO₂eq of Figure 3, which is the wedge at the timely tax. The two measures count locked-in fields at different tax levels. A higher tax draws more fields into their lock-in zone and shortens the lives of those already in it, and between \$57 and \$67 the first effect dominates.

overshoot of about one year of global oil emissions, and equation (11) gives the conversion from one to the other.

How the residual overshoot builds up with the delay. Table 3 reports, for each implementation year, the remaining budget, the lock-in-adjusted tax, the premium, and the residual overshoot left by the naive tax, for delays starting in 2016 and in 2025. The cost of ignoring lock-in grows quickly in the first years of the delay and then more slowly. It reaches 8.2 GtCO₂eq after six years and 17.1 GtCO₂eq after fourteen. Two forces are at work. Each additional year of delay adds new fields that will produce for their whole life after implementation, but it also ages every field already built, so that commitments made earlier are partly emitted during the delay rather than after it. Once the delay is longer than the average producing life of locked-in fields, the two effects balance. The right-hand columns show the same build-up from a later start: ignoring only the five years of lock-in of the 2025–2029 period would still leave 10.2 GtCO₂eq above the budget, 7.1% of what remains, and require a premium of \$1.9. Most of the lock-in measured here is decided in the last years before implementation, mainly by deepwater projects, as Figure 5 shows.

Year	Budget (GtCO ₂ eq)	Adjusted tax (\$/tCO ₂ eq)	From 2016		From 2025	
			Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)	Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)
2016	350.3	57.2	–	–	–	–
2018	320.9	58.3	0.5	2.8 (0.9%)	–	–
2020	291.4	59.4	0.7	5.2 (1.8%)	–	–
2022	264.0	60.6	0.9	8.2 (3.1%)	–	–
2024	235.1	62.8	2.1	11.1 (4.7%)	–	–
2026	205.8	64.8	2.4	13.8 (6.7%)	0.4	1.9 (0.9%)
2028	175.6	67.5	3.2	15.3 (8.7%)	1.3	5.8 (3.3%)
2030	144.9	70.5	3.6	17.1 (11.8%)	1.9	10.2 (7.1%)

Table 3: Lock-in premium and residual overshoot by year of implementation.

Notes: Tax rising at 2% per year. Taxes are 2016-anchored levels. The remaining budget is the initial budget net of emissions released before implementation on the business-as-usual path. Overshoot percentages are relative to the remaining budget. “From 2016” and “From 2025” indicate the year from which the delay is measured, the state of the industry in that year being taken as given.

What the premium measures. The premium rises with the length of the delay in Table 3, because a longer delay lets more projects be screened with a tax that is too low. The length of the delay is not sufficient, however. The premium measures how far the supply curve has moved below the full-cycle calibration, which depends both on how many projects were developed during the delay and on how far the tax their investors used fell below the tax the budget turns out to require (Section 2.4). Delay-induced fields are, by definition,

projects that would not have been worth developing had investors expected the tax: they cover their operating costs after 2030, but will not necessarily recover their development cost. The section below reports the results with a variety of anticipation scenarios by investors, in which the gap between their expectations and the actual tax needed narrows.

4.4 Sensitivity

Figure 7 reports the residual overshoot and the lock-in premium across six groups of assumptions: anticipations, costs and the computation of net present values, carbon intensities, the carbon budget, the increase rate of the tax, and pass-through. Outside the anticipation group, the main specification is not an extreme case: several specifications, such as constant marginal costs, a 15% discount rate, a GWP-20 conversion and a 1% tax increase rate, give a residual overshoot above 19 GtCO₂eq, and outside the anticipation and tax increase groups the premium ranges from \$2.5 to \$4.6. Within the anticipation group, the main specification is by construction the upper bound, but the scenarios closest to observed investor behaviour give results of the same order.

Anticipations. Anticipation changes which fields are developed during the delay, and nothing else. Investors who expect part of the tax stop developing the projects that would not be worth building even at that anticipated tax, and these are the most capital-intensive ones, which carry the deepest lock-in (Section 2.5). The four anticipation scenarios of Figure 7 span two ways of being wrong about the tax. Investors who expect the correct tax but only from 2030 stop developing the late projects that would have carried the largest post-2030 commitments, and the residual overshoot falls to 7.9 GtCO₂eq (and the premium to \$1.9). If investors screen their projects as if the correct tax was implemented five years after projects' startup, the residual overshoot amounts to 12.0 GtCO₂eq and the premium to \$2.7. Investors who expect the tax immediately but underestimate its level leave 8.5 GtCO₂eq and \$1.7 when they expect three quarters of it, and 13.6 GtCO₂eq and \$2.9 when they expect half of it.

Which of these scenarios describes firms best? Oil companies screen investments with internal carbon prices (Fried et al., 2021), most of them between \$23 and \$60 and mainly used to stress-test existing assets rather than to guide new investments (Ahluwalia, 2017). Experimental evidence on firm investment (Dolls et al., 2026) finds that irreversible investment responds to the carbon price firms find credible, which under realistic policy uncertainty is far below the expected price, and that perceived credibility matters more for investment than the price level itself. For a 1.5°C-aligned tax announced in 2016 and not implemented, this credible price is close to zero. The main specification sets it to zero as a normalisation

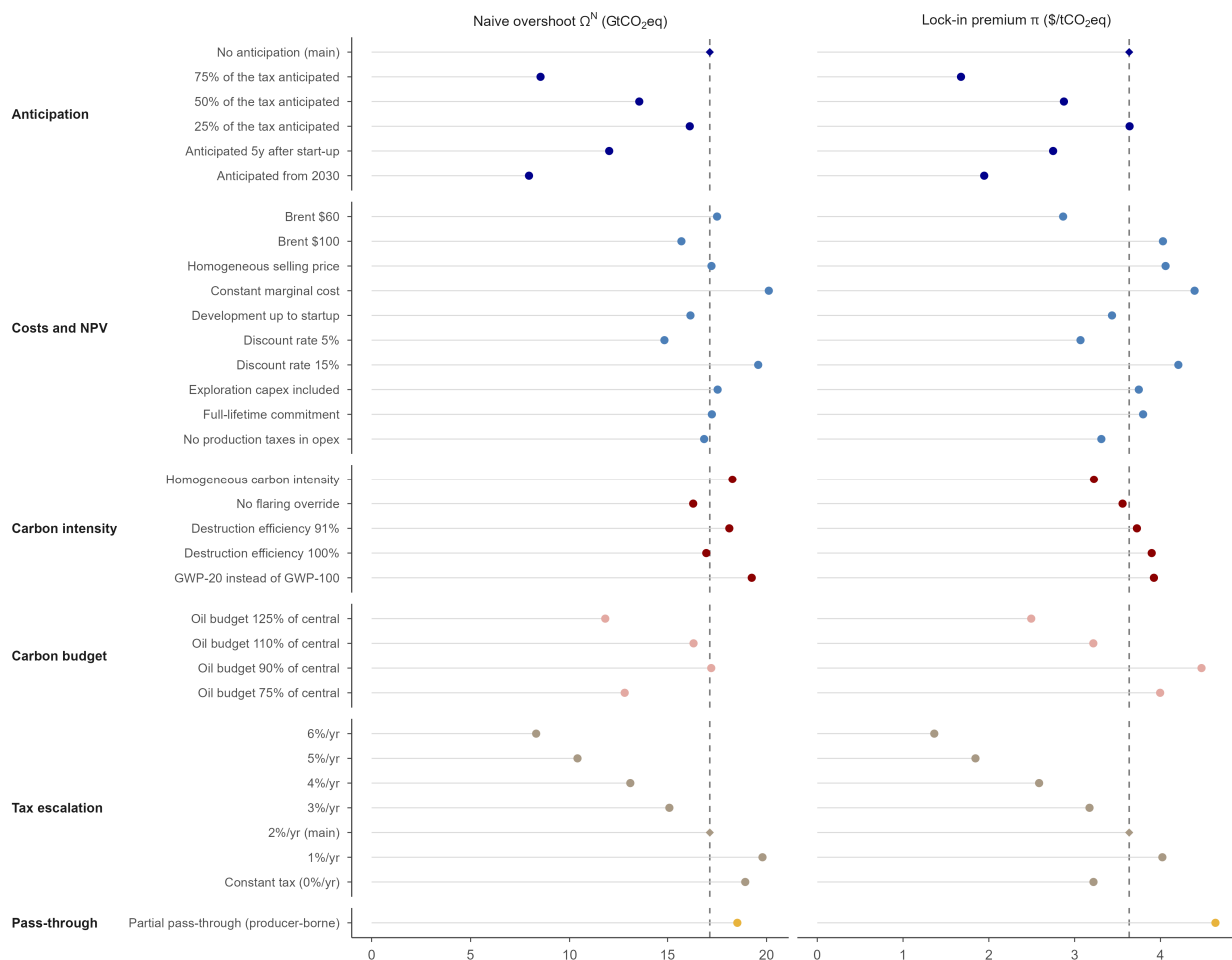


Figure 7: Sensitivity of the residual overshoot and the lock-in premium, by group of assumptions.

Notes: Each row is one specification; the main specification is marked by the diamond and the dashed line. Taxes are 2016-anchored levels of the tax rising at 2% per year. The pass-through row reports the premium per dollar borne by producers at implementation, which makes it comparable with the other rows.

rather than as an estimate, and Figure 8 characterises what that choice costs over the whole range of anticipated taxes.

Figure 8 generalises the second type of scenario to any share of the tax anticipated by investors. The residual overshoot and the premium fall only slowly as long as investors expect less than about half of the tax. The residual overshoot is still 15.3 GtCO₂eq when they expect 30% of it and 13.6 GtCO₂eq when they expect half, and the premium is still \$2.9 at half. The fields removed by weak anticipations are the most capital-intensive ones, but there are few of them, because at these tax levels most projects remain worth developing. The decline is faster above half: the residual overshoot falls to 8.5 GtCO₂eq at three quarters and 2.8 GtCO₂eq at 90%, and both measures reach zero at full anticipation, where investors screen with the tax the budget requires. Reducing the cost of the delay substantially would

thus have required investors to expect a tax close to the one the budget requires, that is a commitment to the tax that was both announced and credible.

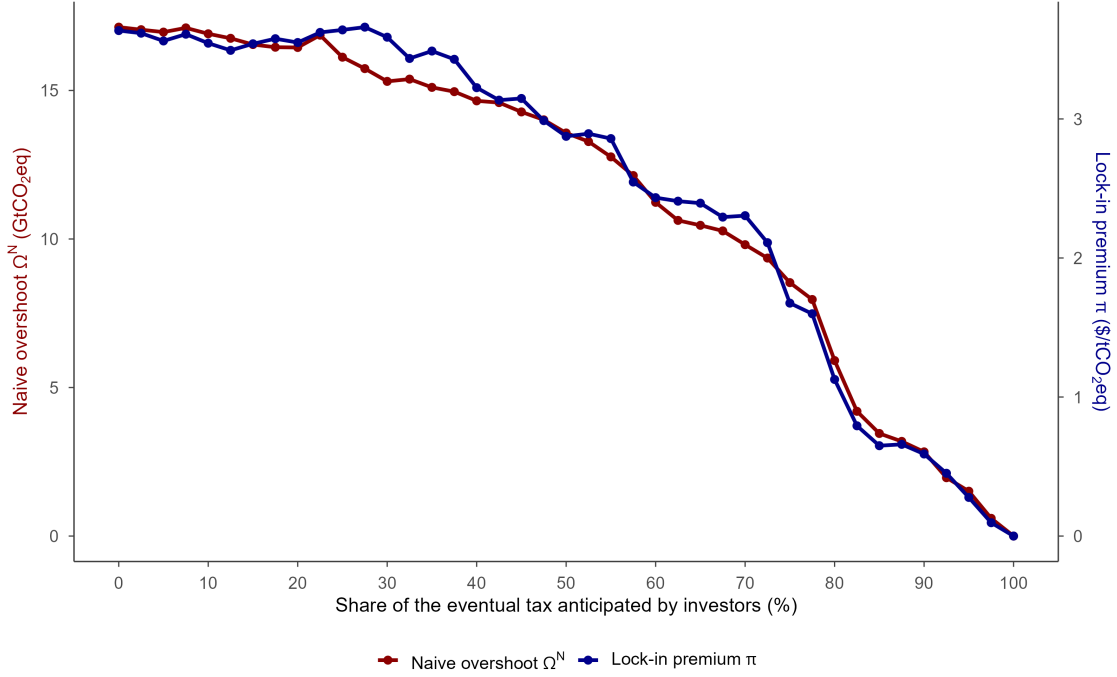


Figure 8: Residual overshoot and lock-in premium, by share of the tax anticipated.

Notes: Share of the eventual tax anticipated by investors on the horizontal axis. Both are zero at full anticipation, where investors screen projects with the tax the budget requires (Section 2.5).

Costs, carbon intensities and budget. The cost assumptions move the results in the direction predicted by the model. A higher discount rate widens every lock-in zone. The residual overshoot is 14.8 GtCO₂eq at 5% and 19.6 GtCO₂eq at 15%, and the premium \$3.1 and \$4.2. Constant marginal costs, which remove the rising cost path that phases out locked-in fields over time, raise the residual overshoot to 20.1 GtCO₂eq. Counting as development only the capital spent up to start-up narrows every lock-in zone and gives a lower bound on lock-in, with a residual overshoot of 16.2 GtCO₂eq and a premium of \$3.4 (Section 3.3): the results do not depend on where exactly the line between development and production capital is drawn. Adding the fields not yet discovered, with their exploration costs treated as development costs, changes little, with a residual overshoot of 17.5 GtCO₂eq, because most of them are not developed before 2030 and are stopped by the tax. The Brent price is the one assumption that moves the two measures in opposite directions. At \$60, the residual overshoot rises to 17.5 GtCO₂eq and the premium falls to \$2.9; at \$100, the residual overshoot falls to 15.7 GtCO₂eq and the premium rises to \$4.0. A higher price raises every production breakeven and makes emissions respond less to the tax near τ^N , so the same

quantity requires a larger premium. Assumptions on carbon intensity move the residual overshoot between 16.3 and 19.2 GtCO₂eq. A common carbon intensity across assets raises it to 18.3 GtCO₂eq, which indicates that locked-in fields are on average cleaner than the rest of the sample. Varying the budget traces the hump of the wedge from another angle: the residual overshoot is 11.8 GtCO₂eq at 125% of the central budget, 16.3 GtCO₂eq at 110%, 17.2 GtCO₂eq at 90% and 12.8 GtCO₂eq at 75%. It changes little around the central value and falls only when the much higher tax required by a small budget starts to shut down locked-in fields themselves.

Instrument design. The constant tax that meets the same budget is \$99.4/tCO₂eq, and the delay then overshoots the budget by 34%, against 19% under the increasing tax. Appendix D.4 reports the full set of constant-tax results. The comparison is a result rather than a robustness check. The total cost of the delay depends heavily on the design of the tax, while its lock-in component does not. Under the constant tax, the residual overshoot is 18.9 GtCO₂eq and the premium \$3.2, against 17.1 GtCO₂eq and \$3.6 under the increasing tax, whereas the total overshoot is almost twice as large. Across increase rates from 0 to 6%, the residual overshoot varies between 8.3 and 19.8 GtCO₂eq. As Section 2.6 explains, the tax increase rate acts on lock-in in two opposite directions: it moves the development boundary and makes more young fields delay-induced, and it shortens the post-implementation life of every locked-in field. The first effect dominates between 0 and 1%, where the residual overshoot rises, and the second above. What is built during a delay without any tax depends on the absence of the tax, not on the shape of the tax that eventually arrives. What the shape of the tax changes is the intensive margin, which is released before implementation and which no tax set in 2030 can recover.

Pass-through. The tax studied so far is entirely borne by producers. I relax this assumption using exogenous pass-through rates from Heal and Schlenker (2019), who solve for the oil price path under a carbon tax using the same field-level data. For a tax rising at 2% per year, consumers initially bear 40% of the tax through a higher oil price and producers 60%, and the share borne by producers then rises until they bear the whole tax. I assume that this share rises at a constant rate and reaches 100% sixty years after the tax is introduced.¹⁰ Only the share borne by producers deters development and retires production, which changes the results in two ways. First, the gross tax required to meet the budget is higher, and so is the gross premium. Second, because the pass-through path starts when the tax is introduced, a tax delayed to 2030 is borne by producers at a lower share in 2030 than a timely tax would

¹⁰Values are drawn from Table A.4 in Heal and Schlenker (2019). A producer share of 60% reaching 100% in sixty years corresponds to a growth rate of 0.855% per year, which is the rate used in the analysis.

have been at the same date. This is a second channel through which the delay raises the tax required, and it is specific to this variant. Expressed per dollar borne by producers at implementation, which makes it comparable with the other rows of Figure 7, the premium is \$4.6 against \$3.6 in the main case, and the residual overshoot is 18.5 GtCO₂eq against 17.1 GtCO₂eq. The gross premium, which also includes the second channel, is \$7.7.

5 Incidence and targeted policy

The previous section measured the lock-in created by the delay and the tax premium it requires. This section asks where the locked-in fields are, who bears the cost of the premium, and what can be done to alleviate the cost of lock-in. The three questions are linked. Locked-in fields sit well below their production breakeven when the tax arrives, so a higher tax removes little of their production (Section 5.1). The premium is therefore paid by other fields, adding a second level of redistribution across fields after the one created by the overproduction during the delay, and this changes who gains and who loses from the delay (Section 5.2). Because the locked-in fields are few and similar, targeted policies can partially achieve what the delayed tax cannot, provided they target the right projects early enough (Section 5.3).

5.1 Where the lock-in sits

Overcommitted emissions are concentrated in a small number of projects, in majority offshore. At the naive tax, the emissions committed beyond 2030 by delay-induced fields come from 694 assets, and 41 of them carry half of the total (Appendix Table 17). Figure 9 shows where these projects are, pooled by province¹¹. Offshore projects account for 75% of these emissions, and deepwater projects alone for 40%. The coasts of Brazil and Guyana and the Gulf of Mexico account for 34% of the total, Brazil alone for 20%, and South America for 31%. The Middle East accounts for 22%, mostly through a few large offshore and onshore redevelopments approved during the delay, the largest single asset of the sample being the expansion of the Marjan field in Saudi Arabia. Capital intensity, rather than carbon intensity or operating cost, is what these projects have in common. This is the sorting result of Section 2.2 as it appears in the data. The timing is also concentrated, since 55% of these emissions are committed by investments made between 2025 and 2029.

The locked-in stock in these assets is also deep. Figure 10 reports, for each level of the

¹¹Province is a level of aggregation available in Rystad smaller than countries, but comprising multiple assets. The median number of assets per dot on the graph is 2, the average 5.1, and the maximum 40 (Western Siberia, Russia).

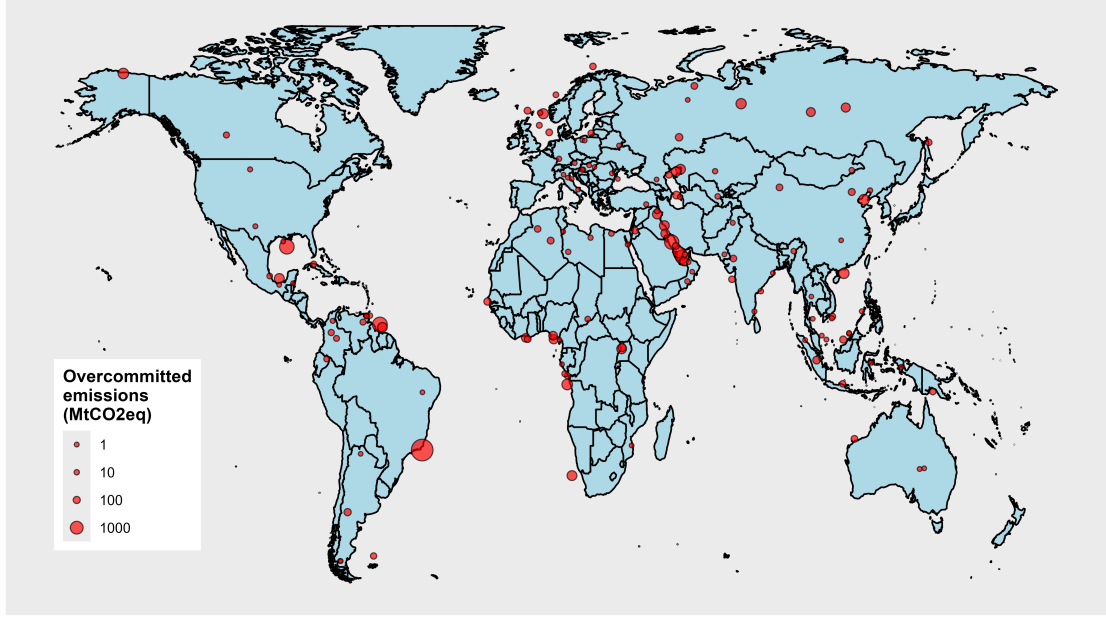


Figure 9: Locked-in projects by province.

Notes: Circle size is the post-2030 emissions of delay-induced fields that survive the naive tax implemented in 2030. A province is a level of aggregation in Rystad's data between the asset and the country. No-anticipation scenario, tax rising at 2% per year.

2030 tax, the quantity of emissions locked in at the naive tax that would still be released, split by supply segment; Appendix Figure 25 gives the same curve for all delay-induced fields together. Raising the tax to the lock-in-adjusted level of \$70.5 leaves 92% of them in place. Even in the case of a \$80 tax, that would undershoot the remaining budget in 2030 by 27%, 70% would survive. The few locked-in shale and oil sands projects, by contrast, are almost all phased out by a tax of \$80, because their lock-in zones, although sometimes wide, sit low. For most locked-in projects, the carbon tax arrives too late. An extra dollar of tax removes little of their production, and meeting the budget requires stranding fields elsewhere.

5.2 Who gains from the delay

The delay changes each producer's post-2030 emissions, relative to the timely policy, through two channels. The overproduction channel runs through fields that a timely tax would also have built. These fields gain the production the delay lets them sell untaxed before 2030, and fields in general lose the production stranded when the tax has to rise from τ_{2016}^* to τ_{2030}^N to compensate for it. The lock-in channel runs through the fields that exist because of the delay. These fields gain their post-2030 production, and other fields lose the production stranded when the tax has to rise further, from τ_{2030}^N to τ_{2030}^* , to compensate for it. The 50.8 GtCO₂eq the delay lets fields sell before 2030 are offset by 50.9 GtCO₂eq

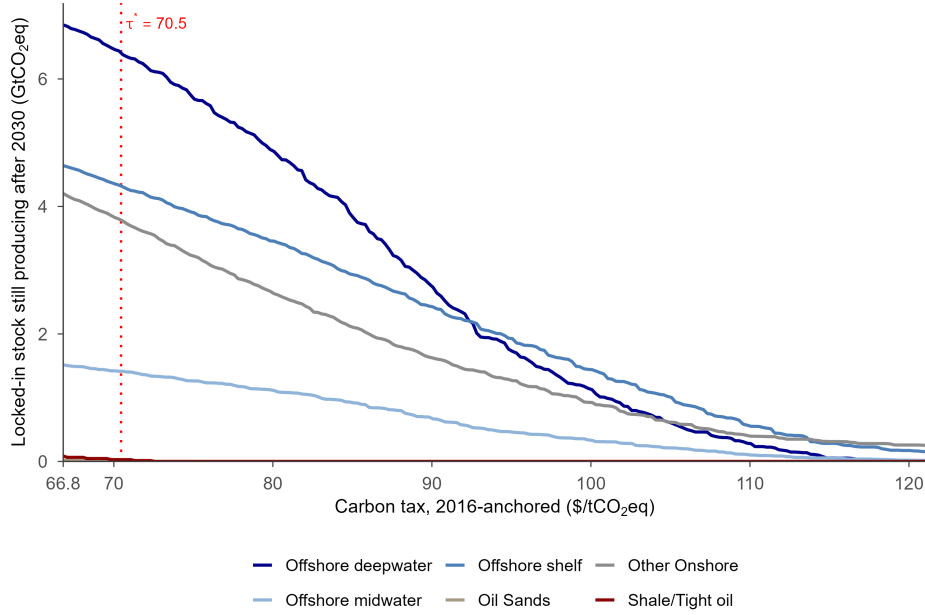


Figure 10: Depth of lock-in, by supply segment.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, by supply segment, in GtCO₂eq. The horizontal axis starts at the naive tax, $\tau^N = \$66.8$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$70.5$. Taxes are 2016-anchored levels of the tax rising at 2% per year.

stranded after 2030 by the first tax increase, and the 15.8 GtCO₂eq that delay-induced fields emit after 2030 are offset by 15.6 GtCO₂eq stranded by the second¹². Table 4 reports the decomposition by continent, in GtCO₂eq and as a share of each continent's own 2016 reserves, and Appendix Table 18 reports it for the largest producing countries.

Three facts stand out. First, the usual view of the delay's incidence, which counts only overproduction, has North America gaining the most and the Middle East paying the most. North American shale and oil sands sell 28 GtCO₂eq more before 2030 than a timely tax would have allowed, while the low-cost fields of the Middle East are the ones a higher tax strands afterwards, 28 GtCO₂eq of them on net. Russia is on the same side as the Middle East, with a net loss of 5 GtCO₂eq. Second, lock-in changes the picture in the middle of the distribution. It accounts for 5% of the net effect for North America, where few projects would have been deterred by a timely tax, but for three quarters of the net effect for South America. It adds to the losses of Russia and the Middle East rather than offsetting them. The production stranded by the premium falls on the fields with the lowest production breakevens, 9.7 GtCO₂eq in the Middle East and 3.0 GtCO₂eq in Russia, 81% of the total. These two regions therefore pay for the lock-in of other regions on top of paying for the overproduction of the delay.

¹²While both channels should balance, small differences between gains and losses can exist, due to steps in the supply curve caused by large assets.

	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% of 2016 reserves)
North America	28.38	1.47	29.85	16.6
South America	1.29	3.98	5.26	10.5
Asia	3.34	1.26	4.60	7.6
Europe	1.10	0.46	1.56	11.2
Africa	-0.39	1.19	0.80	1.6
Oceania	0.03	0.04	0.07	3.2
Russia	-5.41	-2.13	-7.54	-11.8
Middle East	-28.46	-6.09	-34.55	-10.8

Table 4: Incidence of the delay by continent.

Notes: Change in post-2030 emissions relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The overproduction column is the net effect through fields a timely tax would also have built, and the lock-in column the net effect through delay-induced fields; the two sum to the net effect. The last column is the net change in production (in barrels) as a share of the continent's own 2016 reserves; since carbon intensities differ across fields, it can differ in sign from the change in emissions for small net effects.

Third, for some producers the lock-in channel reverses the sign of the total. Africa loses 0.4 GtCO₂eq through overproduction and gains 1.2 GtCO₂eq through the fields it develops because of the delay, which makes it a net winner, at 0.8 GtCO₂eq. The same reversal holds for Brazil, which loses 0.4 GtCO₂eq through overproduction, gains 3.0 GtCO₂eq through its deepwater fields, and ends a net winner at 2.7 GtCO₂eq, and on a smaller scale for Angola, Namibia, Senegal and Vietnam. Producers whose lock-in gain falls short of their overproduction loss, such as Ghana, Kazakhstan, Qatar and Azerbaijan, remain net losers: lock-in makes them smaller losers, not winners. Guyana, often seen as the typical producer created by the delay, does not reverse, because it also gains through overproduction: it gains on both counts.

Figure 11 shows the decomposition as a scatter, normalised by 2016 reserves so that producers of very different size can be compared. The horizontal position is the usual view of the delay's incidence, and the vertical position is what this view misses. Real winners from the delay, accounting for both overproduction and lock-in, lie in the upper-right half of the diagonal dashed line. In the remaining half of the upper-left quarter, producers see their losses diminished by the lock-in, without being net winners. As highlighted in tables 4 and 18, the US, China, and most of Europe and Asia win on both counts, while Russia and Middle Eastern countries see lock-in add a loss of two to four percent of reserves to a loss of eight to fourteen percent through overproduction.

The blue zone highlights producers that the usual view counts as losers, because a higher tax strands their existing fields, but whose investment during the delay is large enough to more than compensate: Brazil, with a gain of 15% of its reserves through lock-in against a

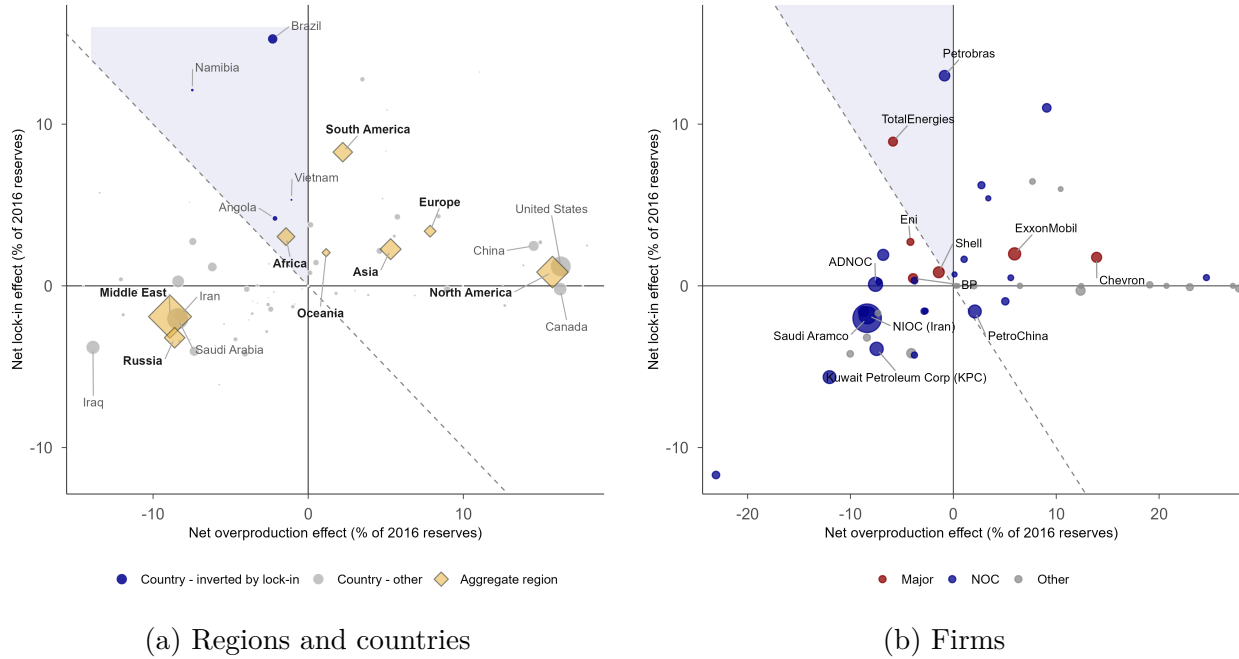


Figure 11: Incidence of the delay as a share of 2016 reserves.

Notes: The horizontal axis is the net effect through overproduction, the vertical axis the net effect through lock-in. (a) Aggregate regions (diamonds) and countries (circles, area proportional to 2016 reserves); countries with less than 500 Mbbl of 2016 reserves are excluded, and Senegal lies outside the frame. (b) The 50 largest firms by 2016 equity reserves; production is attributed to equity holders at their share in each year, and shares after 2026 are held at their 2026 level. EOG Resources, Midland Oil Company and Cenovus Energy lie outside the frame. In both panels, the dashed line is where the net effect is zero, and the shaded area contains the producers that lose through overproduction and gain more through lock-in, whose net sign is reversed by lock-in.

loss of 2% through overproduction; Namibia and Senegal, whose oil industry mostly dates from after 2016; Angola; and Africa as a whole, where these countries outweigh losers on both counts such as Algeria. South America as a region lies just to the right of the vertical axis, with lock-in accounting for most of its gain. The reversals are the same whether the effect is measured in barrels, as in the figure, or in emissions, as in Table 4.

Firms. The same decomposition can be run by company, attributing each year's production to equity holders according to their share in that year (Appendix Table 19) (panel (b) of Figure 11). As expected, national oil companies as a group lose 26 GtCO₂eq through overproduction, Saudi Aramco alone 10.6 GtCO₂eq, and lose again through lock-in. Independent producers, mostly North American and focused on shale and oil sands, gain 15 GtCO₂eq through overproduction and little through lock-in. Majors are split in two groups: American majors win on both counts while European lose from overproduction and gain from lock-in. The firms the usual view misclassifies are those whose investment during the delay is in deep offshore projects. Petrobras loses 0.1 GtCO₂eq through overproduction and gains 2.0 GtCO₂eq through lock-in, 13% of its 2016 equity reserves, and TotalEnergies loses

0.5 GtCO₂eq and gains 1.0 GtCO₂eq. For these firms, the assets that make the delay profitable are those a targeted restriction would have to reach.

5.3 Targeted development restrictions

A worldwide carbon tax on oil producers as envisioned in the paper seems unlikely in the near future. At the international level, the role played by international organisations does not play in favour of supply-side policies in addressing climate change. As [Van Asselt \(2021\)](#) puts it, the “United Nations Framework Convention on Climate Change (UNFCCC) territorial accounting for greenhouse gas emissions emphasizes fossil fuel consumption over supply, so countries are incentivized to pursue demand-side policies to fulfil international agreements”. However, this shortcoming is not the end of the road for production-based policies: a substantial share of the cost arising from infrastructure lock-in could be alleviated through targeted action. The literature on supply-side policy makes the general case for such instruments, highlighting their direct and certain effect on supply ([Erickson et al., 2018](#)) and its greater potential to mobilise public support ([Green and Denniss, 2018](#)).

This subsection studies a variety of targeted policies aiming at alleviating the cost arising from lock-in. All instruments, with the exception of targeted early taxes, act on development only instead of production. A ban prevents the approval of new projects in a given perimeter from a given year. Projects already approved are left alone, so the exercise never cancels capital that is already partly spent. Table 5 reports, for each instrument, the naive tax, the lock-in-adjusted tax, the premium and the residual overshoot, each instrument being evaluated at its own budget-consistent tax, and Appendix D.9 defines each perimeter and each instrument precisely. The remaining budget also changes, because a project that is not developed does not emit during the delay either, and this second effect is reported in the last column: it is what a restriction buys on the intensive margin, as opposed to what it buys on lock-in.

Three lessons come out of the table, and each of them follows from a property of lock-in established earlier. First, what a restriction achieves depends on the width of the lock-in zones it removes. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 40%, from 17.1 to 10.4 GtCO₂eq, and the premium by a third. A ban on new shale approvals cuts the residual overshoot by 1% only and slightly raises the premium, to \$3.8. The ban on fracking, the most politically salient supply-side restriction, is not a bad policy. It enlarges the budget remaining in 2030 by 2.4 GtCO₂eq, because shale projects that are not developed do not produce during the delay either¹³. But it leaves lock-in untouched,

¹³Since a large share of shale assets are already producing by 2016, the reduction in delay-era overproduction remains limited. However, shale production within an asset is maintained or increased through

	τ^N (\$/tCO ₂ eq)	τ^* (\$/tCO ₂ eq)	π (\$/tCO ₂ eq)	Ω^N (GtCO ₂ eq)	Remaining budget (GtCO ₂ eq)
No targeted policy (delay only)	66.8	70.5	3.6	17.1	144.9
<i>Ban on new approvals from 2016</i>					
Deepwater	64.9	67.4	2.5	10.4 (−40%)	149.4
Shale	66.2	70.0	3.8	17.0 (−1%)	147.3
The 50 largest locked-in projects (upper bound)	66.2	68.1	1.9	7.5 (−56%)	147.2
The 20 largest locked-in projects (upper bound)	66.4	69.1	2.7	11.9 (−31%)	145.9
Deepwater, OECD-headquartered firms	65.6	68.7	3.1	13.7 (−20%)	147.5
Deepwater, European firms	66.4	69.9	3.5	15.4 (−10%)	145.8
<i>Ban on new approvals from 2025</i>					
All new development	59.8	63.5	3.7	11.6 (−32%)	145.3
Deepwater	66.3	69.9	3.7	15.3 (−11%)	145.0
Shale	66.8	70.4	3.6	17.1 (−0%)	144.9
<i>Other instruments</i>					
Carbon tax on deepwater from 2016	66.1	68.4	2.4	10.3 (−40%)	150.1
Carbon tax on shale from 2016	66.3	70.0	3.8	16.9 (−1%)	165.4
Carbon tax on OECD deepwater from 2016	66.4	69.3	2.9	13.3 (−22%)	147.4
Development capital cost +50%, deepwater	66.0	70.0	4.0	18.1 (+6%)	145.2
Development capital cost +50%, all projects	63.4	68.2	4.8	19.7 (+15%)	147.2

Table 5: Targeted development restrictions and alternative instruments.

Notes: Each instrument is evaluated at its own budget-consistent tax. Taxes are 2016-anchored levels of the tax rising at 2% per year, in \$/tCO₂eq. The change in the residual overshoot relative to the delay without restriction is in parentheses. The remaining budget rises under a restriction because less is emitted before 2030. Appendix Figure 26 reports the full set of instruments.

since shale lock-in zones are narrow and sit below the tax. The two ban targets pursue different objectives. Policies aimed at shale and oil sands buy budget while policies aimed at capital-intensive offshore projects lower the cost of lock-in.

The second lesson is that the concentration documented in Section 5.1 makes very narrow restrictions effective. Banning the approval of the 50 assets that carry the most locked-in emissions cuts the residual overshoot by 56%, more than a ban on all new deepwater development, and brings the premium down to \$1.9. Banning the 20 largest cuts it by 31%. This is the practical consequence of the result that 41 assets carry half of the locked-in stock. The ranking uses what the delay eventually built, which a planner in 2016 could not know, so these two rows are an upper bound on what targeting by size can achieve. The deepwater ban, which requires no such foresight, is the implementable benchmark.

The third lesson is that a restriction does not have to be worldwide, but it does have to reach the firms that build capital-intensive projects. A ban applying to assets at least half owned and operated by firms headquartered in an OECD country cuts the residual overshoot by 20%, half of what the worldwide deepwater ban achieves, and the same ban restricted to European firms cuts it by 10%. What a coalition can reach is not the deepwater development that takes place on its territory, but the deepwater development its own firms

continuous drilling of new wells. A policy banning the drilling of any new well would hence affect shale overproduction in a much greater magnitude.

own and operate, which is a much larger share. More than 70% of the deepwater projects approved during the delay, and 58% of the reserves they hold, are majority-owned and operated by firms headquartered in an OECD country, so a restriction decided in a handful of jurisdictions reaches projects on three continents.

Timing matters as much as targeting. Applied from 2025 rather than 2016, the deepwater ban cuts the residual overshoot by 11% instead of 40%, and a ban on all new approvals from 2025, which is as broad an instrument as one can write down, cuts it by 32%. Most of the lock-in of the delay is already approved by 2025, consistent with the concentration of deepwater commitments in the 2020–2024 and 2025–2029 periods (Figure 5). Acting later than that means cancelling projects that are already approved, and possibly already under construction, with the compensation issues this raises (Paletta et al., 2025), rather than deterring them.

Two alternative instruments are worth comparing with the bans. Early action through a tax targeting deepwater projects alone from 2016, at the level the budget requires, achieves the same reduction in the residual overshoot as the deepwater ban, 40%. It also enlarges the remaining budget in 2030, 5.2 against 4.5 GtCO₂eq, since it also retires active deepwater assets during the delay. The same instrument applied to shale from 2016 leaves lock-in where it is but enlarges the remaining budget by 20.5 GtCO₂eq, by far the largest budget effect of any instrument in the table: it confirms that segment-specific early taxation is a policy for the intensive margin when it targets shale, and for lock-in when it targets offshore projects. This contrast also points to a limit of the model that works in favour of the second instrument. A tax applied early to shale, or to OECD production more generally, removes a large volume of barrels while the rest of the world is still untaxed, and the price response this would trigger, which the model excludes, would bring part of that production back elsewhere. A restriction on deepwater development is much less exposed to that objection. It barely changes production during the delay, since the projects it stops are not producing yet, and what it removes materialises after 2030, when the tax is in force everywhere, which leaves little room for a price-driven response by other producers. The same argument applies to the ownership-based restrictions of Table 5: an early tax levied by OECD countries on their own deepwater assets cuts the residual overshoot by 22%, close to the ban, but it is the ban that avoids the spillover. A tax on development capital, by contrast, does little against lock-in and can make it worse. Raising the development cost of deepwater projects by 50% raises the residual overshoot by 6% and the premium to \$4.0, and the same shock applied to all new projects raises the residual overshoot by 15% and the premium to \$4.8. Raising development costs lowers the tax a project can survive to before it is built without affecting its operating cost, subsequently enlarging the lock-in zone of all assets. The naive planner

reads the shock as deterring more assets that are in reality already built by 2030, which is why the naive tax falls, from \$66.8 to \$63.4 when the shock applies to all projects, while the emissions these projects still deliver after 2030 rise through the lower tax.

Restrictions also change the incidence described in Section 5.2, and they change both sides of it at once. By removing part of the lock-in, they remove the gain that this lock-in gives to the producers who hold it, and, because the premium falls, they also reduce the production stranded elsewhere to pay for it. The deepwater ban from 2016 cuts the post-2030 emissions of delay-induced fields from 15.8 to 9.8 GtCO₂eq worldwide, and the ban on the 50 largest projects cuts them to 7.1 GtCO₂eq. By the accounting of Section 5.2, the stranding that compensates for them falls by the same amounts, benefiting mainly the Middle East and Russia. South America and Africa, which gain 4.8 and 2.0 GtCO₂eq through lock-in without any restriction, keep 1.0 and 1.1 GtCO₂eq of it under the deepwater ban. The producers that gain from lock-in are therefore those that would oppose a restriction on it, and the producers that pay for it through a higher tax are those that would gain from it. Part of what a restriction removes also comes back elsewhere. Under the deepwater ban the budget constraint is relaxed, the naive tax falls from \$66.8 to \$64.9, and marginal projects in regions with little deepwater exposure now pass the development test, which raises their post-2030 emissions by 4% in Asia and 17% in Russia. This leakage is internal to the model, distinct from the price-driven green paradox excluded in Section 2, and it goes in the same direction. An early tax on shale works on the other channel and therefore has a different incidence. It removes most of the overproduction gain of North America during the delay, which enlarges the remaining budget by 20.5 GtCO₂eq and lowers the tax the budget then requires, so the stranding borne after 2030 by the Middle East and Russia falls while lock-in, and its beneficiaries, are left untouched.

Where public action can act. Convincing large producing countries to adopt such bans may prove difficult, with OPEC countries leading the opposition to any moratorium on new field development (Lujala et al., 2022), but the concentration documented above changes what smaller initiatives can achieve. Most bans adopted so far have come from public campaigns against specific projects (Piggot, 2018; Carter and McKenzie, 2020). With 41 assets carrying half of the post-2030 locked-in emissions, campaigns against specific projects can matter, and Table 5 puts a number on what they can achieve: stopping the approval of the 50 largest of them would have cut the residual overshoot by more than half. The Willow project in Alaska is an example. The project was widely debated in the United States, and was finally approved by the Biden administration in 2023. It embeds 235 MtCO₂eq of locked-in emissions that survive the naive tax, 1.4% of the total. Large offshore develop-

ments can weigh even more: the eight phases of the Buzios deepwater project in Brazil¹⁴ carry 1.4 GtCO₂eq, 8% of the total. These projects are mostly owned and operated by firms headquartered in OECD countries, as shown above, and their development depends on technologies these firms master. OECD countries could therefore restrict these specific projects and encourage the redirection of the corresponding capital. This capital is large. The investment made during the delay in fields that a timely tax would have deterred amounts to 739 billion dollars, discounted to 2025, and it is spread across regions, with about a fifth each in North America, Africa, Asia and South America.

Equity. The two channels of the delay point to different producers, which allows efficiency and equity to be addressed together rather than traded off (Pye et al., 2020). Overproduction during the delay is dominated by North American shale and oil sands, where early action by developed countries is possible within a few jurisdictions (mainly Texas, North Dakota and Alberta) and would be justified on distributive grounds (Lenferna, 2018). Deep lock-in sits in offshore projects largely owned and operated by majors headquartered in developed countries, so restricting these projects and redirecting the committed capital towards clean investment in oil-dependent countries addresses both dimensions at once. The incidence results add one qualification. For Brazil, Angola, Namibia and Senegal, the locked-in projects are what turns the delay into a gain, and a restriction on them is one these countries would experience as a cost. Whether this cost should be compensated depends, as Paletta et al. (2025) show, on contractual provisions and on whether undeveloped reserves count as assets at all.

6 Conclusion

This paper measures what the delay in carbon pricing since the Paris Agreement costs in the oil sector once lock-in from newly developed projects is accounted for. Models of oil supply assign each field a single cost per barrel, which makes the supply curve independent of past investment and rules out lock-in. An oil supply model separating the investment decision from production decision, for 25,336 oil fields, brings lock-in back. A 1.5°C-aligned carbon tax delayed from 2016 to 2030 overshoots the oil carbon budget by 19%. Most of this overshoot is released before 2030, but 16 GtCO₂eq of it is released afterwards, by delay-induced fields that the tax arriving in 2030 cannot stop. A tax set on full-cycle costs, as the usual convention implies, leaves 17 GtCO₂eq unabated, 12% of the budget remaining in 2030, and meeting the budget requires a lock-in premium of \$3.6/tCO₂eq on top of it. The investment behind this lock-in, made during the delay in fields a timely tax would have

¹⁴The Buzios project is split in 13 assets in Rystad data, each corresponding to a phase of the project. Four are already developed by 2016.

deterred, amounts to \$739 billion, discounted to 2025. Most of this cost is decided in the last years before implementation, and it survives to investors screening their projects with a carbon tax as long as it is not sufficiently close to the one the budget requires.

The mechanisms governing locked-in emissions are clear. The depth of a field's lock-in zone is set by its capital intensity, and its height by operating costs and carbon intensity. Consequently, lock-in concentrates in capital-intensive fields with low operating costs, features deepwater offshore projects embody. Conversely, overproduction during the delay concentrates in fields that are cheap to build (or already built in 2016) and expensive to run, shale and oil sands above all. Because locked-in fields barely respond to a higher tax, meeting the budget strands low-cost fields elsewhere, mostly in the Middle East and Russia, and the delay becomes beneficial for producers whose recent investment is in deep offshore projects, such as Brazil and Africa as a whole. The same concentration makes targeted policy effective. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 40% and the premium by a third, while a ban on new shale projects leaves both almost unchanged. Supply-side restrictions on specific projects are therefore not a substitute for carbon pricing, but a complement that addresses the part of the cost of delay that pricing reaches too late.

Three limitations bound these estimates. First, the analysis is confined to oil, so the results should be read as if comparable policies applied to gas and coal at the same time; to the extent that a tax on oil alone leads to substitution towards other fossil fuels, the overshoot is understated. Second, the model excludes the green paradox, the response of producers who anticipate the tax by bringing production and investment forward, so the total cost of the delay computed here is a lower bound. Third, the counterfactual paths of development and production are those forecast by Rystad, which embed current fiscal terms and prices, and the sensitivity analysis varies what can be varied along these paths rather than the paths themselves. Leaving the date of development free would require an equilibrium model of the whole market, whose realism would be open to question, and it would not change the mechanism studied here. Lock-in follows from the presence of a large number of capital-intensive projects in the oil industry, and a delay leaves some of them built whichever path the market takes.¹⁵

¹⁵The model also leaves aside the optimal fiscal response of resource-owning governments to an external carbon price (Section 3.6), and the political economy that produces delays in the first place.

References

- Ahluwalia, M. B. (2017). The business of pricing carbon: how companies are pricing carbon to mitigate risks and prepare for a low-carbon future. *Center for Climate and Energy Solutions, Arlington, VA*.
- Ahlvik, L., Andersen, J. J., Hamang, J. H., and Harding, T. (2022). Quantifying supply-side climate policies.
- Anderson, S. T., Kellogg, R., and Salant, S. W. (2018). Hotelling under pressure. *Journal of Political Economy*, 126(3):984–1026.
- Asker, J., Collard-Wexler, A., De Canniere, C., De Loecker, J., and Knittel, C. R. (2024). Two wrongs can sometimes make a right: The environmental benefits of market power in oil. Technical report, National Bureau of Economic Research.
- Asker, J., Collard-Wexler, A., and De Loecker, J. (2019). (Mis)Allocation, Market Power, and Global Oil Extraction. *American Economic Review*, 109(4):1568–1615.
- Bailey, W., Couët, B., Lamb, F., Simpson, G., and Rose, P. (2000). Taking a calculated risk. *Oilfield Review*, 12(3):20–35.
- Berntsen, M., Bøe, K. S., Jordal, T., and Molnár, P. (2018). Determinants of oil and gas investments on the norwegian continental shelf. *Energy*, 148:904–914.
- Bertram, C., Johnson, N., Luderer, G., Riahi, K., Isaac, M., and Eom, J. (2015). Carbon lock-in through capital stock inertia associated with weak near-term climate policies. *Technological Forecasting and Social Change*, 90:62–72.
- Carter, A. V. and McKenzie, J. (2020). Amplifying “Keep It in the Ground” First-Movers: Toward a Comparative Framework. *Society & Natural Resources*, 33(11):1339–1358.
- Chermak, J. M. and Patrick, R. H. (1995). A well-based cost function and the economics of exhaustible resources: The case of natural gas. *Journal of Environmental Economics and Management*, 28(2):174–189.
- Coulomb, R. and Henriët, F. (2018). The grey paradox: How fossil-fuel owners can benefit from carbon taxation. *Journal of Environmental Economics and Management*, 87:206–223.
- Coulomb, R., Henriët, F., and Reitzmann, L. (2026). “bad” oil, “worse” oil, and carbon misallocation. *Review of Economic Studies*, 93(1):404–437.
- Daubanes, J. X. and Lasserre, P. (2023). How should the use of nonrenewables be taxed under a public budget constraint? *Resource and Energy Economics*, 73:101375.
- Davis, S. J. and Socolow, R. H. (2014). Commitment accounting of CO₂ emissions. *Environmental Research Letters*, 9(8):084018.
- De Canniere, C. (2024). Pump it? market power and the energy transition in the global oil market. Technical report, Working paper.

- Dietz, S. and Venmans, F. (2019). Cumulative carbon emissions and economic policy: in search of general principles. *Journal of Environmental Economics and Management*, 96:108–129.
- Dolls, M., Link, S., Liski, M., and Wolf, G. (2026). When do firms invest green? carbon prices, uncertainty, and policy credibility. Technical report, ifo Institute-Leibniz Institute for Economic Research at the University of
- Draeger, R., Paletta, J., Baptista, L. B., Mortágua, H. D., Rochedo, P. R., Schaeffer, R., and Szklo, A. (2026). The impacts of committed oil projects on remaining oil production: implications for a just transition. *Energy Policy*.
- Elvidge, C. D., Zhizhin, M., Baugh, K., Hsu, F.-C., and Ghosh, T. (2015). Methods for global survey of natural gas flaring from visible infrared imaging radiometer suite data. *Energies*, 9(1):14.
- Erickson, P., Kartha, S., Lazarus, M., and Tempest, K. (2015). Assessing carbon lock-in. *Environmental Research Letters*, 10(8):084023.
- Erickson, P., Lazarus, M., and Piggot, G. (2018). Limiting fossil fuel production as the next big step in climate policy. *Nature Climate Change*, 8(12):1037–1043.
- Fattouh, B. (2007). *OPEC pricing power: The need for a new perspective*. Oxford Institute for Energy Studies.
- Fried, S., Novan, K., and Peterman, W. B. (2021). The Macro Effects of Climate Policy Uncertainty. *Finance and Economics Discussion Series*, 2021(015):1–50.
- Ghosh, S., Jean, L., and Saha, P. (2025). The last good owner effect: Ownership structure and environmental outcomes in oil and gas.
- Gollier, C. (2009). *Pricing the Planet’s Future: The Economics of Discounting in an Uncertain World*. Princeton University Press.
- Green, F., Bois von Kursk, O., Muttitt, G., and Pye, S. (2024). No new fossil fuel projects: The norm we need. *Science*, 384(6699):954–957.
- Green, F. and Denniss, R. (2018). Cutting with both arms of the scissors: the economic and political case for restrictive supply-side climate policies. *Climatic Change*, 150(1-2):73–87.
- Grubb, M., Wieners, C., and Yang, P. (2021). Modeling myths: On DICE and dynamic realism in integrated assessment models of climate change mitigation. *WIREs Climate Change*, 12(3).
- Harstad, B. and Holtsmark, K. (2024). The gas trap: Outcompeting coal vs. renewables. Technical report, Stanford GSB and University of Oslo. December 28, 2024 draft; forthcoming, *Journal of Political Economy*.
- Heal, G. and Schlenker, W. (2019). Coase, Hotelling and Pigou: The Incidence of a Carbon

- Tax and CO Emissions. Technical Report w26086, National Bureau of Economic Research, Cambridge, MA.
- International Monetary Fund (2024). Regional Economic Outlook: Middle East and Central Asia, April 2024: An Uneven Recovery amid High Uncertainty. International Monetary Fund. Working assumptions include Brent crude oil price averaging \$78.61/bbl in 2024 and \$73.68/bbl in 2025.
- IPCC (2006). *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy*. Institute for Global Environmental Strategies (IGES), Hayama, Japan.
- Jahn, F., Cook, M., and Graham, M. (2008). The field life cycle. *Developments in Petroleum Science*, 55:1–7.
- Jakob, M., Luderer, G., Steckel, J., Tavoni, M., and Monjon, S. (2012). Time to act now? Assessing the costs of delaying climate measures and benefits of early action. *Climatic Change*, 114(1):79–99.
- Kellogg, R. (2024). The end of oil. Technical report, National Bureau of Economic Research.
- Kotlikoff, L., Kubler, F., Polbin, A., Sachs, J., and Scheidegger, S. (2021). MAKING CARBON TAXATION A GENERATIONAL WIN WIN. *International Economic Review*, 62(1):3–46.
- Leighty, W. and Lin, C.-Y. C. (2012). Tax policy can change the production path: A model of optimal oil extraction in Alaska. *Energy Policy*, 41:759–774.
- Lemoine, D. and Rudik, I. (2017). Steering the climate system: using inertia to lower the cost of policy. *American Economic Review*, 107(10):2947–2957.
- Lenferna, G. A. (2018). Can we equitably manage the end of the fossil fuel era? *Energy Research & Social Science*, 35:217–223.
- Luderer, G., Pietzcker, R. C., Bertram, C., Kriegler, E., Meinshausen, M., and Edenhofer, O. (2013). Economic mitigation challenges: how further delay closes the door for achieving climate targets. *Environmental Research Letters*, 8(3):034033.
- Lujala, P., Le Billon, P., and Gaulin, N. (2022). Phasing out fossil fuels: Determinants of production cuts and implications for an international agreement. *Global Environmental Politics*, 22(4):95–128.
- Masnadi, M. S., El-Houjeiri, H. M., Schunack, D., Li, Y., Englander, J. G., Badahdah, A., Monfort, J.-C., Anderson, J. E., Wallington, T. J., Bergerson, J. A., et al. (2018). Global carbon intensity of crude oil production. *Science*, 361(6405):851–853.
- McCollum, D., Bauer, N., Calvin, K., Kitous, A., and Riahi, K. (2014). Fossil resource and energy security dynamics in conventional and carbon-constrained worlds. *Climatic change*, 123(3-4):413–426.

- McGlade, C. and Ekins, P. (2015). The geographical distribution of fossil fuels unused when limiting global warming to 2 °C. *Nature*, 517(7533):187–190.
- Meinshausen, M., Meinshausen, N., Hare, W., Raper, S. C., Frieler, K., Knutti, R., Frame, D. J., and Allen, M. R. (2009). Greenhouse-gas emission targets for limiting global warming to 2°C. *Nature*, 458(7242):1158–1162.
- Mercure, J.-F., Pollitt, H., Vinuales, J. E., Edwards, N., Holden, P., Chewpreecha, U., Salas, P., Sognnaes, I., Lam, A., and Knobloch, F. (2018). Macroeconomic impact of stranded fossil fuel assets. *Nature Climate Change*, page 1.
- Metcalf, G. E. (2019). On the economics of a carbon tax for the united states. *Brookings Papers on Economic Activity*, 2019(1):405–484.
- Nordhaus, W. D. (2007). A review of the stern review on the economics of climate change. *Journal of Economic Literature*, 45(3):686–702.
- Okullo, S. J. and Reynès, F. (2016). Imperfect cartelization in opec. *Energy Economics*, 60:333–344.
- Paletta, J., Cunha, B. S., Draeger, R., Schaeffer, R., and Szklo, A. (2025). Stranded assets and compensation in oil and gas upstream projects: Conceptual and practical issues. *Energy and Climate Change*, 6:100178.
- Pierru, A., Smith, J., and Zamrik, T. (2018). Opec’s impact on oil price volatility: The role of spare capacity. *The Energy Journal*, 39(2):173–196.
- Piggot, G. (2018). The influence of social movements on policies that constrain fossil fuel supply. *Climate Policy*, 18(7):942–954.
- Pye, S., Bradley, S., Hughes, N., Price, J., Welsby, D., and Ekins, P. (2020). An equitable redistribution of unburnable carbon. *Nature Communications*, 11(1):3968.
- Riahi, K., Kriegler, E., Johnson, N., Bertram, C., den Elzen, M., Eom, J., Schaeffer, M., Edmonds, J., Isaac, M., Krey, V., Longden, T., Luderer, G., Méjean, A., McCollum, D. L., Mima, S., Turton, H., van Vuuren, D. P., Wada, K., Bosetti, V., Capros, P., Criqui, P., Hamdi-Cherif, M., Kainuma, M., and Edenhofer, O. (2015). Locked into Copenhagen pledges — Implications of short-term emission targets for the cost and feasibility of long-term climate goals. *Technological Forecasting and Social Change*, 90:8–23.
- Semieniuk, G., Holden, P. B., Mercure, J.-F., Salas, P., Pollitt, H., Jobson, K., Vercoulen, P., Chewpreecha, U., Edwards, N. R., and Viñuales, J. E. (2022). Stranded fossil-fuel assets translate to major losses for investors in advanced economies. *Nature Climate Change*, 12(6):532–538.
- Seto, K. C., Davis, S. J., Mitchell, R. B., Stokes, E. C., Unruh, G., and Ürge Vorsatz, D. (2016). Carbon Lock-In: Types, Causes, and Policy Implications. *Annual Review of*

- Environment and Resources*, 41(1):425–452.
- Stefanski, R., Ahlvik, L., Andersen, J. J., Harding, T., and Trew, A. (2025). Extracting wedges: Misallocation and taxation in the oil industry. Technical report, CESifo Working Paper.
- Stern, N. (2007). *The Economics of Climate Change: The Stern Review*. Cambridge University Press.
- Tang, B.-J., Zhou, H.-L., Chen, H., Wang, K., and Cao, H. (2017). Investment opportunity in china’s overseas oil project: An empirical analysis based on real option approach. *Energy Policy*, 105:17–26.
- Tong, D., Zhang, Q., Zheng, Y., Caldeira, K., Shearer, C., Hong, C., Qin, Y., and Davis, S. J. (2019). Committed emissions from existing energy infrastructure jeopardize 1.5 °C climate target. *Nature*, 572(7769):373–377.
- Trout, K., Muttitt, G., Lafleur, D., Van De Graaf, T., Mendelevitch, R., Mei, L., and Meinshausen, M. (2022). Existing fossil fuel extraction would warm the world beyond 1.5 °C. *Environmental Research Letters*, 17(6):064010.
- Unruh, G. C. (2000). Understanding carbon lock-in. *Energy Policy*, 28(12):817–830.
- Van Asselt, H. (2021). Governing fossil fuel production in the age of climate disruption: Towards an international law of ‘leaving it in the ground’. *Earth System Governance*, 9:100118.
- van der Ploeg, F. and Rezai, A. (2020). The risk of policy tipping and stranded carbon assets. *Journal of Environmental Economics and Management*, 100:102258.
- Van Vuuren, D. P., Van Soest, H., Riahi, K., Clarke, L., Krey, V., Kriegler, E., Rogelj, J., Schaeffer, M., and Tavoni, M. (2016). Carbon budgets and energy transition pathways. *Environmental Research Letters*, 11(7):075002.
- Venables, A. J. (2014). Depletion and Development: Natural Resource Supply with Endogenous Field Opening. *Journal of the Association of Environmental and Resource Economists*, 1(3):313–336.
- Wachtmeister, H. and Höök, M. (2020). Investment and production dynamics of conventional oil and unconventional tight oil: Implications for oil markets and climate strategies. *Energy and Climate Change*, 1:100010.
- Weitzman, M. L. (1998). Why the far-distant future should be discounted at its lowest possible rate. *Journal of Environmental Economics and Management*, 36:201–208.
- Weitzman, M. L. (2007). A review of the stern review on the economics of climate change. *Journal of Economic Literature*, 45(3):703–724.
- Welsby, D., Price, J., Pye, S., and Ekins, P. (2021). Unextractable fossil fuels in a 1.5 °C

world. *Nature*, 597(7875):230–234.

Wood Mackenzie (2013). Impacts of Delaying IDC Deductibility (2014-2025).

A Model extensions

A.1 Closed forms for a parametric cohort

This appendix collects the closed-form expressions behind the statements of Section 2. They rely on two simplifications that the body does not use: no discounting where indicated, and a parametric cohort of fields. None of the definitions in the body depend on them.

The discounted profit per barrel. Evaluating the integral in (4) with $a^* = a^*(\tau)$,

$$\bar{v}_\delta(\tau) = \frac{p - c_0 - \theta\tau}{\delta} (1 - e^{-\delta a^*}) - \frac{\gamma}{\delta^2} [1 - e^{-\delta a^*} (1 + \delta a^*)], \quad (13)$$

using $\int_0^{a^*} e^{-\delta a} da = (1 - e^{-\delta a^*})/\delta$ and $\int_0^{a^*} a e^{-\delta a} da = [1 - e^{-\delta a^*} (1 + \delta a^*)]/\delta^2$. By Leibniz, $\partial \bar{v}_\delta / \partial \tau = -\theta \int_0^{a^*} e^{-\delta a} da = -\theta(1 - e^{-\delta a^*})/\delta < 0$, the boundary term vanishing because the margin is zero at $a = a^*$. Since a^* does not depend on δ , $\partial \bar{v}_\delta / \partial \delta = -\int_0^{a^*} a e^{-\delta a} (p - c_0 - \theta\tau - \gamma a) da < 0$, the integrand being positive on $(0, a^*)$. As $\delta \rightarrow 0$, $\bar{v}_\delta(\tau) \rightarrow (p - c_0 - \theta\tau)^2 / 2\gamma$.

Depth without discounting. With $\bar{v}_0(\tau) = (p - c_0 - \theta\tau)^2 / 2\gamma$, the development breakeven, the depth and the width of the zone are explicit:

$$\theta \bar{\tau}_D = p - c_0 - \sqrt{2\gamma\kappa}, \quad \theta D = \sqrt{2\gamma\kappa}, \quad a^*(\bar{\tau}_D) = \sqrt{2\kappa/\gamma}, \quad (\delta = 0). \quad (14)$$

Because $\bar{v}_\delta$ decreases with δ , a higher discount rate lowers $\bar{\tau}_D$ and deepens the zone, so the square root is a lower bound on the depth of lock-in at any positive discount rate. The log-log slope of depth per barrel on capital intensity is 1/2 at $\delta = 0$ and above it at any positive rate. This is the reference for the slope of 0.55 reported in Section 3.7, which, as noted there, is a check on the functional form rather than a test of the model.

The cohort. Let a mass N of fields be developed during the delay, with capital intensity distributed with c.d.f. F_κ on $[0, \kappa_1]$, independent of the age at implementation a_Δ , which is uniform on $(0, \Delta]$. All other characteristics, θ , $\bar{q}$, γ and $p - c_0$, are common across fields. The most capital-intensive field of the cohort is pinned down by the model. A field is developed during the delay only if it is worth developing at the tax investors expect to face, which in the baseline is no tax at all, so the highest capital intensity in the cohort is the one at which development just breaks even without a tax:

$$\kappa_1 = \bar{v}_\delta(0), \quad \text{equivalently} \quad a^*(\bar{\tau}_D(\kappa_1)) = \frac{p - c_0}{\gamma} = a^*(0), \quad (15)$$

with $\kappa_1 = (p - c_0)^2/2\gamma$ at $\delta = 0$. The lock-in zone of this field stays open over its whole untaxed life. Under anticipation, the zero tax is replaced by τ^A . Only one feature of F_κ appears in what follows, the share of delay-induced fields

$$G(\tau) \equiv \Pr [\kappa > \bar{v}_\delta(\tau)] = 1 - F_\kappa(\bar{v}_\delta(\tau)), \quad (16)$$

the fraction of the cohort a timely tax τ would have deterred. Since fields only differ in κ and a_Δ , the development question depends on κ alone and the survival question on a_Δ alone. Every row of the type table in Section 2.3 therefore aggregates into $G(\tau)$ or $1 - G(\tau)$ times an integral over ages, and the expressions below hold for any F_κ . With κ uniform and $\delta = 0$, $G(\tau) = 1 - (1 - \tau/\bar{\tau}_P(0))^2$.

The wedge. Take a tax at which $a^*(\tau) \leq \Delta$, so that the oldest fields of the cohort have already been retired when the tax arrives. A delay-induced field of age a_Δ then produces for $(a^* - a_\Delta)^+$ years after implementation, and integrating over the uniform age gives

$$W(\tau) = \frac{N\theta\bar{q}}{2\Delta} G(\tau) a^*(\tau)^2. \quad (17)$$

At lower taxes, when all fields of the cohort are still producing at implementation, $a^{*2}/2$ is replaced by $\Delta(a^* - \Delta/2)$. The wedge is zero at $\tau = 0$, where $G = 0$, and at $\tau = \bar{\tau}_P(0)$, where $a^* = 0$, and positive in between. With κ uniform, W is proportional to $u(1 - u)$ with $u \equiv (1 - \tau/\bar{\tau}_P(0))^2$, so it is single-peaked.

Actual and naive emissions. In the same range of taxes, the emissions of the cohort after implementation and the emissions computed by the naive planner are

$$E(\tau) = \frac{N\theta\bar{q}}{2\Delta} a^*(\tau)^2, \quad E^N(\tau) = \frac{N\theta\bar{q}}{2\Delta} a^*(\tau)^2 (1 - G(\tau)), \quad E(\tau) - E^N(\tau) = W(\tau), \quad (18)$$

which is (8) for the parametric cohort.

The premium in closed form. At $\delta = 0$, with κ uniform and $a^*(\tau^N) \leq \Delta$, solving $E^N(\tau^N) = E(\tau^*) = B_\Delta$ and eliminating B_Δ gives

$$\begin{aligned} \bar{\tau}_P(0) - \tau^N &= \sqrt{\bar{\tau}_P(0) (\bar{\tau}_P(0) - \tau^*)}, & \text{equivalently} \\ \frac{\pi}{\bar{\tau}_P(0)} &= \sqrt{x} - x, & x \equiv 1 - \frac{\tau^*}{\bar{\tau}_P(0)} \quad (\delta = 0). \end{aligned} \quad (19)$$

Three properties follow. The premium depends on the primitives only through $\bar{\tau}_P(0)$ and τ^* : two cohorts with different cost slopes, carbon intensities and capital-intensity distributions but the same production breakeven and the same budget-consistent tax have the same premium. It is strictly positive whenever $\tau^* > 0$, since $\bar{\tau}_P(0) - \tau^N$ is the geometric mean of $\bar{\tau}_P(0)$ and $\bar{\tau}_P(0) - \tau^*$ and lies strictly between them. Relative to the production breakeven, it is largest at $x = 1/4$, when the budget-consistent tax is three quarters of the production breakeven.

Anticipation. When investors screen projects with an anticipated tax τ^A , the cohort ends at $\kappa_1 = \bar{v}_\delta(\tau^A)$. With κ uniform and $\delta = 0$, G and a^* keep the same form once $\bar{\tau}_P(0)$ is replaced by $\bar{\tau}_P(0) - \tau^A$ and taxes are measured from τ^A , so (19) holds with these substitutions as long as $\tau^N \geq \tau^A$. The premium is zero when investors screen with τ^* or with a higher tax, since no field developed during the delay is then delay-induced at τ^* .

A.2 An increasing tax

This appendix gives the algebra behind Section 2.6. The tax increases at rate r from its 2016-anchored level τ . A field that is a_Δ years old at implementation earns, x years after implementation, a margin per barrel of

$$p - c_0 - \gamma(a_\Delta + x) - \theta\tau e^{r(\Delta+x)}. \quad (20)$$

Three objects of the constant-tax model are affected.

The survival boundary. The field is phased out when (20) reaches zero. The corresponding age has no closed form but is unique, since the margin decreases with x , and it is lower than under a constant tax because the tax keeps rising. To first order in r , the margin falls by $\gamma + r\theta\tau e^{r\Delta}$ per year after implementation: the operating cost rises by γ and the tax in force at implementation, $\tau e^{r\Delta}$, rises by $r\tau e^{r\Delta}$. This is the statement used in Section 2.6. Whether a field survives implementation depends on its margin at $x = 0$ only, so the survival boundary is still $\bar{\tau}_P(a_\Delta)$ once the tax is measured at its implementation level $\tau e^{r\Delta}$.

The development boundary. Under a timely policy, a field developed at date s faces the tax $\tau e^{r(s+a)} = (\tau e^{rs}) e^{ra}$ at age a . Its discounted profit per barrel of capacity therefore depends on τ and s only through τe^{rs} , the tax in force at its start-up, and the development condition is a threshold on that product. Writing $\bar{\tau}_D^0$ for the development breakeven of the field if it were developed at $s = \Delta$, the development breakeven of a field that is $a_\Delta = \Delta - s$

years old at implementation is exactly $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{ra_\Delta}$. Fields developed early in the delay would have faced a lower tax under a timely policy and are more likely to have been developed anyway. Because the tax keeps rising after start-up, $\bar{\tau}_D^0$ depends on the whole tax path and not only on the margin at start-up, so the depth of the zone now depends on the tax. Figure 2 shows the resulting partition.

Aggregation. The four types of Section 2.3 and the identity $W = E - E^N$ are unchanged, and so is the zero overcommitted emissions of intensive fields: a field that the timely tax would also have developed produces for the same period under both policies, whatever the tax path. What is lost is the factorisation of the overshoot into $G(\tau)$ times an integral over ages, because whether a field is delay-induced now depends on its age at implementation as well as on its capital intensity. The wedge and the premium are computed numerically over (κ, a_Δ) :

$$W(\tau) = \frac{N\theta\bar{q}}{\Delta} \int_0^\Delta \int_{\bar{v}_\delta(\tau, a_\Delta)}^{\kappa_1} \left(a^*(\tau, a_\Delta) - a_\Delta\right)^+ dF_\kappa(\kappa) da_\Delta, \quad (21)$$

where $\bar{v}_\delta(\tau, a_\Delta)$ is the discounted profit per barrel of capacity of a field of age a_Δ at implementation under the timely tax path, and $a^*(\tau, a_\Delta)$ is its phase-out age. E , E^N , τ^N , τ^* and π follow as in Section 2.4.

Numerical check. Equation (21) and the code used for the empirical analysis have been checked against each other on a synthetic cohort of 3,000 fields that satisfy the model exactly: flat production, an operating cost rising linearly with age, development capital spent in the year before start-up, a common carbon intensity and start-up margin, capital intensity uniformly distributed, and start-up years spread over the delay. Run through the empirical code at taxes from 0 to 100 and at increase rates of 0 and 2%, the cohort reproduces the analytic development decisions, the phase-out year of every field, given by the root of (20), and the wedge of (21), all exactly.

Increasing tax and the premium: an illustration. The same synthetic cohort, with no discounting and a budget set at 60% of its untaxed emissions, shows how the tax increase rate affects the objects of Section 2.4. Table 6 reports them for a constant tax and for a tax rising at 2% per year. A rising tax that meets the same budget starts lower, so the premium falls in 2016-anchored terms, from 19.1 to 12.0, and in dollars of the implementation year, from 19.1 to 15.8. As a share of the tax, however, it does not change: π/τ^* is 0.44 in both cases, and π/τ^N is 0.80 and 0.78. The residual overshoot falls by 15%. These magnitudes describe a cohort developed entirely during the delay, with no field already in production,

and are not comparable with those of Section 4.

	Constant tax	Tax rising at 2%
Naive tax τ^N	23.9	15.3
Lock-in-adjusted tax τ^*	43.0	27.3
Premium π , 2016-anchored	19.1	12.0
Premium π , at implementation	19.1	15.8
Premium as a share of τ^*	0.44	0.44
Residual overshoot Ω^N	4,560	3,901

Table 6: Increasing tax and the lock-in premium in a synthetic cohort.

Notes: Naive and lock-in-adjusted taxes, premium and residual overshoot for a constant tax and a tax rising at 2% per year. 3,000 fields satisfying the model exactly, no discounting, budget set at 60% of untaxed emissions. Taxes and emissions are in the units of the synthetic cohort.

A.3 A development period

In Section 2 development is instantaneous. In the data it takes several years, and a field may be partly built when the tax arrives. Let development take B years, with development spending spread evenly over them, so that a field whose development began at date $s > \Delta - B$ has paid a share $(\Delta - s)/B$ of its development cost at implementation, and a field whose development began earlier has paid all of it. Production starts at $s + B$, so a completed field is $a_\Delta = \Delta - s - B$ years old at implementation.

Nothing changes for a field completed before implementation: its lock-in zone at implementation is $(\bar{\tau}_D, \bar{\tau}_P(a_\Delta)]$ and Section 2.2 applies with the shifted age. A field under construction at implementation is completed if its discounted profit per barrel of capacity covers the development cost that remains to be paid, rather than the full development cost:

$$\bar{v}_\delta(\tau) \geq \kappa \left(1 - \frac{\Delta - s}{B}\right), \quad \text{equivalently} \quad \tau \leq \tau_L(s), \quad \bar{v}_\delta(\tau_L(s)) = \kappa \left(1 - \frac{\Delta - s}{B}\right). \quad (22)$$

The completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$, when nothing remains to be paid and the field is completed at any tax that leaves it a positive margin, to $\bar{\tau}_D$ at $s = \Delta$, when nothing has been paid and the full development condition applies. Without discounting, $\theta \tau_L(s) = p - c_0 - \sqrt{2\gamma\kappa(s - \Delta + B)/B}$, so the boundary falls steeply as soon as part of the development cost remains to be paid, and flattens as s approaches Δ . The threshold on remaining development cost used in Section 3.4 is its empirical counterpart.

Figure 12 shows the partition in (s, τ) at a given capital intensity. Two types are added to the four of Section 2.3. Fields under construction at implementation whose completion

satisfies (22) are completed and locked in (Type IV-a). They produce over their whole life after implementation, and they are the only fields that can be locked in by a tax above the survival boundary of their age. Fields under construction whose completion does not satisfy (22) are abandoned, with their capital stranded and no emissions (Type IV-b). A development period can only add fields to the locked-in set, since a partly built field is kept at any tax below $\tau_L(s) \geq \bar{\tau}_D$, so the case of instantaneous development in the body is a lower bound. Lock-in is largest for fields whose development began just before $\Delta - B$, which have just started producing when the tax arrives.

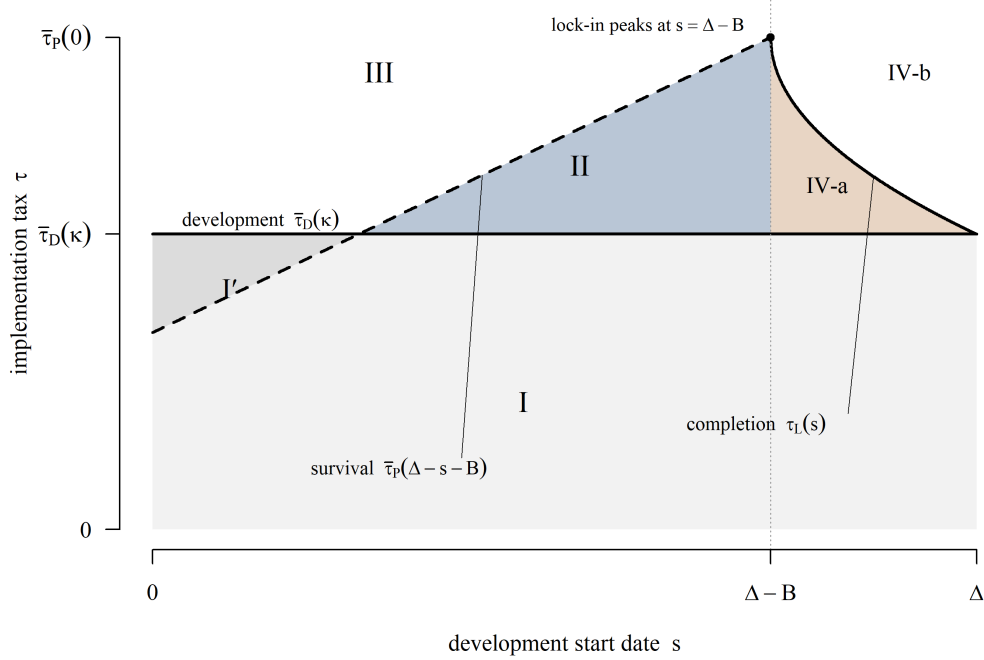


Figure 12: The taxonomy with a development period, at a given capital intensity.

Notes: The horizontal axis is the date s at which development starts; fields starting after $\Delta - B$ are under construction at implementation. The development boundary $\bar{\tau}_D$ is horizontal; the survival boundary $\bar{\tau}_P(\Delta - s - B)$ rises with s because later fields are younger at implementation; the completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$ to $\bar{\tau}_D$ at $s = \Delta$. Region II is the locked-in set among completed fields, IV-a among partly built fields that are completed, IV-b the partly built fields abandoned at implementation. Lock-in is largest at $s = \Delta - B$.

B Data appendix

B.1 Rystad cost categories

Figure 13 reproduces Rystad's definition of each cost item in the UCube database. Development costs in this paper are the facility and well capex incurred up to two years after production start-up; production costs are production, transportation and SG&A opex, taxes

reported as opex, and all facility and well capex incurred later. Exploration capex and government take other than taxes reported as opex are excluded from the main specification.

Economics Category	The Economics Category splits the Economics variable (=Revenue) into higher granularity than Economics Group, e.g. capital expenses is subdivided into facility and well expenses. The sum of all items equals the revenue. Use Economics Category to analyze breakdown of cost models or specific cost elements.
Free Cash Flow	FCF is the incurred cash flow available to investors and creditors, thus largely used in DCF valuation analysis. $FCF = \text{revenue} - \text{capex} - \text{opex} - \text{govt take}$. FCF can be compared with cash from operations and investments activity in the cash flow statement from companies (only upstream). The present value of (all discounted) FCF equals NPV (Net Present Value also called Gross Asset Value) that can be compared with Enterprise Value (=Market cap+Debt for upstream activities), or compared with asset transaction costs. The discount factor in UCube is 10% flat for all assets.
Government Profit Oil	Gvt PO is the PSA equivalent to petroleum taxes, but paid in kind. Gvt PO reduces the company's entitlement production and is thus treated as a royalty effect in company reports.
Royalty effects	Royalty effects is the sum of all gross taxes, including royalty and oil and export duties.
Income Tax	Income tax is a sum of all profit based taxes, ie corporate taxes, where the tax rate is equal to the country's corporate tax rate, and special petroleum taxes. The tax base is the net sales revenue minus opex and depreciation. For petroleum taxes there may be applied an additional deduction called uplift. In many cases, some profit based taxes can be deducted in the tax base for other profit taxes. The income taxes in the UCube assumes there is no interest expenses.
Bonuses	Signature bonus for the license at award, as officially reported.
Well Capex	Well capex is capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials.
Facility Capex	Development capex except well construction costs, includes costs to develop, install, maintain and modify surface installations and infrastructure. As reported by operators, field partners or officials, or modelled. Distribution of costs over time is largely modelled, one exception is Norway, where all historic costs are as reported in national budgets.
Exploration Capex	Costs incurred to find and prove hydrocarbons: seismic, wildcat and appraisal wells, general engineering costs, based on reports and budgets or modelled.
Abandonment cost	Abandonment cost represents the costs associated with shutting down and dismantling the surface and subsea facilities such as surface equipment, platforms, subsea structures and pipelines. Activities included is topside and jacket removal as well as pipeline cleaning and decommissioning, onshore disposal, surveys and monitoring, and project management costs. Note that the Abandonment cost only includes the remaining P&A work for wells, and not full cycle P&A work. Abandonment cost is mostly modelled and assumed to be incurred and expensed over couple of years after the production has ceased. The estimate is proportional to initial greenfield capex.
Production Opex	Represents operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment lease costs and operation related salaries. Depreciation and other non-cash items are not included.
Transportation Opex	Represents the costs of bring the oil and gas from the production site/processing plant to the pricing point (only upstream transportation). The category includes transport fees and blending costs.
SG&A Opex	Represent operation expenses not directly associated with field operations. The category includes administrative staff costs, office leases, related benefits (stocks and stock option plans) and professional expenses (legal, consulting, insurance). Only E&P related SG&A are included.
Taxes Opex	Represent gross tax reported by companies under operational costs and applies for US and Brazil. For US the category represent ad valorem taxes (county based) and severance taxes (state based).

Figure 13: Rystad economic data, split items.

Notes: Descriptions are provided by Rystad alongside the UCube database.

B.2 Asset-specific price discounts

Each asset sells its oil at Rystad's long-term discount to Brent, which depends on crude quality. Table 7 reports the reserve-weighted average discount by oil type. All types sell on average at a discount to Brent, which reflects the high quality of the crudes constituting the

Brent stream; heavier oils, which are more difficult to refine into the same products, carry larger discounts, and NGL sells at roughly half the price of oil.

Oil type	Average discount	2016 reserves (Gbbl)
Regular	−5.4%	760
NGL	−50.6%	222
Light	−2.7%	203
Condensate	−5.9%	113
Sour (> 3%)	−4.6%	52
Bitumen	−29.0%	47
Heavy oil 15–19	−14.0%	34
Heavy oil 20–22	−10.4%	22
Extra heavy oil	−12.4%	21
Synthetic crude	−7.4%	19
Total	−12.9%	1,493
Excluding NGL	−6.3%	1,271

Table 7: Discounts relative to the Brent price by oil type.

Notes: Reserve-weighted averages of the discounts observed and forecast by Rystad over 2016–2100, estimation sample. Discount data are missing for a small part of the sample’s production, so the reserves in the table (1,493 Gbbl) fall slightly short of the total of Table 1 (1,555 Gbbl).

B.3 Anticipation scenarios

The main specification assumes investors do not anticipate the tax during the delay. The alternative scenarios reported in Section 4.4 are defined as follows. In the *tax in 2030* scenario, investors anticipate the correct level of tax to be implemented in 2030, and thus predict the planner’s behaviour exactly. In the *five years after start-up* scenario, investors anticipate the correct level of tax to be implemented five years after production starts in the asset, a screening rule used in the industry. In the *share of the tax* scenarios, investors anticipate the tax to be implemented immediately after investing, on the totality of the asset’s lifetime production, but underestimate its level; the sensitivity panel reports 25, 50 and 75%, and Figure 8 a grid in steps of 2.5 percent. Scenarios with heterogeneous beliefs across firms, reported in earlier versions, are not used. In every scenario the anticipated tax enters only the investment decision (12), through T_τ and τ ; production decisions after implementation are taken under the implemented tax. The limit case, screening with the implemented tax τ_{2030}^* , reproduces $\pi = 0$ exactly. Screening with the timely tax τ_{2016}^* , anticipation of the announced timely policy, is a different scenario and is not reported.

C Field-level carbon intensities

While Rystad’s databases provide detailed asset-level data on oil production, they lack comprehensive carbon intensity (CI) estimates. This appendix fills this gap by combining engineering-based models (OPGEE, PRELIM, OPEM), satellite-derived flaring data (VIIRS), and predictive regressions to estimate lifecycle CI for all assets, including those not yet in production. Carbon emissions related to oil can be split into 3 stages (Gordon et al., 2015). The upstream stage includes all emissions arising from extraction processes to the transport of crude oil to the refinery inlet. The midstream stage corresponds to refining processes and are heavily dependent on the refinery configuration. Finally, the downstream stage comprises emissions from the transport of oil products to their end-user and its final use. While downstream emissions usually represents around 80% of total emissions, the upstream stage features important variability from one field to the other.

For each of these stages, I rely on a state-of-the-art model developed by the Oil-Climate Index on a sample of currently produced oil to estimate their carbon intensity. I then fit a regression model for each stage to assign a carbon intensity to all other Rystad assets.

Figure 14 reports cumulative 2016 reserves ordered by lifecycle carbon intensity. Most reserves lie between 400 and 500 kgCO₂eq/bbl, with a short tail of cleaner assets and a longer tail of more carbon-intensive ones. Fields developed during the delay and fields already producing in 2016 have similar distributions.

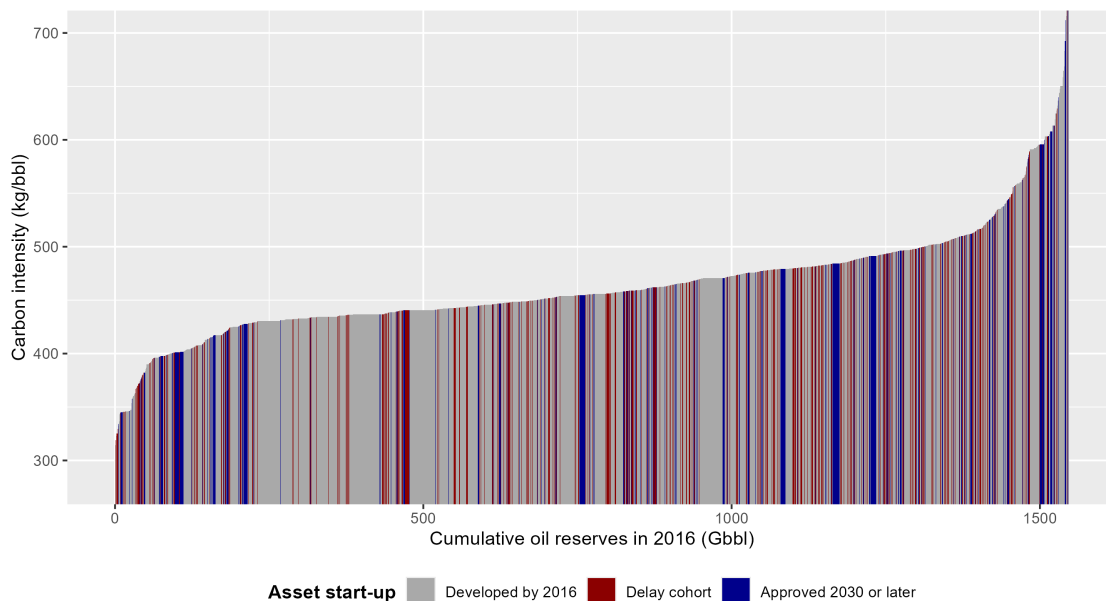


Figure 14: Cumulative 2016 oil reserves ordered by lifecycle carbon intensity, by development timing.

C.1 Upstream CI

C.1.1 Flaring data

Upstream emissions exhibit substantial heterogeneity across assets, and the major contributor to this heterogeneity is the flaring-to-oil ratio (FOR). Flaring corresponds to the combustion on-site of co-produced gas¹⁶, and represents more than 150 billion cubic meters worldwide in 2024 (World Bank, 2025). To compute precise estimates of upstream CI, I use satellite data on flares to derive field-specific flaring-to-oil ratios.

Satellite flares observations and imputed emissions. Rystad UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report flared volumes. Moreover, under-reporting by fields operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, I use geocoded flaring volumes calculated by the Earth Observation Group (EOG) at the Payne Institute for Public Policy, within the Colorado School of Mines. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), on board of satellites launched in 2012 and 2017, which detects heat from gas flares at night. Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by Elvidge et al. (2015) and further refined by Zhizhin et al. (2021). I use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad’s database.

Rystad assets geographic information. In order to assign flaring observations to Rystad assets, I need geographic information on these assets. I complement data from the UCube with two sets of geocoded data obtained from the same company, one in 2022 and the other in 2025.

First, geocoded boundaries were shared by Rystad in 2022 for 41,762 assets, among which 16,801 were generic (i.e. a 500m radius circle). Since the universe of Rystad assets has evolved between 2022 and 2025, 1,916 generic shapes were added for assets missing in the shapefile, based on their geocoded centroid.

¹⁶Co-produced gas is often used to power equipments on-site, or sold, if the asset is connected to the local gas market or possess equipments to liquefy and ship it. If not, excess gas is then either reinjected, to increase pressure in the reservoir, flared, or directly vented. Given the higher global warming potential of methane, flaring is preferable to venting, and venting is banned in most countries except for rare emergencies. Gas reinjection is often costly, making flaring an easier decision in many oil fields.

Secondly, data was obtained on the location of wells completed between 2000 and 2025. This dataset contains the latitude and longitude of 1,733,698 producer, injector or wildcat wells, from 39,494 assets. In some regions, the location of wells correspond to the asset centroid, meaning that Rystad probably does not have its precise location. In total, 857,900 unique locations exist in the dataset. Crucially, wells locations are precisely defined for onshore US and Canada assets, where all assets boundaries are generic, making this dataset a useful complement to the first one.

Matching satellite observations to Rystad assets. The assignment of satellite observations to Rystad assets relies on both assets’ eligibility (the existence or not of surface wells, excluding for example subsea tiebacks and not yet completed wells) and geographic distances to existing wells and assets’ boundaries. For each observation, distances to all completed wells and assets in a 20km radius were computed. A score reflecting the probability of a detected flare to originate from each nearby asset is then computed. For observations with existing wells in a 2km radius, only information on wells was used to compute this score. For other observations, distances to both assets and wells were incorporated in the score. When scores were too close to allow to decide between assets, flared volumes were assigned based on each eligible asset’s score and production capacity. [Ghosh et al. \(2025\)](#) provides more detailed information on the 5-stage matching procedure used to assign flaring observations to Rystad assets.

Overall, all observations with at least one eligible well or asset in a 20km radius are matched, leaving 7,261 observations unmatched (corresponding to 4.7% of total flared volumes between 2012 and 2024). Among matched observations, around two thirds were assigned to a unique asset (70.4% of matched flared volumes) and the rest split between multiple assets.

Computing flaring-to-oil ratios (FOR). Once the matching procedure is done, total flared volumes over the 2012-2024 period are computed and divided by total oil production to obtain asset-specific FOR. For not-yet-active assets over this period, an average FOR is assigned. Two categories of assets producing over the period were also attributed averages: assets with at least 75% of years in the sample with positive production outages, mainly due to wars in Lybia and Syria (109 assets), and assets which production started less than 2 years before the end of the period (911 assets). The average FOR is computed at the lowest scale possible, for asset of similar size if possible¹⁷, and always by oil type. Averages are

¹⁷Assets are split in 10 categories depending on their reserve size, with thresholds at 1, 3, 10, 30, 100, 300, 1000, 3000 and 10000 MMboe.

capped at 12,000 scf/bbl, corresponding to twice as more gas flared than oil produced in barrel of oil equivalent. Table 8 reports the number and type of average FORs assigned to Rystad assets, and the corresponding oil reserves.

FOR origin	Number of assets	Share of oil reserves in 2016
From satellite matching	20,571	64.3%
Area x Oil Type x Field Size	5,133	4.2%
Province x Oil Type x Field Size	7,350	7.3%
Province x Oil Type	10,497	8.9%
Country x Oil Type x Field Size	6,662	3.3%
Country x Oil Type	4,250	4.8%
Region x Oil Type x Field Size	2,948	3.4%
Region x Oil Type	3,136	3.1%
World x Oil Type x Field Size	946	0.9%
World x Oil Type	2	0.0%

Table 8: Origin of FOR values.

Table 9 shows the binned distribution of computed FOR. Most assets display relatively low levels of flaring with around a third of them having a FOR equal to 0 (representing around 22% of oil reserves). In these assets, the 0 value can be explained by either complete use of co-produced gas or particularly low gas-to-oil ratio, but it can also be due to flared volumes so low (or so intermittent) that it falls below the satellite detection threshold. Among assets producing mainly oil (assets with gas-to-oil ratios below 10,200 scf/bbl), the reserves weighted average FOR is 104.4 scf/bbl, and the median is 13.4 scf/bbl.

FOR (kscf/bbl)	Assets count	Reserves (Gbbl)
0	11,651	345.6
0–0.1	10,820	748.7
0.1–1	10,974	435.4
1–10	1,069	21.0
>10	291	2.5

Table 9: Distribution of flaring-to-oil ratios.

C.1.2 OPGEE model

The Oil Production Greenhouse Gas Emissions Estimator (OPGEE) is an engineering-based model developed by the Oil-Climate Index team (Gordon et al., 2015)¹⁸. It measures

¹⁸Latest versions of the model are available at: <https://eao.stanford.edu/research-project/opgee-oil-production-greenhouse-gas-emissions-estimator>

GHG emissions from extraction, processing and transport of crude oil in grams of CO₂ equivalent per megajoule of oil produced.

Data inputs and matching with Rystad assets. Running the model requires a wide range of inputs for each field, from extraction methods to gas composition. When information is missing, the model computes smart default values (deduced from other variables when possible). I recover publicly available input data from [Masnadi et al. \(2018\)](#) on 958 oil fields worldwide, and match it to Rystad assets manually. Among the 958 original fields, 737 were matched to 1,136 assets in the Rystad database. Most unmatched fields correspond to marginal heavily-depleted oil fields in the US. The difference between the number of OPGEE fields and matched Rystad assets comes from differing level of aggregation. In many instances, a unique OPGEE field is split into multiple assets in Rystad data. In rare occasions, multiple OPGEE fields were matched to a unique Rystad asset.

After completing the matching procedure, I complement the 737 fields dataset with information on 6 variables from Rystad data in order to improve the accuracy of my estimates. These variables are: production level, field age, gas-to-oil ratio, flaring-to-oil ratio, number of producer wells and number of injection wells. Since Rystad databases do not contain information on the quantity of gas reinjected, used on site, or vented, I assume the amount of gas produced to be equal to the sum of flared and sold gas in order to compute gas-to-oil ratios. Finally, extraction methods reported in [Masnadi et al. \(2018\)](#) input data were adjusted if necessary to match information retrieved from Rystad¹⁹.

Computing CI. The resulting input dataset was fed into the OPGEE model to export results. Two distinct carbon emissions accounting methods are available within the model: energy-based allocation (EBA) and carbon product displacement (CPD). In the EBA approach CI are defined such that $CI_{EBA} = \frac{TotalEmissions_{field}}{Prod_{oil} + Prod_{gas}}$, with $Prod_{oil}$ and $Prod_{gas}$ in MJ. This means that total emissions from the oil field are computed, and allocated to oil and gas sold on the market according to their energy content. In the CPD approach, the definition becomes $CI_{CPD} = \frac{TotalEmissions_{field} - Prod_{gas} \cdot \overline{CI}_{gas}}{Prod_{oil}}$. Co-produced gas gives rise to carbon credits corresponding to the energy content of gas produced multiplied by $\overline{CI}_{gas}$, a worldwide average carbon intensity of gas production computed from the CA-GREET model. The latter approach can lead to negative oil CI, which is why the retained approach was the EBA one.

¹⁹This is for example the case for water reinjection. If Rystad data reports existing injection wells, at least water or gas injection must be indicated in OPGEE input data. When this was not the case, water reinjection was assumed to happen, since it more common in oil fields than gas reinjection.

Figure 15 displays CI computed using the model. The production-weighted average carbon intensity is equal to 59 kgCO₂eq/bbl (indicated by the dashed red line on the figure). The variance among fields is considerable, with 10% and 90% weighted deciles ranging from 35 kgCO₂eq/bbl to 113 kgCO₂eq/bbl. Three key features influencing CI are indicated on the graph:

- **FOR:** as expected, large CI values often originate in large quantities of co-produced gas flared. The influence of FOR on CI largely depends on the assumptions made on destruction removal efficiency (DRE). This value measures to the efficiency of flares, i.e. the share of gas effectively combusted and thus transformed into CO₂. Since the global warming potential of methane is 28 times larger than the one of carbon dioxide, assumptions on DRE greatly affects computed CI. I keep the default value of 96.1% retained by the model to compute baseline CI, and provide in sensitivity analysis results using values of 91.1% and 98% to cover the range of estimates found in the literature²⁰(Plant et al., 2022; Jackson and Smith, 2014; Gvakharia et al., 2017; Zavala-Araiza et al., 2021).
- **Steam injection:** Heavier oils (API below 20) tend to be thicker than lighter ones, and often require the injection of steam in order to lower its viscosity and allow its extraction. This is captured by the steam-to-oil ratio (SOR), corresponding to the number of barrels of vaporized water injected to extract one barrel of oil. High SOR values imply larger quantities of liquids extracted, and therefore larger energy consumed to extract oil.
- **Upgrading into synthetic oil:** some particularly viscous oil (bitumen or extra heavy) are extracted without necessitating reducing its viscosity first (for instance using steam injection). This is mainly the case for Canadian oil sands, which are often directly extracted in open air mines. However, they are often too viscous to flow through pipelines and be handled by existing refineries. These oils are therefore "upgraded" into synthetic crude on site. This requires using chemical treatments and/or distillation in order to break long hydrocarbon chains into shorter ones, thus reducing viscosity. These processes are often very energy intensive, translating into large CI for fields using it (Brandt and Farrell, 2007). Produced synthetic crude is however easier to refine, translating into relatively lower midstream CI²¹.

²⁰While playing a much smaller role than FOR, the quantity of co-produced gas (GOR) can still influence extraction emissions through fugitive emissions of methane related to its processing and transport on site.

²¹Midstream CI for synthetic crudes are on average 30% lower when compared to other extra heavy or bitumen oils, and still 8% lower when compared to all types of oil.

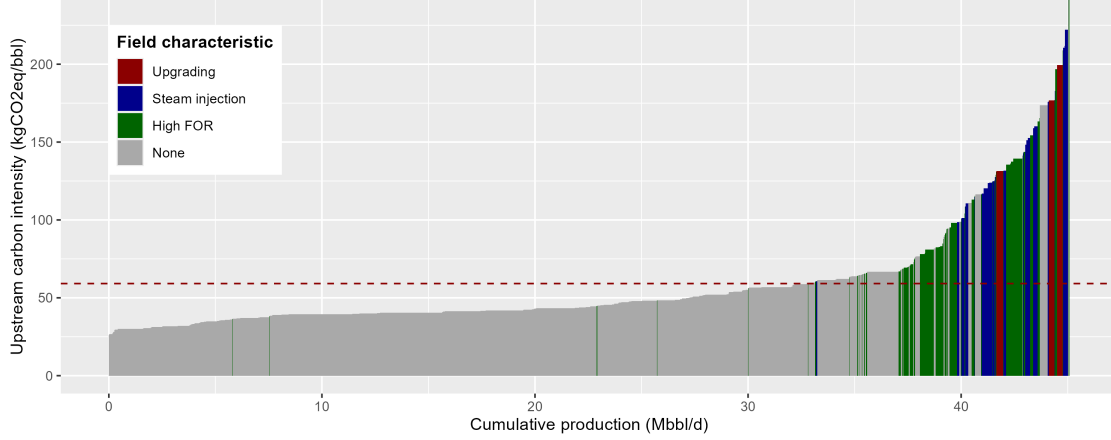


Figure 15: OPGEE-computed CI.

Notes: The figure depicts CI computed in the OPGEE model, with cumulative production in 2015 on the x-axis. The horizontal dashed red line indicates the production-weighted average.

C.1.3 Estimation model

Model and sample restrictions. Once the 737 OPGEE-computed CI are attributed to the corresponding 1,136 Rystad assets, I fit a regression model to explain across-field variations. Following the logic of the EBA approach, the dependent variable in the model thus consists of the field carbon intensity divided by its "energy denominator" $ED_f = \frac{Prod_{oil,f}}{Prod_{oil,f} + Prod_{gas,f}}$. The retained model is:

$$CI_f^{OPGEE} * ED_f = \sum_0^{10} \beta_i^C ShareOilType_f + \beta_{11} Offshore_f + \beta_{12} SOR_f + \beta_{13} GOR_f + \beta_{14} FOR_f + \epsilon_f, \quad (23)$$

where f denotes a Rystad asset, $ShareOilType_f$ the share of each oil type in its production²², $Offshore_f$ a dummy indicating if the field is offshore, SOR_f its steam-to-oil ratio (in bbl of water/bbl of oil), GOR_f its gas-to-oil ratio (in scf/bbl) and FOR_f its flaring-to-oil ratio (in scf/bbl).

Oil types variables, together with the offshore dummy, capture various aspects of the extraction methods and facility characteristics on site. This is especially relevant for synthetic crude, where upgrading processes such as distillation or chemical treatments significantly increase CI. SOR , GOR and FOR have a more straightforward and linear impact on carbon

²²An asset produces a unique type of crude oil, among 8 possible (synthetic, bitumen, extra heavy, heavy (15-19), heavy (20-22), regular, light and sour). It can however produce simultaneously condensate and/or natural gas liquids (NGL), which are sold as oil, similar to light crude.

emissions, and explain most of the remaining variance. SOR data used for the estimation is the one used as input in the OPGEE model since precise values are not available in Rystad data.

The sample used in the regressions was restricted to fields with water-to-oil ratio below 40 bbl of water/bbl of oil, and gas-to-oil ratio below 40,000 scf/bbl. The first restriction excludes fields almost fully depleted, characterized by exponentially increasing WOR²³. The second one is motivated by the fact that my focus is on oil producing fields. Oil fields are defined by Rystad as reservoirs with GOR inferior to 10,200 scf/bbl. I exclude fields with GOR higher than four times this threshold in order to keep all fields producing non-negligible oil quantities. Altogether, these two restrictions exclude 20 Rystad assets, making the final sample 1,116 assets (associated to 719 OPGEE fields). Weights were added to the regressions in order to keep each unique OPGEE-computed values equivalent. If a unique OPGEE value was matched to multiple Rystad assets, weights were computed based on oil reserves size and sum up to 1.

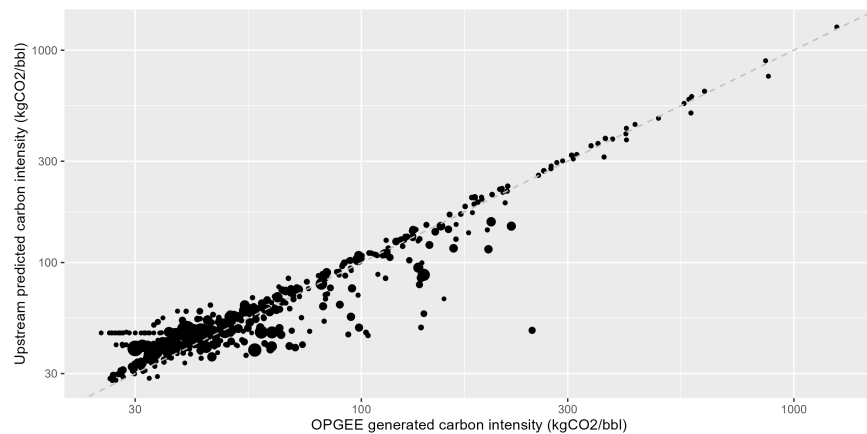


Figure 16: Predicted vs OPGEE-computed upstream CI.

Notes: The figure displays upstream CI computed using the OPGEE model on the x-axis and predicted using the model on the y-axis. Only assets matched with OPGEE fields are included. Both axis are transformed using the decimal logarithm. The size of each point reflects the log volume of oil reserves for each asset.

Regression results. Table 10 shows coefficient estimates from the regression for two sets of CI: the first corresponds to my baseline estimates, using a 100-years horizon for global warming potential, and the second one use a 20-years horizon (associated with higher global warming potential for methane as compared to carbon dioxide, due to its shorter lifetime).

²³I exclude these fields because Rystad data does not contain information on this ratio. The threshold of 40bbl/bbl is particularly high and only excludes 7 fields from the sample.

Variables	EBA - AR6 100y	EBA - AR6 20y
Oil type: Synthetic	157.096*** (6.514)	192.358*** (9.943)
Oil type: Bitumen	52.854*** (4.555)	78.442*** (6.952)
Oil type: Extra Heavy	48.122*** (4.169)	78.368*** (6.363)
Oil type: Heavy (15-19)	54.778*** (3.73)	96.841*** (5.693)
Oil type: Heavy (20-22)	58.861*** (3.561)	97.629*** (5.436)
Oil type: Regular	46.522*** (1.105)	79.708*** (1.687)
Oil type: Light	46.939*** (1.862)	77.204*** (2.842)
Oil type: Sour (> 3%)	43.308*** (5.14)	71.654*** (7.846)
Oil type: Condensate	43.535*** (4.235)	72.446*** (6.465)
Oil type: NGL	48.05*** (6.641)	83.603*** (10.137)
Offshore	-6.048*** (1.273)	-8.435*** (1.943)
SOR (bbl water/bbl oil)	26.317*** (0.983)	27.369*** (1.501)
GOR (kscf/bbl)	4.601*** (0.154)	8.7*** (0.236)
FOR (kscf/bbl)	83.908*** (0.533)	146.98*** (0.813)
Corrected R ²	0.970	0.977
Predicted R ²	0.936	0.943

Table 10: Regression results from upstream emissions estimation.

Notes: The table displays coefficients from the regressions using equations 23, with standard errors indicated between parenthesis. R² are corrected in order to account for the absence of intercept, as well as predicted R².

R-squared values are close to 1, and predicted R-squared are also quite close to 1, indicating that the model's predictive power is relatively high.

As expected, synthetic crude production is associated with large CI, and coefficients for heavier crudes tend to be larger than the ones of lighter crudes. While coefficients associated with bitumen and extra heavy crudes are not significantly higher than others, production of these two types often implies the use of steam injection, which means that most of the variation is captured by the SOR variable. The negative coefficient associated with the offshore dummy reflects the absence of land-use transformation, source of GHG

emissions. The three remaining variables explain most of the variance. Steam injection greatly increases emissions (the average steam-to-oil ratio is 3 bbl of water/bbl of oil). While having a relatively low magnitude, GOR still plays an important role in explaining the variance in carbon intensity across fields. Fields producing gassy oils, even when selling their co-produced gas, are subject to higher quantities of fugitive emissions. Finally, FOR plays the most crucial role in explaining the variance across fields, as noted by [Masnadi et al. \(2018\)](#).

Regression using 20-years horizon CI as dependent variable display similar results. By construction, reducing the time horizon can only increase CI: while carbon dioxide emissions are not affected, methane emissions lead to higher global warming potential. Consequently, variables associated at least partially with methane emissions see their associated coefficients increase in large proportions. This is particularly visible when comparing the SOR, GOR and FOR variables. While steam injection does not entail methane emissions, gas production leads to fugitive methane emissions and flaring does not fully convert methane into carbon dioxide (depending on the retained DRE value as explained above). The large increase in the coefficients associated to GOR and FOR, along with the SOR coefficient remaining unchanged is thus consistent with this mechanism.

Sensitivity analysis. Figure 16 highlights the strong correlation between OPGEE-computed CI and CI predicted using the model for the same assets sample. To verify the influence of outliers, I re-estimate coefficients of equation 23 excluding each observation one by one. Figure 17 displays estimated coefficients, with the baseline estimate indicated in red. While heavier oils types coefficients can vary due to the low number of assets producing it, results indicate that my estimates are overall robust to changes in the sample.

Finally, I test the sensitivity of estimated coefficients to the exclusion of the most influential observations. I compute the Cook’s distance of each observation, defined by $CD_i = \sum_{j=1}^N (\hat{y}_j - \hat{y}_{j(i)})^2 / ps^2$, where $\hat{y}_j$ and $\hat{y}_{j(i)}$ are the fitted response values from using the full sample and after excluding field i , respectively; s^2 is the mean squared error of the regression model, p the number of coefficients, and N the number of observations. Table 11 displays re-estimated coefficients excluding the 1, 10 and 50 most influential observations. Heavier oil types see their estimate vary slightly, due to the small number of assets producing it in my OPGEE sample, while SOR, GOR and FOR coefficients are particularly stable. Predicted R-squared stay high, even when removing the 50 most influential observations. Overall, re-estimated coefficients remain consistent with my original estimates using the full sample.

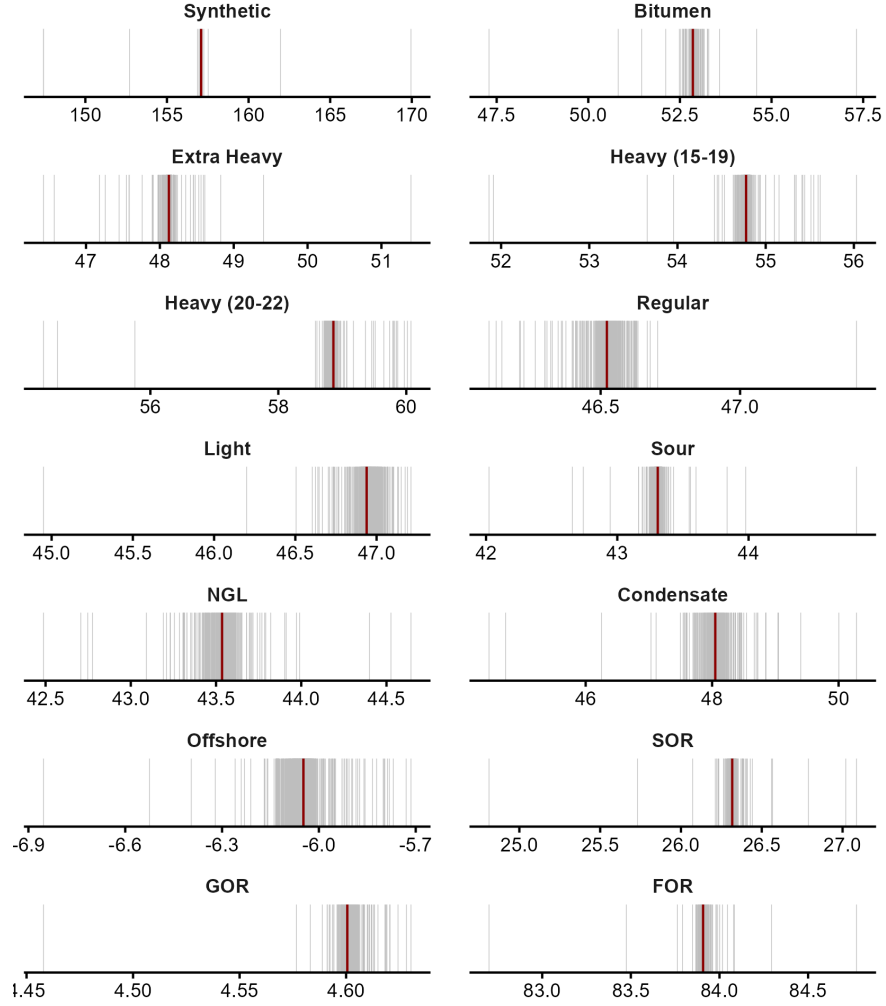


Figure 17: Upstream CI estimation: Estimates of the coefficients on the explanatory variables removing the OPGEE fields one-by-one.

Notes: The panels show the estimated coefficients on each of the explanatory variables in equation (23) removing observations one-by-one. Each grey line corresponds to a different estimation with the same number of observations: $N(OPGEE_{sample}) - 1$. The thick red lines show the original estimates obtained with the full sample of OPGEE publicly-available fields (OPGEE sample).

C.1.4 Predicting upstream carbon intensity for all Rystad assets

Using estimated coefficients, predicted upstream CI were computed for each Rystad asset. Data on reserves by oil type, offshore status, and GOR was directly available in Rystad databases. Asset-specific FOR were either derived from satellite observations or predicted for not-yet-producing assets, as described in the Flaring data section. Values for other needed variables were assigned as described below:

- SOR: on top of assets matched to OPGEE fields, assets using steam injection were identified and assigned a SOR value of 3, in line with both the default value in the

	Synthetic	Bitumen	Extra Heavy	Heavy 15–19	Heavy 20–22
0	157.1*** (6.51)	52.85*** (4.55)	48.12*** (4.17)	54.78*** (3.73)	58.86*** (3.56)
1	157.12*** (6.29)	53.14*** (4.40)	48.48*** (4.02)	55.42*** (3.60)	59.49*** (3.44)
10	133.06*** (6.51)	56.86*** (3.70)	51.35*** (3.37)	53.23*** (3.07)	50.40*** (2.99)
50	154.73*** (14.33)	51.56*** (2.88)	47.81*** (2.50)	49.14*** (2.40)	49.60*** (2.64)
	Regular	Light	Sour (>3%)	Condensate	NGL
0	46.52*** (1.11)	46.94*** (1.86)	43.31*** (5.14)	43.54*** (4.24)	48.05*** (6.64)
1	47.42*** (1.07)	46.87*** (1.80)	43.98*** (4.96)	44.40*** (4.09)	49.04*** (6.41)
10	47.62*** (0.90)	45.37*** (1.50)	44.17*** (4.12)	45.48*** (3.41)	53.37*** (5.37)
50	44.59*** (0.65)	41.86*** (1.07)	44.73*** (3.05)	43.83*** (2.51)	46.00*** (3.87)
	Offshore	SOR	GOR	FOR	Pred. R ²
0	-6.05*** (1.27)	26.32*** (0.98)	4.60*** (0.15)	83.91*** (0.53)	0.934
1	-6.85*** (1.23)	26.22*** (0.95)	4.62*** (0.15)	82.70*** (0.53)	0.931
10	-7.39*** (1.03)	24.97*** (0.81)	4.51*** (0.13)	82.67*** (0.54)	0.918
50	-5.21*** (0.74)	26.02*** (0.67)	4.54*** (0.09)	83.93*** (0.46)	0.910

Table 11: Upstream CI estimation: coefficient stability to influential observations (Cook’s distance).

Notes: Each panel shows re-estimated coefficients using equation 23 after excluding the 0, 1, 10, and 50 most influential observations. Cook’s distances were computed to determine the influence of each observation. The last panel indicates predicted R-squared for each of the re-estimated models.

OPGEE model and the average SOR value in OPGEE fields data from [Masnadi et al. \(2018\)](#). All assets containing "Steam", "Thermal" or "SAGD" in their name were identified as using steam injection. In addition, the list of Enhanced Oil Recovery projects made available by the EIA was used to manually identify remaining steam injecting assets ([McGlade et al., 2018](#)).

- ED_f : energy denominators for each asset were approximated by $ED_f = 1 + \frac{GOR_f - FOR_f}{6000}$, using a 6,000MJ average energy content per barrel. This approximation is remarkably close to energy denominators derived from the OPGEE model, with a correlation coefficient of 0.975 within the OPGEE matched assets sample.

When a unique Rystad asset was matched with a unique OPGEE field, the carbon intensity computed using the OPGEE model was kept. For all remaining assets, predicted values prevail. Figure 18 shows the 2012-2024 production-weighted distribution of CI for both samples, with weighted means depicted by the vertical lines. Although averages are remarkably similar, the predictive model does not seem to fully replicate the variance across OPGEE-computed values. The general shape of the distribution remains coherent with OPGEE outputs.

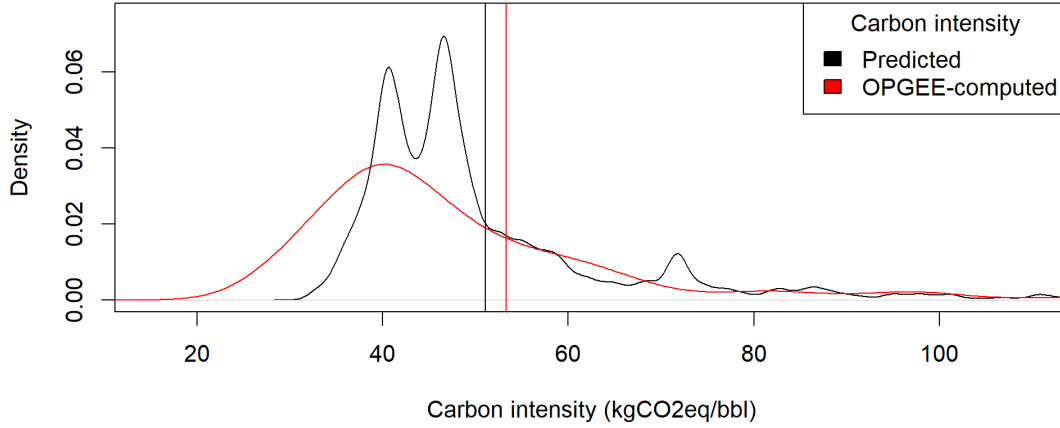


Figure 18: Distribution of CI: Predicted vs OPGEE-computed.

Notes: The figures shows in red the distribution of baseline CI (100-years horizon, EBA) for the OPGEE sample, i.e. assets matched with OPGEE fields and for which the OPGEE value was kept, and in black predicted CI for all remaining assets. Densities are weighted by oil production volumes over the 2012-2024 period. Vertical lines indicate weighted averages for both samples.

C.2 Midstream-downstream CI

Midstream emissions correspond to emissions from refining crude oil, while downstream emissions include emissions from both transport to end-users and final use (often combustion) of the different oil products obtained from refining. Refining crude oil yields multiple oil products depending on its characteristics and refinery configurations. Midstream and downstream emissions are typically computed at the crude stream level, which corresponds to the actual crude stream arriving to the refinery inlet. In some instances, a crude stream originates in a single oil field, while the majority of crude streams consists of multiple fields outputs blended together. For instance, the Brent crude used as reference in oil pricing is a blend of crude oils produced in 13 separate assets in the North Sea.

C.2.1 PRELIM and OPEM models

Midstream and downstream CI can be computed using the Petroleum Refinery Life Cycle Inventory Model (PRELIM) and Oil Products Emissions Module (OPEM) respectively²⁴. The versions used in this study to compute CI are PRELIM 1.6 and OPEM 1.1. The PRELIM model requires detailed information on crude assays, such as API gravity, sulphur

²⁴Information and most recent versions of the models can be found at: <https://ociplus.rmi.org/methodology>.

content, and various boiling points. This data is typically not made available by producers, and publicly available data usually does not follow the same format and is therefore unusable in the models. Models authors thus included in the model data on 657 unique crude assays, that we match to Rystad assets they originate from.

Running the model. The PRELIM model allows for a large array of refinery configurations, designed to handle different types of oils and aimed at maximizing the production of specific oil products. Since I do not have sufficient information to match Rystad assets to the refineries treating their oil production, I follow the assumptions of the model on refinery configurations. Among the three main configurations used worldwide, the model selects the adapted configuration based on crude oil characteristics in terms of API gravity and sulphur content. All heavy crudes (API gravity below 22) are assigned the "deep conversion" configuration, entailing additional treatments aimed at reducing the length of carbon chains yielded. Light crudes with low sulphur levels (API gravity above 32 and sulphur content below 0.5%) are assigned the "hydroskimming" configuration, associated with less treatment stages and in turn less emissions. All other crudes are refined using the "medium conversion" configuration. The importance of refinery configuration is illustrated in figure 19, depicting PRELIM-computed midstream CI. Intensive treatment of crude in the case of deep conversion is associated with higher emissions, while hydroskimming units yield much smaller emissions.

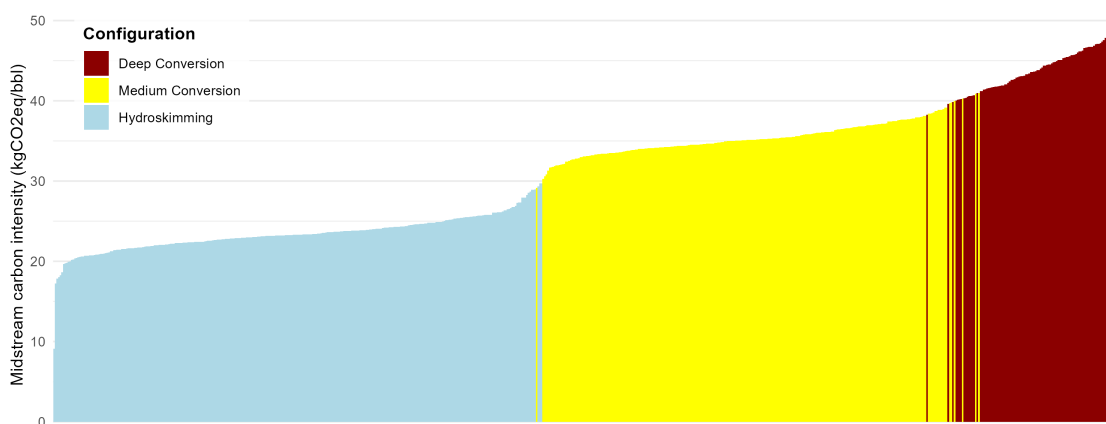


Figure 19: PRELIM-computed midstream CI.

Once midstream emissions are computed, PRELIM output sheets are imported to the OPEM model in order to estimate downstream CI of each crude stream. Emissions from transport and final combustion (for oil products used as fuel) are computed, giving me for a

given barrel of crude oil its final products yield and associated emissions. Final combustion represents the larger share of emissions along oil lifecycle, and displays smaller variability across crudes.

C.2.2 Matching PRELIM crude assays to Rystad assets

Crude streams available in the PRELIM model are matched to Rystad assets using data made available by Rystad on crude streams and associated assets. The matching is performed using fuzzy matching methods on crudes names. After cleaning crudes names from both sources, Jaro-Winkler distances were calculated and capped at 0.25. Matches were then manually verified, leaving me with 493 crude assays from PRELIM (from the original 657) matched to 392 crude streams from Rystad²⁵. Additionally, 25 unmatched PRELIM assets were manually matched to 41 Rystad assets with no specified crude streams. Overall, 518 PRELIM crude assays were linked to 8,386 Rystad assets, representing 64.4% of world production in 2024.

A unique crude stream often consists of a blend of crude oils from multiple assets, but in certain cases (mainly in Canada), production in a unique asset is split between different crudes. For these assets, midstream and downstream CI from PRELIM/OPEM were averaged weighting by the share of production being allocated to each crude stream.

C.2.3 Estimation models

Model and sample restrictions. Using PRELIM output, I fit a regression model to explain the variance in midstream and downstream CI separately. In both cases, I use the two most influential variables, API gravity and sulphur content, which are also available at the asset level in Rystad data. Refinery configuration plays the most important role in explaining variations. I follow the assumptions made in the PRELIM model and assign "deep conversion", "hydroskimming" and "medium conversion" based on the same rules on API gravity and sulphur content. Both variables (in log form for sulphur content) are subsequently introduced in quadratic form to account for as much of the residual variation as possible. Higher-order terms are iteratively dropped to avoid overfitting based on predicted R-squared. The final models used to explain midstream and downstream carbon emissions are as follow:

²⁵The difference in number comes from unique crudes streams being split into different crude assays in PRELIM differing by the data source. For instance, the "Hibernia" crude in Canada is split into 3 assays, corresponding to information on this specific crude obtained from ExxonMobil, Statoil and Chevron.

$$CI_f^{PRELIM} = \sum_1^3 \beta_i^C RefineryConfigf + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 \log(Sulphur) + \beta_8 \log(Sulphur)^2 + \epsilon_f \quad (24)$$

$$CI_f^{OPEM} = \sum_1^3 \beta_i^C RefineryConfigf + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 API_f^4 + \beta_8 \log(Sulphur) + \beta_9 \log(Sulphur)^2 + \beta_{10} \log(Sulphur)^3 + \epsilon_f \quad (25)$$

Regression results. Table 12 shows results from both estimations. All coefficients are significant and predicted R-squared are 0.95 and 0.88 respectively. As expected, refinery configuration explains a large share of variations across crudes, especially in the case of midstream emissions. Adding quadratically API gravity and log sulphur content improves the variance explained marginally, as the relatively low within R-squared indicates. On the contrary, API gravity and sulphur content play a much more important role in explaining the variance in downstream emissions.

Sensitivity analysis. The distribution of midstream and downstream CI are compared to predicted values using the model in figure 20. Midstream values appear grouped according to each refinery configuration, illustrating the crucial role played by the dummies in the regression results. While the inclusion of API gravity and sulphur content in the model explains a large share of within-group variation for medium and deep conversion, it does not appear to be the case for crudes using the hydroskimming configuration (bottom-left of panel A), explaining the low within R-squared value. Downstream CI are more evenly distributed across the graph, and predicted values remain close to OPEM-computed ones.

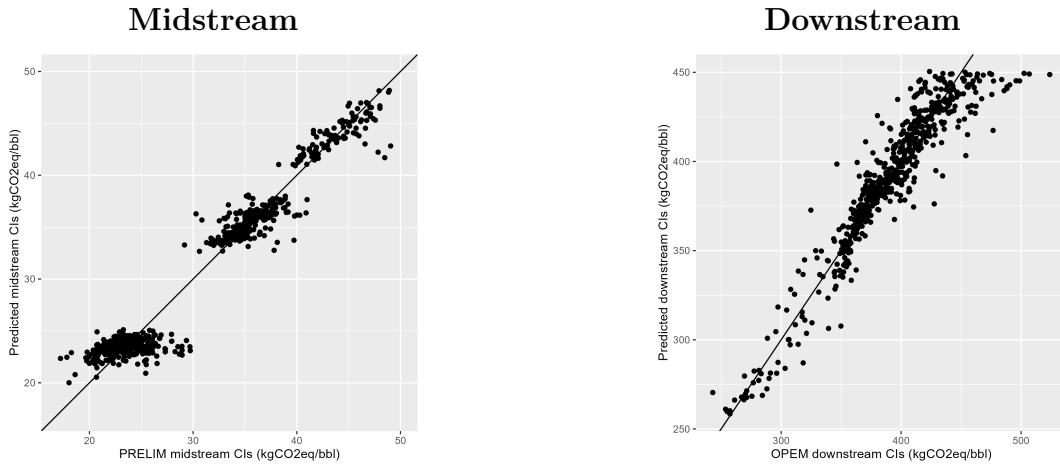


Figure 20: Comparison between PRELIM/OPEM CI and predicted ones.

Variables	Midstream	Downstream
Configuration: Deep conversion	56.552*** (1.374)	530.269*** (17.341)
Configuration: Hydroskimming	40.093*** (1.608)	587.056*** (17.764)
Configuration: Medium conversion	51.361*** (1.604)	515.626*** (18.037)
API	-0.933*** (0.124)	-16.122*** (2.159)
API ²	0.016*** (0.003)	0.781*** (0.096)
API ³	-9.62e-05*** (2.74e-05)	-0.017*** (0.002)
API ⁴		1.11e-04*** (1.16e-05)
log(Sulphur)	-0.988*** (0.103)	-7.784*** (0.832)
log(Sulphur) ²	-0.135*** (0.025)	-2.404*** (0.474)
log(Sulphur) ³		-0.19*** (0.072)
Corrected R ²	0.948	0.887
Predicted R ²	0.947	0.883
Within R ²	0.300	0.875

Table 12: Regressions results from midstream and downstream CI estimations.

Notes: The table displays coefficients from the regressions using equations 24 and 25, with standard errors indicated between parenthesis. R² are corrected in order to account for the absence of intercept, as well as predicted R². Within R² are computed using fixed-effects regressions to capture the effect of configuration dummies.

Contrary to upstream CI, midstream and downstream CI computed using PRELIM and OPEM models do not contain visible outliers. The risk of overfitting particular values is thus low. Tables 13 and 14 report re-estimated coefficients excluding most-influential observations for midstream and downstream CI respectively. In both cases, estimates remain significant and stable enough to exclude the risk of overfitting.

C.2.4 Predicting midstream/downstream CI

Rystad assets matched to PRELIM crudes are assigned midstream and downstream CIs computed using the PRELIM and OPEM models. These assets represent 59% of oil reserves in 2016. For remaining assets, I use the coefficients estimated in both regressions to predict

	Deep Conversion	Hydroskimming	Medium Conversion	API
0	56.55*** (1.37)	40.09*** (1.61)	51.36*** (1.60)	-0.93*** (0.12)
1	56.38*** (1.30)	40.05*** (1.52)	51.22*** (1.51)	-0.91*** (0.12)
10	56.69*** (1.23)	40.58*** (1.44)	51.73*** (1.44)	-0.93*** (0.11)
50	56.56*** (1.05)	40.80*** (1.22)	51.91*** (1.21)	-0.89*** (0.10)
	API ²	API ³	log(Sulphur)	log(Sulphur) ²
0	0.0159***	-9.62e-05***	-0.99*** (0.10)	-0.13*** (0.03)
1	0.0152***	-9.03e-05***	-1.00*** (0.10)	-0.13*** (0.02)
10	0.0153***	-8.77e-05***	-1.10*** (0.09)	-0.18*** (0.02)
50	0.0126***	-5.57e-05***	-1.18*** (0.08)	-0.22*** (0.02)

Table 13: Midstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

	Deep Conversion	Hydroskimming	Medium Conversion	API	API ²
0	530.27*** (17.34)	587.06*** (17.76)	515.63*** (18.04)	-16.12*** (2.16)	0.78*** (0.10)
1	543.82*** (18.09)	599.01*** (18.32)	527.93*** (18.62)	-18.39*** (2.33)	0.90*** (0.11)
10	544.71*** (16.65)	600.12*** (16.83)	528.63*** (17.12)	-18.46*** (2.15)	0.91*** (0.10)
50	555.93*** (14.05)	609.59*** (14.18)	540.02*** (14.41)	-19.59*** (1.83)	0.94*** (0.09)
	API ³	API ⁴	log(Sulphur)	log(Sulphur) ²	log(Sulphur) ³
0	-1.66e-02***	1.11e-04***	-7.78*** (0.83)	-2.40*** (0.47)	-0.19*** (0.07)
1	-1.92e-02***	1.30e-04***	-7.81*** (0.83)	-2.24*** (0.48)	-0.16** (0.07)
10	-1.93e-02***	1.31e-04***	-7.60*** (0.76)	-2.40*** (0.45)	-0.15** (0.07)
50	-1.95e-02***	1.30e-04***	-7.31*** (0.60)	-2.74*** (0.36)	-0.21*** (0.06)

Table 14: Downstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses.

their CI, at the crude level in the case of crude streams unmatched to PRELIM crude assays (17.3% of reserves) and at the asset level if the crude stream is unspecified (11.7%).

Several assets, representing 12% of world reserves, lack information on either API gravity or sulphur content. For these assets, reserve-weighted averages from similar oils were computed, according to the following order: reservoir, sub-basin, similar API within the same country, same main oil type within the same country, and finally same main oil type. The vast majority of assets are attributed values computed within the same reservoir or sub-basin. Geographically close assets having a greater probability to be blended together before refining, averages computed from assets nearby is preferable. Table 15 sums up the methods and granularity levels of midstream and downstream CI predictions.

Method	Granularity level	Share of 2016 reserves
PRELIM/OPEM	Crude stream	58.9%
Predicted	Crude stream	17.3%
	Asset-specific	11.7%
Average	Reservoir	5.4%
	Sub-basin	5.7%
	API category $\times$ country	0.7%
	Oil type $\times$ country	0.2%
	Oil type	$< 0.1\%$

Table 15: Data sources and aggregation levels for midstream and downstream carbon intensity values.

D Additional results

D.1 Overshoot against baseline emissions

Figure 21 draws the bands of Figure 3 on top of cumulative emissions under the timely policy. Baseline emissions fall from 635 GtCO₂eq at a zero tax to the 350.3 GtCO₂eq budget at \$57.2, so that at the timely tax the overshoot adds 19% to what the budget allows. The top band, the part a tax set in 2030 can still act on, is a thin layer on top of a much larger stock, which is why a small error in the calibration of the 2030 tax matters.

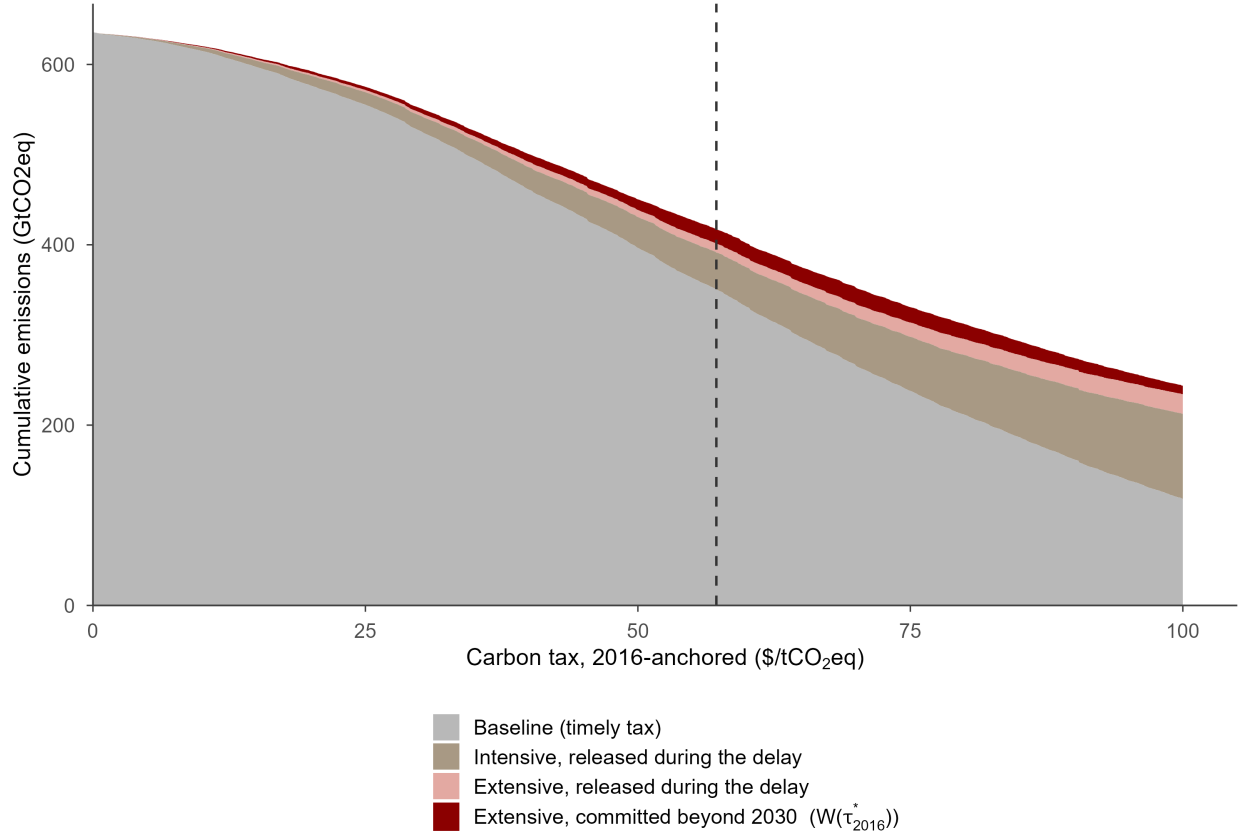


Figure 21: Overshoot caused by the delay and emissions under a timely policy.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay, by margin and date of emission. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.2 Overshoot by supply segment and region

Figure 22 shows the overshoot bands of Figure 4 without the baseline, which makes the post-2030 band of the onshore and shelf segments visible. Figure 23 shows the same decomposition by region, which Section 4.1 discusses.

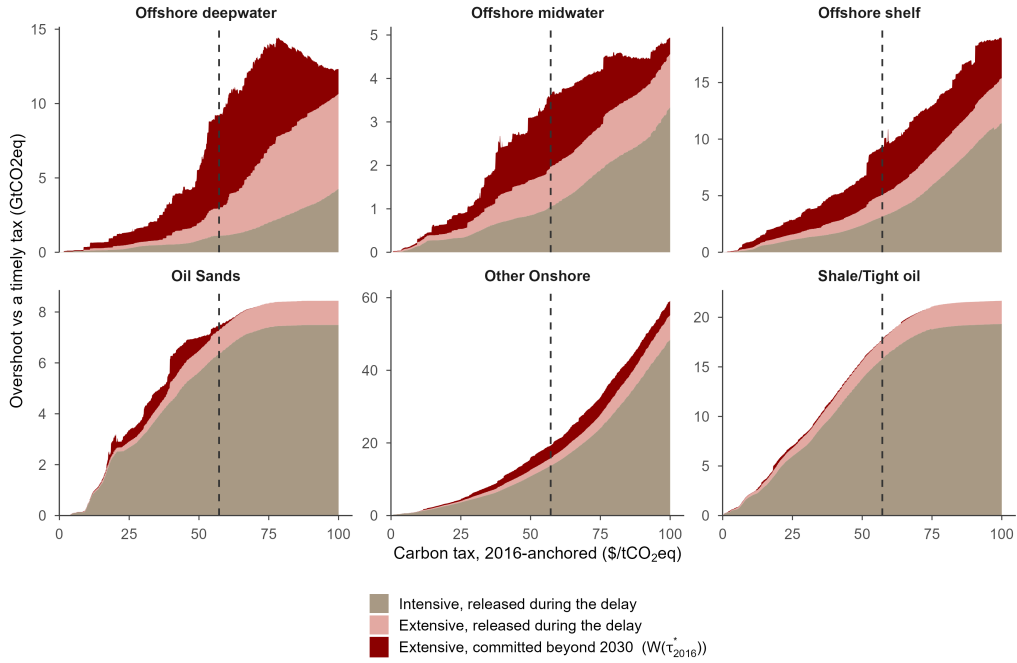


Figure 22: Overshoot caused by the delay by supply segment, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

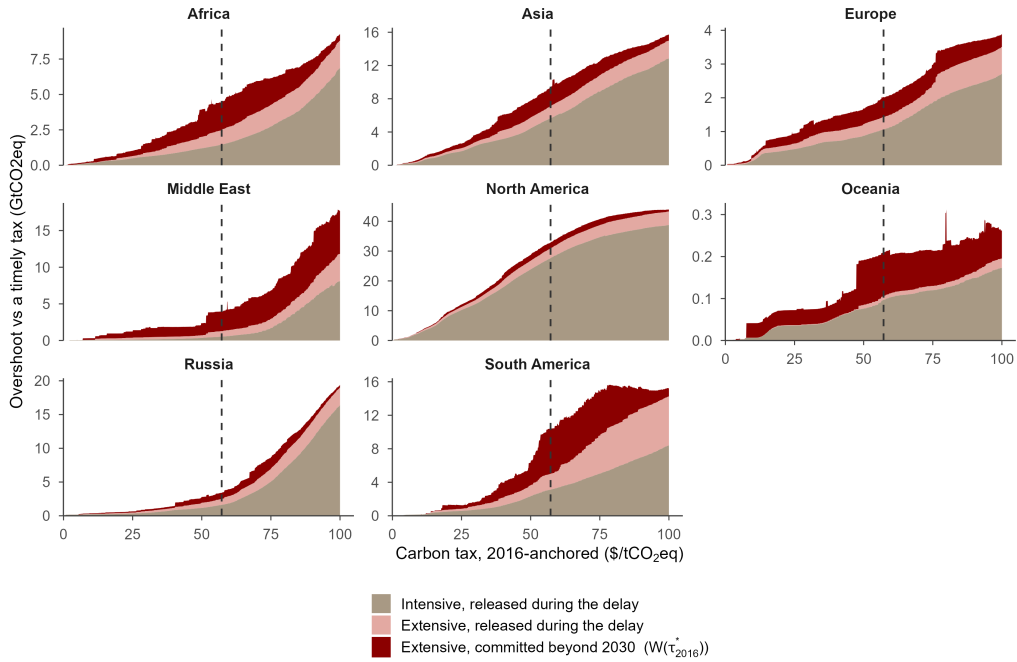


Figure 23: Overshoot caused by the delay by region, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.3 Capital intensity and the depth of lock-in, field by field

Figure 24 plots the depth of each field’s lock-in zone against its capital intensity. Panel (a) measures depth in tax units, $\bar{\tau}_P(0) - \bar{\tau}_D$, and shows that fields keep the ordering of Table 1 individually: shale fields, at the left of the distribution, have narrow zones, and offshore fields, at the right, have wide ones, despite the variation in carbon intensity and operating costs across fields. Panel (b) measures depth per barrel, $\theta(\bar{\tau}_P(0) - \bar{\tau}_D)$, which removes carbon intensity, on logarithmic scales. The fitted slope is 0.55, estimated on the 3,023 fields of the delay cohort with a defined capital intensity. The model of Section 2.2 predicts 0.5 without discounting and a higher value at any positive discount rate, so the estimate lies on the side the calibration implies. Because development capital enters both the horizontal axis and the full-cycle breakeven, part of the fit reflects this common term, and the goodness of fit should not be read as evidence for the model; the slope, which describes how the depth of lock-in scales with capital intensity, is the informative statistic.

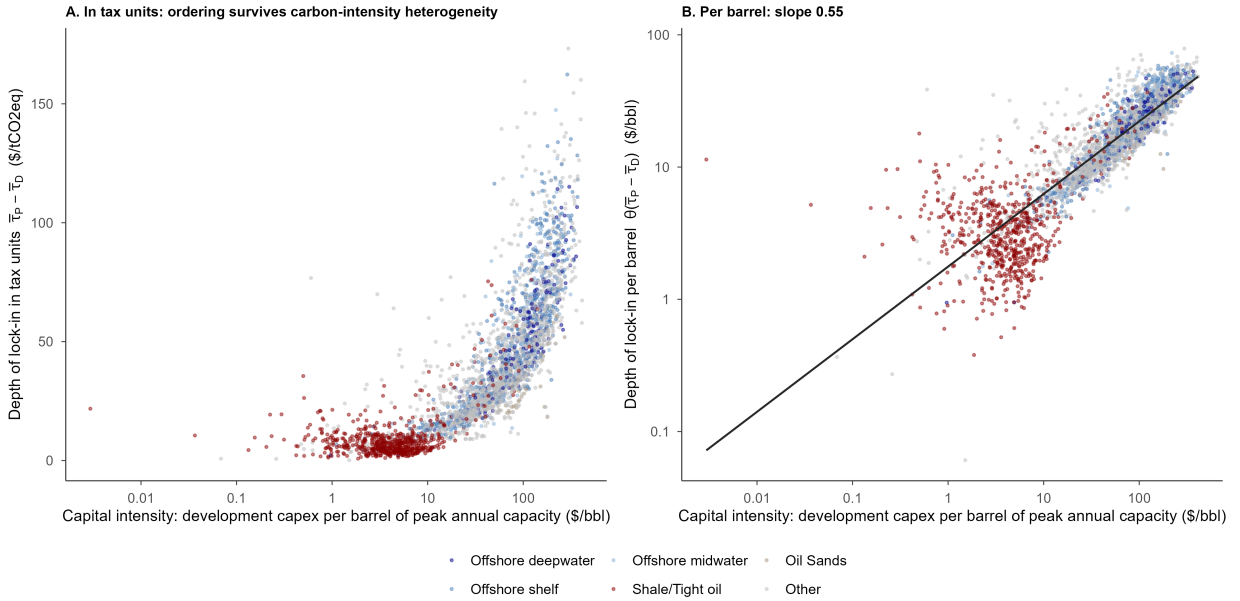


Figure 24: Capital intensity and the depth of lock-in, field by field.

Notes: Each point is one field of the delay cohort. Capital intensity is development capital expenditure per barrel of peak annual capacity. (a) Depth of the lock-in zone in tax units; (b) depth per barrel, logarithmic scales, with the fitted line.

D.4 Constant carbon tax

This appendix reports the results for a constant tax that meets the same budget under a timely policy, \$99.4/tCO₂eq. The delay then overshoots the budget by 119 GtCO₂eq, 34% of the budget, of which 78 GtCO₂eq on the intensive margin and 41 GtCO₂eq on the extensive

margin. The intensive margin is larger than under the increasing tax because a constant tax meeting the same budget is higher in the early years, so the timely policy removes more production during the delay. Table 16 is the counterpart of Table 3. In 2030, the lock-in-adjusted tax is \$133.5, the premium \$3.2 and the residual overshoot 18.9 GtCO₂eq, 13.1% of the remaining budget. The premium does not rise steadily with the length of the delay, because large fields enter and leave their lock-in zone one at a time, but the residual overshoot does, as under the increasing tax. The lock-in component of the cost of delay is therefore of the same size under both designs, while the total cost of the delay is not (Section 4.4).

Year	Budget (GtCO ₂ eq)	Adjusted tax (\$/tCO ₂ eq)	From 2016		From 2025	
			Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)	Premium (\$/tCO ₂ eq)	Residual overshoot (GtCO ₂ eq)
2016	350.3	99.4	–	–	–	–
2018	320.9	105.2	2.8	7.4 (2.3%)	–	–
2020	291.4	109.7	2.4	9.9 (3.4%)	–	–
2022	264.0	112.8	2.1	12.3 (4.7%)	–	–
2024	235.1	118.0	3.4	15.6 (6.6%)	–	–
2026	205.8	125.6	5.6	15.8 (7.7%)	0.1	2.6 (1.3%)
2028	175.6	129.8	2.5	19.4 (11.0%)	0.0	7.3 (4.2%)
2030	144.9	133.5	3.2	18.9 (13.1%)	2.4	10.3 (7.1%)

Table 16: Lock-in premium and residual overshoot by year of implementation, constant carbon tax.

Notes: Taxes in 2016 dollars. Overshoot percentages are relative to the remaining budget. “From 2016” and “From 2025” indicate the year from which the delay is measured.

D.5 Depth of lock-in

Figure 25 reports, for all delay-induced fields together, the emissions locked in at the naive tax that survive each level of the 2030 tax. The stock falls to 92% of its initial level at the lock-in-adjusted tax, and still stands at 70% at \$80 and 43% at \$90. Section 5.1 reads the same object by supply segment. The curve falls slowly over the range the budget makes relevant and steeply above it, which is the aggregate counterpart of the fact that a locked-in field stops producing only when the tax exceeds what its operating costs can bear.

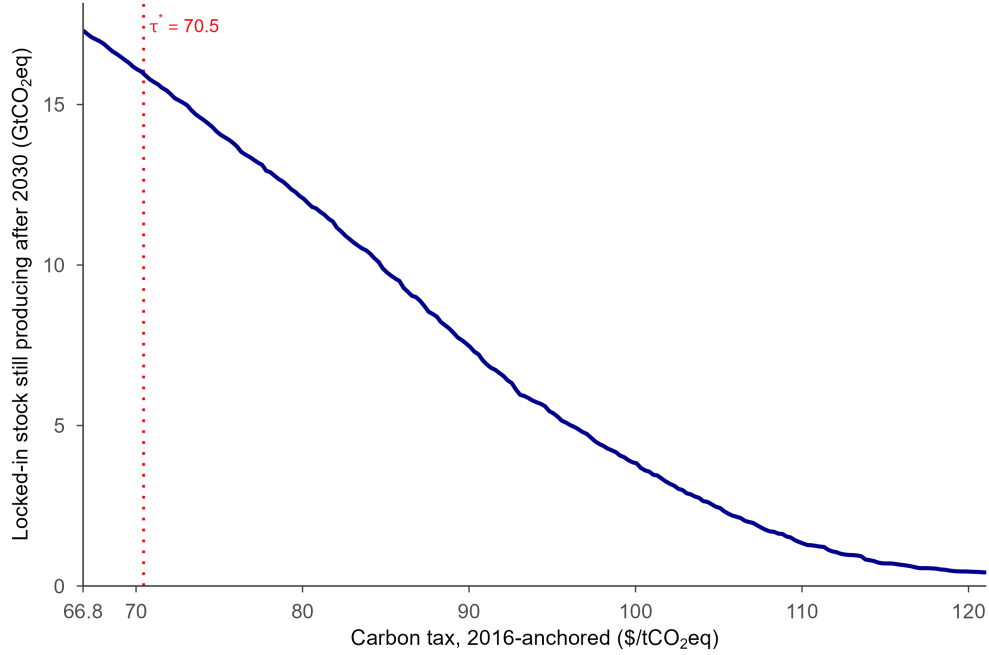


Figure 25: Depth of lock-in, all delay-induced fields.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, all delay-induced fields, in GtCO₂eq. The horizontal axis starts at the naive tax, $\tau^N = \$66.8$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$70.5$. Taxes are 2016-anchored levels of the tax rising at 2% per year.

D.6 The largest locked-in assets

Table 17 lists the twenty assets that carry the most post-2030 emissions at the naive tax, out of the 694 that carry any. They account for 31.9% of the total, and fifty assets account for 56.1%, which is what the size-based bans of Section 5.3 remove. The year of responsibility is the year from which the asset would be completed even if the tax were implemented immediately (Section 3.4).

Asset	Country	Segment	Year of responsibility	Emissions (GtCO ₂ eq)	Share	Cumulative
Marjan (Expansion)	Saudi Arabia	Offshore shelf	2023	0.87	5.0%	5.0%
Atapu (P-84)	Brazil	Offshore deepwater	2027	0.34	1.9%	7.0%
Venus Phase 1	Namibia	Offshore deepwater	2029	0.33	1.9%	8.9%
Tengiz FGP	Kazakhstan	Onshore conventional	2019	0.31	1.8%	10.7%
Satah Al Razboot (SARB)	UAE	Offshore shelf	2016	0.27	1.6%	12.2%
Buzios XI (P-83)	Brazil	Offshore deepwater	2025	0.27	1.5%	13.8%
Sepia (P-85)	Brazil	Offshore deepwater	2028	0.26	1.5%	15.3%
Kirkuk (redevelopment)	Iraq	Onshore conventional	2028	0.26	1.5%	16.8%
Buzios IX (P-80)	Brazil	Offshore deepwater	2026	0.25	1.5%	18.2%
Buzios X (P-82)	Brazil	Offshore deepwater	2026	0.25	1.4%	19.7%
Willow (GMT)	United States	Onshore conventional	2026	0.24	1.4%	21.0%
Bacalhau Phase 2	Brazil	Offshore deepwater	2029	0.23	1.3%	22.4%
Longtail	Guyana	Offshore deepwater	2028	0.23	1.3%	23.7%
Argos (Mad Dog Phase 2)	United States	Offshore deepwater	2020	0.23	1.3%	25.0%
Al Shaheen Ruya 3 (Batch 1)	Qatar	Offshore shelf	2026	0.22	1.3%	26.3%
Hammerhead	Guyana	Offshore midwater	2029	0.21	1.2%	27.5%
Hail (Sour Gas)	UAE	Offshore shelf	2027	0.20	1.1%	28.6%
Bacalhau Phase 1	Brazil	Offshore deepwater	2022	0.20	1.1%	29.7%
Azadegan North Phase 2	Iran	Onshore conventional	2028	0.19	1.1%	30.9%
Trion	Mexico	Offshore deepwater	2027	0.19	1.1%	31.9%

Table 17: The twenty largest locked-in assets.

Notes: Post-2030 emissions of delay-induced fields surviving the naive tax implemented in 2030, no-anticipation scenario, tax rising at 2% per year. Asset names are those of the Rystad database.

D.7 Incidence by country

Table 18 reports the decomposition of Table 4 for the largest producers of each region and for the producers discussed in Section 5.2.

	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% of 2016 reserves)
<i>Net winners through overproduction mainly</i>				
United States	18.63	1.40	20.03	17.5
Canada	9.36	−0.10	9.27	16.0
China	3.99	0.67	4.66	17.0
Norway	0.47	0.33	0.80	10.0
United Kingdom	0.50	0.09	0.58	−
Mexico	0.38	0.17	0.55	−
<i>Net winners through lock-in mainly</i>				
Guyana	0.19	0.73	0.92	16.3
Nigeria	0.04	0.31	0.36	3.9
<i>Sign reversed by lock-in</i>				
Brazil	−0.38	3.04	2.66	13.0
Angola	−0.08	0.24	0.15	2.0
Namibia	−0.13	0.23	0.10	4.6
Senegal	−0.02	0.09	0.07	−
Vietnam	−0.00	0.04	0.04	−
<i>Smaller losers through lock-in</i>				
Ghana	−0.11	0.05	−0.06	−
Kazakhstan	−0.83	0.34	−0.50	−4.7
Qatar	−1.21	0.27	−0.94	−5.0
Azerbaijan	−0.43	0.01	−0.41	−
UAE	−3.23	0.09	−3.14	−8.1
<i>Net losers on both counts</i>				
Saudi Arabia	−10.60	−2.53	−13.13	−10.4
Iraq	−7.35	−1.98	−9.34	−17.7
Russia	−5.41	−2.13	−7.54	−11.8
Iran	−3.93	−0.83	−4.76	−10.0
Kuwait	−1.65	−0.95	−2.60	−11.4
Algeria	−0.21	−0.01	−0.22	−4.2

Table 18: Incidence of the delay by country.

Notes: Change in post-2030 emissions relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The last column is the net change in production (in barrels) as a share of the country’s 2016 reserves.

D.8 Incidence by firm

Table 19 reports the decomposition of Table 4 for the twenty largest companies by 2016 equity reserves. Production is attributed to equity holders at their share in each year, so a firm keeps what it lifted while it held the asset, and ownership beyond 2026 is held at its 2026 level. Figure 11 plots the same decomposition for the fifty largest firms, normalised by their own reserves.

Company	Type	Headquarters	Reserves (Gbbl)	Overproduction (GtCO ₂ eq)	Lock-in (GtCO ₂ eq)	Net (GtCO ₂ eq)	Net (% res.)
Saudi Aramco	NOC	Saudi Arabia	289.7	-10.62	-2.50	-13.13	-10.4
NIOC	NOC	Iran	99.1	-3.92	-0.82	-4.74	-10.2
ADNOC	NOC	UAE	64.8	-2.08	0.02	-2.06	-7.5
Kuwait Petroleum Corp	NOC	Kuwait	56.1	-1.79	-0.97	-2.77	-11.4
PetroChina	NOC	China	52.7	0.56	-0.40	0.16	0.5
Rosneft	NOC	Russia	52.3	-2.90	-1.37	-4.28	-17.7
ExxonMobil	Major	United States	46.4	1.43	0.44	1.87	7.9
QatarEnergy	NOC	Qatar	37.5	-1.05	0.35	-0.70	-4.9
Shell	Major	United Kingdom	36.2	-0.15	0.15	0.00	-0.6
Petrobras	NOC	Brazil	33.1	-0.09	2.00	1.91	12.1
Chevron	Major	United States	29.9	2.09	0.23	2.32	15.7
Gazprom	NOC	Russia	29.8	-1.11	-0.19	-1.30	-10.3
Lukoil	Integrated	Russia	24.7	-0.47	-0.49	-0.97	-8.2
BP	Major	United Kingdom	24.4	-0.44	0.06	-0.38	-3.4
TotalEnergies	Major	France	22.5	-0.53	0.98	0.45	3.1
CNOOC	NOC	China	20.5	0.95	1.08	2.04	20.1
ConocoPhillips	Independent	United States	16.8	2.23	0.26	2.50	32.3
Basra Oil Company	NOC	Iraq	14.8	-1.64	-0.82	-2.46	-34.8
Pemex	NOC	Mexico	14.2	0.36	-0.07	0.29	4.1
Canadian Natural Resources	Independent	Canada	13.0	2.12	-0.01	2.11	27.6

Table 19: Incidence of the delay by firm.

Notes: Change in post-2030 emissions relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The twenty largest companies by 2016 equity reserves, excluding one residual category of unattributed United States partners. The last column is the net change in production (in barrels) as a share of the firm’s own 2016 equity reserves.

D.9 Definition of the targeted instruments

Each instrument of Table 5 restricts or taxes a set of assets from a starting year, and is then evaluated at the tax the remaining budget requires under that restriction.

Perimeters. A segment restriction applies to all assets of the segment, using the classification of Table 1. An ownership restriction applies to the assets that firms headquartered in the perimeter own at least half of *and* operate; Europe is the European Union together with Norway and the United Kingdom. A size restriction applies to the assets that carry the most post-2030 emissions at the naive tax in the delay without restriction, ranked as in Table 17; the twenty largest carry 31.9% of the residual overshoot and the fifty largest 56.1%.

Instruments. A ban on new approvals removes from the analysis every asset of the perimeter approved from the starting year onwards, so assets already approved are left alone and no capital already committed is cancelled. An early carbon tax applies to the assets of the perimeter from the starting year, at the level the budget requires, while the rest of the industry faces the tax only from 2030; it therefore acts on the development of new assets and on the production of assets already in operation. A shock to development capital raises the development cost of every asset of the perimeter by the stated percentage from the starting year, leaving operating costs unchanged. In every case, the naive planner observes the

restriction and its effect on development costs when calibrating the 2030 tax.

D.10 Other targeted instruments

Figure 26 reports the naive tax, the lock-in-adjusted tax, the premium between them and the residual overshoot for every instrument evaluated in Section 5.3. It covers the development bans, by segment, by perimeter of ownership and by size of project, the early carbon taxes applied to the same perimeters, and the two shocks to development capital, each of them starting in 2016 or in 2025 and each evaluated at its own budget-consistent tax.

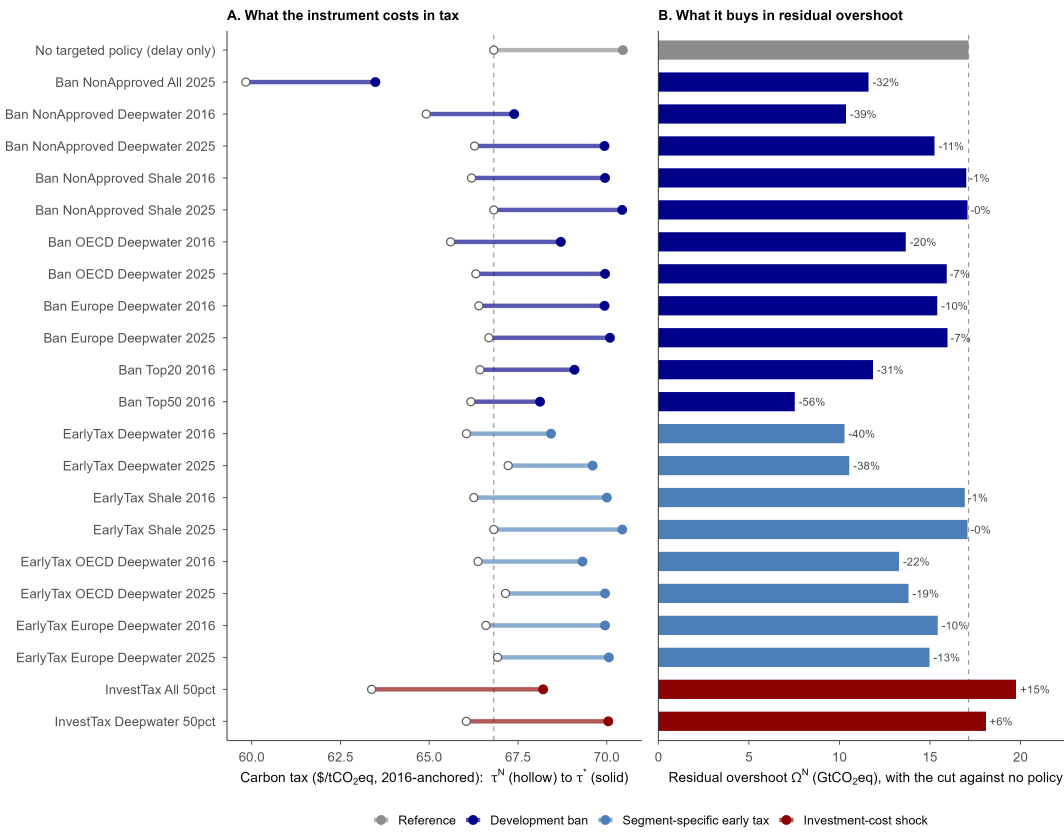


Figure 26: Targeted instruments.

Notes: Panel (a) runs from the naive tax (hollow) to the lock-in-adjusted tax (solid), so that the length of each segment is the premium. Panel (b) gives the residual overshoot and its change relative to the delay without restriction. Each instrument is evaluated at its own budget-consistent tax. Taxes are 2016-anchored levels of the tax rising at 2% per year.

References

- Brandt, A. R. and Farrell, A. E. (2007). Scraping the bottom of the barrel: greenhouse gas emission consequences of a transition to low-quality and synthetic petroleum resources. *Climatic Change*, 84(3):241–263.
- Elvidge, C. D., Zhizhin, M., Baugh, K., Hsu, F.-C., and Ghosh, T. (2015). Methods for global survey of natural gas flaring from visible infrared imaging radiometer suite data. *Energies*, 9(1):14.
- Ghosh, S., Jean, L., and Saha, P. (2025). The last good owner effect: Ownership structure and environmental outcomes in oil and gas.
- Gordon, D., Brandt, A. R., Bergerson, J., and Koomey, J. (2015). *Know your oil: creating a global oil-climate index*. Carnegie Endowment for International Peace Washington, DC.
- Gvakharia, A., Kort, E. A., Brandt, A., Peischl, J., Ryerson, T. B., Schwarz, J. P., Smith, M. L., and Sweeney, C. (2017). Methane, black carbon, and ethane emissions from natural gas flares in the bakken shale, north dakota. *Environmental Science & Technology*, 51(9):5317–5325.
- Jackson, P. M. and Smith, L. K. (2014). Exploring the undulating plateau: the future of global oil supply. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 372(2006).
- Masnadi, M. S., El-Houjeiri, H. M., Schunack, D., Li, Y., Englander, J. G., Badahdah, A., Monfort, J.-C., Anderson, J. E., Wallington, T. J., Bergerson, J. A., et al. (2018). Global carbon intensity of crude oil production. *Science*, 361(6405):851–853.
- McGlade, C., Sondak, G., and Han, M. (2018). Whatever happened to enhanced oil recovery?
- Plant, G., Kort, E. A., Brandt, A. R., Chen, Y., Fordice, G., Gorchov Negron, A. M., Schwietzke, S., Smith, M., and Zavala-Araiza, D. (2022). Inefficient and unlit natural gas flares both emit large quantities of methane. *Science*, 377(6614):1566–1571.
- World Bank (2025). Global gas flaring tracker report, july 2025. Technical report, World Bank.
- Zavala-Araiza, D., Omara, M., Gautam, R., Smith, M. L., Pandey, S., Aben, I., Almanza-Veloz, V., Conley, S., Houweling, S., Kort, E. A., et al. (2021). A tale of two regions: methane emissions from oil and gas production in offshore/onshore mexico. *Environmental Research Letters*, 16(2):024019.
- Zhizhin, M., Matveev, A., Ghosh, T., Hsu, F.-C., Howells, M., and Elvidge, C. (2021). Measuring gas flaring in russia with multispectral viirs nightfire. *Remote Sensing*, 13(16):3078.